CS1231S · Condensed Revision

Lectures 1–13, Tutorials 1–11, the AY23–25 midterms and the AY25/26 final. Definitions, theorems and worked examples are integrated by topic; Appendix A is complete. Theorem and appendix labels follow the lecture slides. Parts I–V (§1–25) are the midterm material; Parts VI–X (§26–52) cover functions, induction, cardinality, counting and probability, graphs and trees.

Find material: topic locator §63; theorem locator §65; question locators §57–58; formula sheet §56. **Final exam format (AY25/26):** 2 hours, open book, 100 marks: 23 MCQ (46) + 3 structured questions (54).

Conventions: $\mathbb{N} = \{0, 1, 2, ...\}$; $\overline{A}$ is complement; $A \setminus B$ is difference; use $\subseteq$ for subset. Largest/smallest = greatest/least. Blue boxes define terms; grey boxes contain proofs and worked examples; green boxes state results; amber boxes solve past papers.

Part I · Propositional Logic

1 · Statements, connectives, truth tables

DEFINITION Statement (2.1.1) · Statement form (2.1.5)

A **statement** (proposition) is a sentence that is true or false, but not both. A **statement form** is an expression built from statement variables and connectives that becomes a statement when statements are substituted for its variables.

DEFINITION Negation (2.1.2) · Conjunction (2.1.3) · Disjunction (2.1.4)

$\sim p$: "not p"; $p \wedge q$: "p and q"; $p \vee q$: inclusive "p or q", true when at least one is true.

p	q	$\sim p$	$p \wedge q$	$p \vee q$	$p \to q$	$p \leftrightarrow q$	$p \oplus q$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	T
F	T	T	F	T	T	F	T
F	F	T	F	F	T	T	F

Order of operations. $\sim$ first; then $\wedge$ and $\vee$, which are **coequal**, so $p \wedge q \vee r$ is ambiguous and must be bracketed; then $\to, \leftrightarrow$ last. (CS2100 puts $\wedge$ before $\vee$; CS1231S does not.)

Exclusive or: $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q)$.

A $\to$ is false in exactly one row: **T** $\to$ **F**.

DEFINITION Tautology, contradiction (2.1.7–8) · Logical equivalence (2.1.6)

A **tautology** is a statement form that is true for every assignment; a **contradiction** is false for every assignment. A statement whose form is a tautology (contradiction) is a **tautological** (**contradictory**) statement. $P \equiv Q$ iff P and Q have identical truth values under every assignment; equivalently, iff $P \leftrightarrow Q$ is a tautology.

2 · Logical equivalences and set identities

Theorem 2.1.1 (logic) and Theorem 6.2.2 (sets) side by side (Lecture 2, slides 25–26; Lecture 5, slides 42–43). Read $\wedge$ as $\cap$, $\vee$ as $\cup$, $\sim$ as complement, $\boldsymbol{t}$ as U, $\boldsymbol{c}$ as $\varnothing$. Cite laws **by name**.

Law	Logic (for statement variables)	Sets (subsets of U)
Commutative	$p \wedge q \equiv q \wedge p \quad p \vee q \equiv q \vee p$	$A \cap B = B \cap A \quad A \cup B = B \cup A$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(A \cap B) \cap C = A \cap (B \cap C)$
	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(A \cup B) \cup C = A \cup (B \cup C)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity	$p \wedge \boldsymbol{t} \equiv p \quad p \vee \boldsymbol{c} \equiv p$	$A \cap U = A \quad A \cup \varnothing = A$
Negation / Complement	$p \vee \sim p \equiv \boldsymbol{t} \quad p \wedge \sim p \equiv \boldsymbol{c}$	$A \cup \overline{A} = U \quad A \cap \overline{A} = \varnothing$
Double negation / complement	$\sim(\sim p) \equiv p$	$\overline{\overline{A}} = A$
Idempotent	$p \wedge p \equiv p \quad p \vee p \equiv p$	$A \cap A = A \quad A \cup A = A$
Universal bound	$p \vee \boldsymbol{t} \equiv \boldsymbol{t} \quad p \wedge \boldsymbol{c} \equiv \boldsymbol{c}$	$A \cup U = U \quad A \cap \varnothing = \varnothing$
De Morgan's	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\overline{A \cap B} = \overline{A} \cup \overline{B}$
	$\sim(p \vee q) \equiv \sim p \wedge \sim q$	$\overline{A \cup B} = \overline{A} \cap \overline{B}$
Absorption	$p \vee (p \wedge q) \equiv p \quad p \wedge (p \vee q) \equiv p$	$A \cup (A \cap B) = A \quad A \cap (A \cup B) = A$

Negations of $\boldsymbol{t}$, $\boldsymbol{c}$	$\sim\boldsymbol{t} \equiv \boldsymbol{c} \quad \sim\boldsymbol{c} \equiv \boldsymbol{t}$	$\overline{U} = \varnothing \quad \overline{\varnothing} = U$
Set difference	—	$A \setminus B = A \cap \overline{B}$

THEOREM **Further equivalences**

- **Implication law:** $p \to q \equiv \sim p \vee q$
- **Contrapositive:** $p \to q \equiv \sim q \to \sim p$
- **Negated conditional:** $\sim(p \to q) \equiv p \wedge \sim q$
- **Biconditional:** $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
- **Variant absorption:** $p \wedge (\sim p \vee q) \equiv p \wedge q$
- **Variant absorption:** $p \vee (\sim p \wedge q) \equiv p \vee q$
- **Exportation:** $(p \wedge q) \to r \equiv p \to (q \to r)$
- **Cases:** $(p \vee q) \to r \equiv (p \to r) \wedge (q \to r)$
- $p \to (q \vee r) \equiv (p \wedge \sim q) \to r$

The left column is proved in Lecture 2. The right column comes from tutorials and assignments; each follows from the table in a line or two, so derive it if a question says "use the laws" (the AY24 answer key cites variant absorption as an assignment result).

PAST PAPER **AY24 Q4 · Simplify** $(((p \to q) \to (q \to r)) \to (r \to s)) \vee p) \wedge (p \to (p \to q)) \wedge p$
Call the long first conjunct $X \vee p$. Answer: $\boldsymbol{p \wedge q}$ (option C).

$(X \vee p) \wedge (p \to (p \to q)) \wedge p$
$\equiv (X \vee p) \wedge (p \wedge (p \to (p \to q)))$ *(associative, commutative)*
$\equiv ((X \vee p) \wedge p) \wedge (p \to (p \to q))$ *(associative)*
$\equiv p \wedge (p \to (p \to q))$ *(commutative ×2, absorption)*
$\equiv p \wedge (\sim p \vee (\sim p \vee q))$ *(implication law ×2)*
$\equiv p \wedge (\sim p \vee q)$ *(associative, idempotent)*
$\equiv p \wedge q$ *(variant absorption)*

METHOD **Checking tautologies and equivalences**
To test whether $P \to Q$ is a tautology, try to make it false: set Q false and P true, and follow the consequences. If this forces a contradiction, $P \to Q$ is a tautology. **AY24 Q3(iii)** $(p \vee q \vee \sim r) \wedge (s \vee \sim(p \vee q)) \to (s \vee \sim r)$: falsifying needs $s = F, r = T$; the premise becomes $(p \vee q) \wedge \sim(p \vee q)$, a contradiction, so the statement is a tautology. **To show $P \not\equiv Q$,** one row is enough. AY23 Q3(i): $\sim(p \to q) \equiv p \vee (p \to \sim q)$? At $p = q = T$ the left side is F and the right side is T, so they are not equivalent.

WORKED EXAMPLE **Logic laws**
Without a truth table, show $(p \to q) \wedge (p \to \sim q) \equiv \sim p$.
$(p \to q) \wedge (p \to \sim q)$
$\equiv (\sim p \vee q) \wedge (\sim p \vee \sim q)$ *(implication law ×2)*
$\equiv \sim p \vee (q \wedge \sim q)$ *(distributive law)*
$\equiv \sim p \vee \boldsymbol{c}$ *(negation law)*
$\equiv \sim p$ *(identity law).* ■

3 · Conditional statements

DEFINITION **Conditional (2.2.1) and its relatives (2.2.2–4)**
$p \to q$ ("if p then q", "p implies q") is false only when p is true and q is false; when p is false it is **vacuously true**. p is the **hypothesis** (antecedent) and q the **conclusion** (consequent). From $p \to q$ form the **contrapositive** $\sim q \to \sim p$, the **converse** $q \to p$, and the **inverse** $\sim p \to \sim q$.

THEOREM
A conditional ≡ its contrapositive. Its converse ≡ its inverse (they are contrapositives of each other). A conditional is **not** equivalent to its converse or its inverse.

DEFINITION **Only if (2.2.5) · Biconditional (2.2.6) · Necessary/sufficient (2.2.7)**
The table below translates each phrase. "p iff q" is true exactly when their truth values agree. A necessary condition must hold; a sufficient condition guarantees the conclusion.

English	Symbolic	English	Symbolic
if p then q; p implies q	$p \to q$	p only if q	$p \to q$
q if p; q whenever p	$p \to q$	p unless q	$\sim q \to p \equiv p \vee q$
p is sufficient for q	$p \to q$	p is necessary for q	$q \to p$
p if and only if q	$p \leftrightarrow q$	neither p nor q	$\sim p \wedge \sim q$

TRAP **Everyday "if" is often an "iff"**
"If you behave, you will get ice-cream" is usually understood as a biconditional. In mathematics it is only $b \to i$; misbehaving and still getting ice-cream does not make it false.

4 · Arguments and rules of inference

DEFINITION **Argument, validity (2.3.1) · Critical row · Syllogism · Sound, unsound (2.3.2)**
An **argument** (**argument form**) is a sequence of statements (statement forms); all but the last are **premises** (assumptions, hypotheses), the last (after $\therefore$) is the **conclusion**. An argument form is **valid** iff, whatever statements are substituted for its variables, if the premises are all true then the conclusion is true. A **critical row** is a truth-table row in which all premises are T: the form is valid iff the conclusion is T in every critical row. A **syllogism** is an argument form with two premises and a conclusion (e.g. modus ponens). An argument is **sound** iff it is valid **and** all its premises are true; otherwise it is **unsound**.

Rule	Form	Rule	Form
Modus ponens	$p \to q, p \therefore q$	Generalization	$p \therefore p \lor q$
Modus tollens	$p \to q, \sim q \therefore \sim p$	Specialization	$p \land q \therefore p$
Conjunction	$p, q \therefore p \land q$	Elimination	$p \lor q, \sim q \therefore p$
Transitivity	$p \to q, q \to r \therefore p \to r$	Contradiction rule	$\sim p \to \boldsymbol{c} \therefore p$
Division into cases	$p \lor q, p \to r, q \to r \therefore r$		

Fallacies (invalid): *converse error* $p \to q, q \therefore p$; *inverse error* $p \to q, \sim p \therefore \sim q$.

PAST PAPER **AY24 Q5 · Which conclusions follow from $p \to (q \land r)$, $\sim p \to \sim r$, $\sim p \land (r \to \sim p)$?**
1. $\sim p$ *(specialization, premise 3)* then $\sim r$ *(modus ponens with premise 2)*.
2. So every critical row has $p = r = F$ (q free). Then (i) $p \to q$ is T (vacuous); (ii) $\sim p \lor \sim r$ is T; (iii) $q \to (p \to r)$ is T since $p \to r$ is T; (iv) $\sim p \land \sim q \land r$ is F since $r = F$.

Valid: **(i), (ii), (iii)**. Answer C.

PAST PAPER **AY25 Q13 · Sound vs valid**
Sound $\Rightarrow$ valid by definition, so "$\forall x(Sound(x) \to Valid(x))$" is **true**; the converse fails (valid with a false premise), so "$\exists x(\sim Sound(x) \land Valid(x))$" is **true**; the other two are false. Answer: **only I and IV**.

PAST PAPER **AY23 Q4 · Knights and knaves**
Method: let A, D, N mean "is a knight". A statement S made by X gives $X \leftrightarrow S$. Here: $A \leftrightarrow D$; $D \leftrightarrow \sim N$; "Aiken is not my type" gives $N \leftrightarrow \sim(A \leftrightarrow N)$; "Aiken lies" gives $N \leftrightarrow \sim A$. From $N \leftrightarrow \sim A$, A and N differ, so $\sim(A \leftrightarrow N)$ is T, hence N is T. Then $A = F$, $D = A = F$, and $D \leftrightarrow \sim N$ checks out. **Aiken and Dueet are knaves; Noasu is a knight.** Answer B.

PAST PAPER **AY25 Q16 (6 marks) · Four animals, four colours, nine clues**
Names: cat (3 letters), sheep (5), rabbit (6), penguin (7).
1. (2) six letters $\leftrightarrow$ black: **rabbit** is black; cat, sheep, penguin are not.
2. (3) an "e" $\to$ white: **sheep** and **penguin** are white.
3. (4) a repeated letter $\to$ not grey: sheep (ee), rabbit (bb), penguin (nn) are not grey. (5) brown $\to$ has an "i": cat and sheep are not brown.
4. (7) sheep has exactly one colour: **sheep = {white}**. (9) someone is grey, and only the cat can be: **cat = {grey}** (exactly one, by (7)).
5. (6) penguin has exactly two colours: white, and not black or grey, so **penguin = {white, brown}**.
6. (8) some animal is exactly {brown, black}; only the rabbit is black: **rabbit = {brown, black}**. (1) holds.

Model derivations

PROOF **Simplify $\sim(\sim p \land q) \land (p \lor q)$ using the laws**

$\sim(\sim p \land q) \land (p \lor q) \equiv (\sim\sim p \lor \sim q) \land (p \lor q)$ *(De Morgan's law)*
$\equiv (p \lor \sim q) \land (p \lor q)$ *(double negative law)*
$\equiv p \lor (\sim q \land q)$ *(distributive law)*
$\equiv p$ ■ *(commutative, negation and identity laws)*

PROOF **Derive variant absorption $p \wedge (\sim p \vee q) \equiv p \wedge q$ from the table**
$p \wedge (\sim p \vee q)$
$\equiv (p \wedge \sim p) \vee (p \wedge q)$ *(distributive)*
$\equiv \boldsymbol{c} \vee (p \wedge q)$ *(negation)*
$\equiv (p \wedge q) \vee \boldsymbol{c}$ *(commutative)*
$\equiv p \wedge q$ *(identity)*. ■

PROOF **A deduction with named rules: from $p \vee q$, $q \to r$, $(p \wedge s) \to t$, $\sim r$, $\sim q \to (u \wedge s)$, conclude t**
1. $\sim q$ *(from $q \to r$ and $\sim r$, by modus tollens)*
2. $u \wedge s$ *(from $\sim q \to (u \wedge s)$ and step 1, by modus ponens)* so s *(specialization)*
3. p *(from $p \vee q$ and step 1, by elimination)*
4. $p \wedge s$ *(steps 2, 3, by conjunction)* so t *(from $(p \wedge s) \to t$, by modus ponens)*. ■

p	q	r	$p \to q$	$q \to r$	$p \to r$
T	T	T	**T**	**T**	**T**
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	**T**	**T**	**T**
F	T	F	T	F	T
F	F	T	**T**	**T**	**T**
F	F	F	**T**	**T**	**T**

METHOD **Validity by critical rows**
For $p \to q, q \to r \therefore p \to r$ (transitivity): the **critical rows** are those where every premise is T: rows 1, 5, 7 and 8 (bold). The conclusion is T in each, so the argument is **valid**. A critical row with a false conclusion would make the argument invalid; that row is a counterexample.

TRAP **$\to$ is not associative**
$(p \to q) \to r \not\equiv p \to (q \to r)$: at $p = q = r = F$ the left is $T \to F = F$, the right is T. Always bracket chains of $\to$.

WORKED EXAMPLE **Validity**
Decide validity: (a) $p \to q$, $q \to r$, $\sim r \therefore \sim p$. (b) $p \to q$, $\sim p \therefore \sim q$.
(a) **Valid:** $p \to r$ *(transitivity)* with $\sim r$, $\sim p$ *(modus tollens)*.
(b) **Invalid** (inverse error): the critical row $p = F$, $q = T$ makes both premises true and the conclusion $\sim q$ false.

Part II · Quantified Statements

5 · Predicates and quantifiers

DEFINITION **Predicate (3.1.1) · Truth set (3.1.2)**
A **predicate** is a sentence with finitely many variables that becomes a statement when values are substituted. The **domain** of a variable is the set of values that may be substituted for it (also called the domain of discourse, universe of discourse, universal set, or universe). The **truth set** of $P(x)$, $x \in D$, is the set of all elements of D that make $P(x)$ true: $\{x \in D : P(x)\}$ (also written $\{x \in D \mid P(x)\}$).

DEFINITION **Kinds of mathematical statements (Lecture 1)**
Universal: "all/every/any" ($\forall$); **existential:** "some/at least one" ($\exists$); **conditional:** "if … then" ($\to$).
Combined forms:
- **Universal conditional:** $\forall x\,(P(x) \to Q(x))$.
- **Universal existential:** $\forall x\,\exists y\,P(x, y)$.
- **Existential universal:** $\exists x\,\forall y\,P(x, y)$.

DEFINITION **Universal (3.1.3) and existential (3.1.4) statements · Uniqueness**
$\forall x \in D, Q(x)$ is true iff $Q(x)$ is true for **every** $x \in D$, and false iff $Q(x)$ is false for at least one x, called a **counterexample**.
$\exists x \in D$ such that $Q(x)$ is true iff $Q(x)$ is true for **at least one** $x \in D$, and false iff $Q(x)$ is false for all $x \in D$.
$\exists! x \in D, Q(x) \quad \equiv \quad \exists x \in D\,(Q(x) \wedge \forall y \in D\,(Q(y) \to y = x))$, read "there is exactly one".

METHOD **Translating English**
Use → under ∀ and ∧ under ∃. "All A are B" is $\forall x(A(x) \to B(x))$; "some A is B" is $\exists x(A(x) \wedge B(x))$. Writing $\forall x(A(x) \wedge B(x))$ claims **everything** is an A; writing $\exists x(A(x) \to B(x))$ is true as soon as one non-A exists.
Implicit quantification: "If $x > 2$ then $x^2 > 4$" in a real-number context means $\forall x \in \mathbb{R}(x > 2 \to x^2 > 4)$.

PAST PAPER **AY24 Q2 · "All birds cannot fly"**
$\forall x(Bird(x) \to \sim Fly(x))$, option C. Option B, $\forall x(Bird(x) \wedge \sim Fly(x))$, asserts that everything is a bird; D quantifies a second, unrelated y.

6 · Negation, vacuous truth, variants

THEOREM **Theorem 3.2.1 · Theorem 3.2.2 · Negation of quantified statements**
$\sim(\forall x \in D, P(x)) \equiv \exists x \in D, \sim P(x) \qquad \sim(\exists x \in D, P(x)) \equiv \forall x \in D, \sim P(x)$
Universal conditional: $\sim\forall x \in D(P(x) \to Q(x)) \equiv \exists x \in D(P(x) \wedge \sim Q(x))$.
Nested: push $\sim$ inward, flipping each quantifier: $\sim\forall x\exists y\, P(x,y) \equiv \exists x\forall y \sim P(x,y)$.

DEFINITION **Vacuous truth**
$\forall x \in D(P(x) \to Q(x))$ is **vacuously true** when no $x \in D$ satisfies $P(x)$. In particular, every universal statement over an **empty** domain is true, and every existential statement over an empty domain is false.

DEFINITION **Variants (3.2.1) · Necessary, sufficient, only if (3.2.2)**
For $\forall x \in D(P(x) \to Q(x))$: **contrapositive** $\forall x(\sim Q(x) \to \sim P(x))$ (equivalent), **converse** $\forall x(Q(x) \to P(x))$, **inverse** $\forall x(\sim P(x) \to \sim Q(x))$.
"$r(x)$ is sufficient for $s(x)$": $\forall x(r(x) \to s(x))$. "$r(x)$ is necessary for $s(x)$": $\forall x(s(x) \to r(x))$. "$r(x)$ only if $s(x)$": $\forall x(r(x) \to s(x))$.

PAST PAPER **AY23 Q2 · Given $\forall x \in S(SoC(x) \wedge CS1231(x))$ is true**
(i) All computing students took CS1231S: **true**. (ii) Every student is a computing student: **true**, because of the $\wedge$. (iii) $S \neq \varnothing$: **false**, since the statement is vacuously true when $S = \varnothing$. (iv) Some non-computing student took it: **false**, as there is no such student. Only (i) and (ii) hold; no option lists this, so the answer is **E**.

PAST PAPER **AY25 Q2 · $\forall x \in D(Red(x) \wedge Flying(x))$ for the objects in LT11**
A: if the statement is true, every object is red. True. B: if true, every object flies. True. C: it is true when LT11 is empty. True, vacuously. D: if it is false, there is a counterexample object, so LT11 is non-empty. True.
Options A–D are all correct, so the false option is **E**.

7 · Multiple quantifiers

THEOREM **Order of quantifiers**
Adjacent quantifiers **of the same kind** commute: $\forall x\forall y \equiv \forall y\forall x$ and $\exists x\exists y \equiv \exists y\exists x$. Swapping a $\forall$ past an $\exists$ usually changes the meaning: in $\forall x\exists y\, P(x,y)$ the witness y **may depend on** x; in $\exists y\forall x\, P(x,y)$ one y must work for all x. $\quad \exists y\forall x\, P \Rightarrow \forall x\exists y\, P$, never conversely in general.

METHOD **Evaluating and proving nested statements**
Read the quantifiers from left to right. A $\forall$ variable may take any value; an $\exists$ variable is chosen afterwards and may depend on the values chosen before it.
To **prove** $\forall x\exists y\, P$: "Let x be arbitrary. Let $y = f(x)$. Then $P(x, f(x))$ because …". To **disprove** it, prove the negation $\exists x\forall y \sim P$: give one x and show that every y fails for it.

PAST PAPER **AY23 Q5 · Domain $\mathbb{R}$**
(i) $\forall x\exists y(xy = 1)$ is **false**: at $x = 0$, $0 \cdot y = 0 \neq 1$ for every y.
(ii) $\exists x\forall y(x^2 \geq y^2)$ is **false**: for any x, $y = |x| + 1$ gives $y^2 > x^2$.
(iii) $\exists x\forall y\exists z(xy^2 = yz)$ is **true**: take $x = 0$; for any y choose $z = 0$. (Or $x = 1$, $z = y$.) Answer **B**.

PAST PAPER **AY23 Q16 (6 marks) · Solving for the witnesses**
(a) $A=\{1,2,3\}$, $B=\{4,5,6\}$; smallest positive integer that can lie in C with $\forall x\in A\,\forall y\in B\,\forall z\in C\,(|x-y|\le|y-z|)$. For each y the worst x is 1, so we need $|y-z|\ge y-1$ for $y=4,5,6$: at $z=1$, $|y-1|=y-1$. So $\boldsymbol{z=1}$.
(b) $\exists y\in\mathbb{Z}^+\,\forall x\in\{10,\ldots,50\}\,\forall z\in\{3,5,7,9\}\,(x\le y^2\wedge y\le z^2)$: need $y^2\ge 50$, so $y\ge 8$, and $y\le 3^2=9$. So $\boldsymbol{y\in\{8,9\}}$.
(c) $\forall x\in\mathbb{Z}\,\exists y\in\mathbb{R}\setminus\{0\}\,(xy<1)$: take $\boldsymbol{y=-1}$ **if** $\boldsymbol{x\ge 0}$, $\boldsymbol{y=1}$ **if** $\boldsymbol{x<0}$; then $xy=-x\le 0$ or $xy=x<0$, both <1. (The key lists several others, e.g. $y=1$ at $x=0$ and $y=-x$ otherwise.)

TRAP **Hidden domain changes**
$\forall x\in\mathbb{Z}^+\,\exists y\in\mathbb{Z}(x+y=0)$ is true but becomes false over $y\in\mathbb{Z}^+$. Check every domain again after negating or reordering quantifiers.

WORKED EXAMPLE **Nested quantifiers**
Decide and justify: (a) $\forall x\in\mathbb{Z}\,\exists y\in\mathbb{Z}\,(y>x^2)$; (b) $\exists y\in\mathbb{Z}\,\forall x\in\mathbb{Z}\,(y>x^2)$.
(a) **True.** Let $x\in\mathbb{Z}$ be arbitrary; take $y=x^2+1\in\mathbb{Z}$ *(closure)* then $y>x^2$.
(b) **False.** Its negation $\forall y\in\mathbb{Z}\,\exists x\in\mathbb{Z}\,(y\le x^2)$ holds: given y, take $x=y$; then $y\le y^2$ for every integer y (if $y\le 0$ then $y\le 0\le y^2$; if $y\ge 1$ then $y^2=y\cdot y\ge y$).

8 · Arguments with quantified statements

DEFINITION **Valid argument form (3.4.1)**
Valid iff every substitution of predicates that makes all premises true also makes the conclusion true.
An argument is valid iff its form is valid.

Rule	Form	Rule	Form
Universal instantiation	$\forall x\in D\,P(x);\ a\in D\therefore P(a)$	Universal generalization	$P(a)$ for an **arbitrary** $a\in D\therefore\forall x\in D\,P(x)$
Universal modus ponens	$\forall x(P(x)\to Q(x));\ P(a)\therefore Q(a)$	Universal modus tollens	$\forall x(P(x)\to Q(x));\ \sim Q(a)\therefore\sim P(a)$
Existential instantiation	$\exists x\,P(x)\therefore P(c)$ for some **new** c	Existential generalization	$P(a);\ a\in D\therefore\exists x\in D\,P(x)$

Invalid: quantified **converse error** $\forall x(P(x)\to Q(x));\ Q(a)\therefore P(a)$ and **inverse error** $\forall x(P(x)\to Q(x));\ \sim P(a)\therefore\sim Q(a)$.
Validity with diagrams: draw "all P are Q" as the P-disc inside the Q-disc; the argument is valid iff **every** consistent drawing forces the conclusion.

PAST PAPER **AY25 Q6 (number theory in quantifier form) · n a perfect square**
(i) "n is prime" is false ($4=2^2$). (ii) "n has an odd number of positive factors" is true for $n\ge 1$ (divisors pair up as $d\leftrightarrow n/d$, except $d=\sqrt{n}$), but $\boldsymbol{n=0}$ **is a perfect square with infinitely many factors**, so the key accepted both C and D. (iii) "$p\mid n\Rightarrow p^2\mid n$" is true: if $n=m^2$ and $p\mid m^2$ then $p\mid m$ (unique factorization), so $p^2\mid m^2$.

Model solutions with nested quantifiers

PROOF **Negating the definition of a limit (Lecture 1)**
$\lim_{n\to\infty}a_n=L$ means $\forall\varepsilon\in\mathbb{R}^+\,\exists N\in\mathbb{Z}\,\forall n\in\mathbb{Z}\,(n>N\to L-\varepsilon<a_n<L+\varepsilon)$.
Its negation, one quantifier at a time: $\exists\varepsilon\in\mathbb{R}^+\,\forall N\in\mathbb{Z}\,\exists n\in\mathbb{Z}\,(n>N\wedge(a_n\le L-\varepsilon\vee a_n\ge L+\varepsilon))$. ■

PROOF $\forall x\in\mathbb{Z}\,\exists y\in\mathbb{Z}\,(x+y=0)$ **is true;** $\exists y\in\mathbb{Z}\,\forall x\in\mathbb{Z}\,(x+y=0)$ **is false**
1. Let x be an arbitrary integer and let $y=-x$. Then $y\in\mathbb{Z}$ *(closure)* and $x+y=0$.
2. For the second, prove the negation $\forall y\in\mathbb{Z}\,\exists x\in\mathbb{Z}\,(x+y\neq 0)$: given any y, take $x=1-y$; then $x+y=1\neq 0$. ■

English	Symbolic ($L(x,y)$: x loves y)
Everybody loves somebody.	$\forall x\,\exists y\,L(x,y)$
Somebody is loved by everybody.	$\exists y\,\forall x\,L(x,y)$
Nobody loves everybody.	$\sim\exists x\,\forall y\,L(x,y)\equiv\forall x\,\exists y\sim L(x,y)$
There is exactly one person everybody loves.	$\exists!y\,\forall x\,L(x,y)$

PROOF **Formal negation of a nested quantified conditional (Lecture 3, Tarski's world)**
"There is a square x such that for all triangles y, x is to the right of y":
$\exists x\,(Square(x) \wedge \forall y\,(Triangle(y) \to RightOf(x,y)))$. Negate step by step:

$$\begin{aligned}
&\sim\exists x\,(Square(x) \wedge \forall y\,(Triangle(y) \to RightOf(x,y))) \\
&\equiv \forall x \sim(Square(x) \wedge \forall y\,(Triangle(y) \to RightOf(x,y))) && \textit{(negation of } \exists\textit{)} \\
&\equiv \forall x\,(\sim Square(x) \vee \sim\forall y\,(Triangle(y) \to RightOf(x,y))) && \textit{(De Morgan)} \\
&\equiv \forall x\,(\sim Square(x) \vee \exists y \sim(Triangle(y) \to RightOf(x,y))) && \textit{(negation of } \forall\textit{)} \\
&\equiv \forall x\,(\sim Square(x) \vee \exists y\,(Triangle(y) \wedge \sim RightOf(x,y))) && \textit{(negated conditional)}
\end{aligned}$$

"Every object is either not a square, or has some triangle it is not to the right of." ■

DRILL **Translate, then negate (answers after the arrow)**
- "Every student has a friend": $\forall s\,\exists f\,F(s,f)$ → negation $\exists s\,\forall f \sim F(s,f)$.
- "No integer is both even and odd": $\forall n \sim(E(n) \wedge O(n))$ → negation $\exists n\,(E(n) \wedge O(n))$.
- "All primes greater than 2 are odd": $\forall p\,(P(p) \wedge p > 2 \to O(p))$ → $\exists p\,(P(p) \wedge p > 2 \wedge \sim O(p))$.
- "x is a **minimal** element": $\forall y\,(y \preccurlyeq x \to y = x)$ → $\exists y\,(y \preccurlyeq x \wedge y \neq x)$.

Part III · Number Theory and Methods of Proof

9 · Writing proofs

Term	Meaning (Lecture 1)
Definition	precise meaning of a term: all and only the required properties
Axiom / postulate	assumed true without proof (Peano, Euclid's postulates)
Theorem	a major result, proved rigorously
Lemma	a small result used to prove a theorem
Corollary	a simple deduction from a theorem
Conjecture	believed true, unproved (Goldbach)

METHOD **Proof format**
- **Number the steps;** nest sub-arguments $(1.1, 1.2, \ldots)$.
- **Justify every step** in brackets: "by definition of odd", "by closure of integers under +", "by basic algebra", "by Theorem 4.3.3".
- Open a universal proof with "Let x be a **arbitrary** element of D".
- Give each existential witness a **fresh** name.
- End with the claim, restated, then ■.

DEFINITION **Basic properties of integers (Lecture 1, Appendix A)**
For all $x, y, z \in \mathbb{Z}$: **closure** ($x+y, x-y, xy \in \mathbb{Z}$), **commutativity**, **associativity**, **distributivity**, **trichotomy** (exactly one of $x < y$, $x = y$, $x > y$). Appendix A's field and order axioms for $\mathbb{R}$ may also be cited; the complete Appendix A appears in §64.
Assumption 1: every integer is even or odd, but not both. **Assumption 2:** every rational number can be reduced to lowest terms.

TRAP **Common mistakes in proofs**
Arguing from examples · reusing a letter ($m = 2r$ and $n = 2r$ makes $m = n$) · assuming what is to be proved · "therefore" without a reason · writing "$a \mid b = 3$" ($a \mid b$ is a statement, not a number).

WORKED EXAMPLE **T1 Q11 · Odd square iff odd integer · Proposition 4.6.4**
For $n \in \mathbb{Z}$, a proposed "contradiction proof" shows only that n even implies n^2 even. **Faults:** it uses contraposition, proves only one direction, and omits justifications. A complete proof is:
1. ($\Rightarrow$) Suppose n is even: $n = 2k$, $k \in \mathbb{Z}$. Then $n^2 = 2(2k^2)$ is even *(definition; closure)*. By contraposition, n^2 odd implies n odd.
2. ($\Leftarrow$) If $n = 2k+1$, $k \in \mathbb{Z}$, then $n^2 = 2(2k^2+2k)+1$ is odd *(algebra; closure)*. Hence n^2 is odd iff n is odd. ■

Proposition 4.6.4: n^2 even implies n even, by contraposition of line 2 and the even/odd dichotomy.

10 · Definitions and results in number theory

DEFINITION Even, odd · Prime, composite
$n \in \mathbb{Z}$ is **even** $\Leftrightarrow \exists k \in \mathbb{Z}\,(n = 2k)$; **odd** $\Leftrightarrow \exists k \in \mathbb{Z}\,(n = 2k+1)$.
n is **prime** iff $n > 1$ and $\forall r, s \in \mathbb{Z}^+\,(n = rs \to (r = 1 \wedge s = n) \vee (r = n \wedge s = 1))$.
n is **composite** iff $n > 1$ and $\exists r, s \in \mathbb{Z}^+\,(n = rs \wedge 1 < r < n \wedge 1 < s < n)$. 1 is neither.
An equivalent definition of prime: $n > 1 \wedge \forall r, s \in \mathbb{Z}\,(r > 1 \wedge s > 1 \to rs \neq n)$.

DEFINITION Rational, irrational · Lowest terms
$r \in \mathbb{R}$ is **rational** $\Leftrightarrow \exists a, b \in \mathbb{Z}\,(r = a/b \wedge b \neq 0)$; otherwise **irrational**. A fraction a/b is in **lowest terms** if the largest integer dividing both a and b is 1.

DEFINITION Divisibility
For $n, d \in \mathbb{Z}$: $d \mid n \Leftrightarrow \exists k \in \mathbb{Z}\,(n = dk)$. Read: d divides n; n is a multiple of d; d is a factor/divisor of n.
Consequences: every d divides 0; 0 divides only 0; $1 \mid n$ and $n \mid n$ for all n. No division is performed: $3 \mid 12$ is **true** and $3 \mid 10$ is **false**.

DEFINITION Absolute value (Lecture 1) · Colorful (Lecture 1, CS1231S only)
For $x \in \mathbb{R}$, the **absolute value** of x is $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.
An integer n is **colorful** iff there exists some integer k such that $n = 3k$. (Non-standard term, used only in the lecture: $-1353 = 3 \cdot (-451)$ and $0 = 3 \cdot 0$ are colorful; 7 is not.)

THEOREM Theorem 4.4.1 · Quotient–Remainder · div and mod
For $n \in \mathbb{Z}$ and $d \in \mathbb{Z}^+$ there exist **unique** $q, r \in \mathbb{Z}$ with $n = dq + r$ and $0 \leq r < d$. We write n div $d = q$ and $n \bmod d = r$. Hence every integer is of exactly one form $dq, dq+1, \ldots, dq+(d-1)$. This is the basis for **division into cases**.

Slide-numbered results and proofs: §12; **theorem locator:** §65; **Appendix A:** §64.

11 · Choosing the method

To show	Method	Skeleton
$\exists x \in D\,P(x)$	Construction	Name a witness $x = \ldots$; verify $x \in D$ and $P(x)$. There is no need to explain how it was found.
$\sim\forall x \in D(P(x) \to Q(x))$	Counterexample	One x with $P(x)$ true and $Q(x)$ false. One suffices.
$\forall x \in D\,P(x)$, D finite	Exhaustion	Check every case explicitly.
$\forall x \in D(P(x) \to Q(x))$	Direct (generic particular)	Let $x \in D$ be arbitrary with $P(x)$. Unpack definitions; rebuild $Q(x)$.
same	Division into cases	Split by parity / by $x \bmod d$ (QR theorem) / by sign; prove each case.
same	Contraposition	Prove $\forall x(\sim Q(x) \to \sim P(x))$ directly.
any statement S	Contradiction	Suppose $\sim S$. Derive a contradiction. Conclude S.
$P \Leftrightarrow Q$	Two directions	Prove $P \Rightarrow Q$ and $Q \Rightarrow P$ separately.
$\exists! x\,P(x)$	Existence + uniqueness	Give x with $P(x)$; then if $P(y)$, show $y = x$.

METHOD Contradiction or contraposition?
Use **contraposition** when the negated conclusion is easier to work with; "n is odd" is easier to use than "n^2 is not odd". Use **contradiction** for statements such as "there is no …", "there are infinitely many …" and "… is irrational", whose negations assert that some object exists.

WORKED EXAMPLE T2 Q8 · Cases inside a conditional
Prove $\forall x \in \mathbb{R}\,((x^2 > x) \to ((x < 0) \vee (x > 1)))$.
1. Let $r \in \mathbb{R}$ with $r^2 > r$. Then $r(r-1) > 0$ *(algebra)*.
2. By Appendix A, Theorem T25, r and $r-1$ have the same sign. If both are positive, $r - 1 > 0$, so $r > 1$. If both are negative, $r < 0$.
3. Thus $r < 0$ or $r > 1$; the statement follows since r was arbitrary. ■

The case $x^2 \leq x$ needs no proof: the implication is vacuously true. A graph alone is not a proof.

PAST PAPER **T2 Q11 · A factor is at most $\sqrt{n}$**
For $n, a, b \in \mathbb{Z}^+$, prove $n = ab \to (a \leq \sqrt{n} \vee b \leq \sqrt{n})$. **Contraposition:** suppose $a > \sqrt{n}$ and $b > \sqrt{n}$. Then $ab > \sqrt{n} \cdot \sqrt{n} = n$ *(Appendix A, Theorem T27)*, hence $n \neq ab$. ■
Contradiction: suppose $n = ab$ and both factors exceed $\sqrt{n}$. The same inequality gives $ab > n$, contradicting $ab = n$. ■

TRAP **T2 Q4 · Closure claims**
$\mathbb{Z}$ is **not** closed under division ($3/2 = 1.5 \notin \mathbb{Z}$). Division is not a binary operation on **all** of $\mathbb{Q} \times \mathbb{Q}$, since $r/0$ is undefined; division of a rational by a **nonzero** rational does give a rational. In proving $\mathbb{Q}$ closed under $+$, the line "$bd \neq 0$ *(zero product property)*" is required, and is commonly omitted.

12 · Model proofs

PROOF **Sum of two even integers is even (Lecture 4, Ex. 4)**
1. Let m and n be arbitrary even integers.
   1.1. Then $m = 2r$ and $n = 2s$ for some integers r and s *(by definition of even)*.
   1.2. $m + n = 2r + 2s = 2(r + s)$ *(by basic algebra)*.
   1.3. $r + s$ is an integer *(by closure of integers under $+$)*, so $m + n$ is even *(by definition of even)*.
2. Therefore the sum of any two even integers is even. ■

Lecture 3.4.3: "by definition" and "by closure" apply universal modus ponens: instantiate the general rule at the chosen integers, then use its hypothesis.

WORKED EXAMPLE **Odd products and contraposition · Lecture 1, Ex. 1**
Prove that the product of two odd integers is odd; deduce that an even product has an even factor.
1. Let $a = 2r + 1$ and $b = 2s + 1$, where $r, s \in \mathbb{Z}$ *(definition of odd)*.
2. $ab = 2(2rs + r + s) + 1$, with $2rs + r + s \in \mathbb{Z}$ *(algebra; closure)*. Thus ab is odd.
3. By contraposition and the even/odd dichotomy, ab even implies a even or b even. ■

In particular, two consecutive odd integers have odd product. Taking $s = r + 1$ gives $ab = 2(2r^2 + 4r + 1) + 1$.

PROOF **Theorem 4.2.2 · The sum of any two rational numbers is rational**
1. Let r and s be arbitrary rational numbers.
   1.1. Then $r = a/b$ and $s = c/d$ for some integers a, b, c, d with $b \neq 0$, $d \neq 0$ *(by definition of rational)*.
   1.2. $r + s = (ad + bc)/(bd)$ *(by basic algebra)*.
   1.3. $ad + bc$ and bd are integers *(by closure under $+$ and $\times$)*, and $bd \neq 0$ *(zero product property)*, so $r + s$ is rational *(by definition of rational)*.
2. Therefore the sum of any two rational numbers is rational. ■

Corollary 4.2.3: if $r \in \mathbb{Q}$, then $2r = r + r \in \mathbb{Q}$ by this theorem.

PROOF **Theorem 4.3.1 · For all $a, b \in \mathbb{Z}^+$, if $a \mid b$ then $a \leq b$**
1. Let a, b be positive integers with $a \mid b$.
   1.1. Then $b = ak$ for some integer k *(by definition of divisibility)*.
   1.2. Since a and b are positive, k is positive, i.e. $k \geq 1$.
   1.3. Therefore $a \leq ak = b$. ■

PROOF **Theorem 4.3.2 · The only divisors of 1 are 1 and -1 (division into cases)**
1. Let m be any integer with $m \mid 1$. Then $1 = mk$ for some integer k *(by definition of divisibility)*.
   1.1. Since $mk > 0$, m and k are both positive or both negative.
   1.2. **Case 1:** both positive. Then $m \leq 1$ *(by Theorem 4.3.1)*, so $m = 1$.
   1.3. **Case 2:** both negative. Then $1 = (-m)(-k)$, so $-m$ is a positive divisor of 1; as in Case 1, $-m = 1$, i.e. $m = -1$.
2. Therefore the only divisors of 1 are 1 and -1. ■

PROOF **Theorem 4.3.3 · Transitivity of divisibility**
1. Suppose a, b, c are integers with $a \mid b$ and $b \mid c$.
   1.1. Then $b = ar$ and $c = bs$ for some integers r, s *(by definition of divisibility)*.
   1.2. $c = bs = (ar)s = a(rs)$ *(by substitution and associativity)*.
   1.3. rs is an integer *(by closure under $\times$)*, so $a \mid c$ *(by definition of divisibility)*. ■

PROOF **Every integer n has $n^2 \bmod 4 \in \{0, 1\}$ (division into cases, via parity)**
1. Let n be any integer. n is even or odd *(Assumption 1)*.

1.1. **Case 1:** $n = 2k$ for some integer k. Then $n^2 = 4k^2 = 4(k^2) + 0$, and $k^2 \in \mathbb{Z}$ *(closure)*.
1.2. **Case 2:** $n = 2k + 1$. Then $n^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1$, and $k^2 + k \in \mathbb{Z}$ *(closure)*.
2. In either case $n^2 = 4q + r$ with $r \in \{0, 1\}$, so $n^2 \bmod 4 \in \{0, 1\}$ *(by the quotient–remainder theorem, r is unique)*. ■

PROOF **Theorem 4.6.1 · There is no greatest integer (contradiction)**
1. Suppose not; that is, there is a greatest integer g, so $g \geq n$ for every integer n.
1.1. Let $G = g + 1$. Then G is an integer *(by closure under +)* and $G > g$.
1.2. This contradicts that g is the greatest integer.
2. Hence the supposition is false, and there is no greatest integer. ■

PROOF **Theorem 4.7.1 · $\sqrt{2}$ is irrational (contradiction)**
1. Suppose not; that is, $\sqrt{2}$ is rational.
1.1. Then $\sqrt{2} = a/b$ for some integers a, b with $b \neq 0$ *(by definition of rational)*.
1.2. Reduce a/b to lowest terms m/n *(Assumption 2)* then $m^2 = 2n^2$ *(by basic algebra)*.
1.3. So m^2 is even *(by definition of even, as $n^2 \in \mathbb{Z}$)*, hence m is even *(by Proposition 4.6.4)*.
1.4. Let $m = 2k$. Then $4k^2 = 2n^2$, so $n^2 = 2k^2$ is even, hence n is even *(by Proposition 4.6.4)*.
1.5. So 2 divides both m and n, contradicting that m/n is in lowest terms.
2. Therefore $\sqrt{2}$ is irrational. ■

PROOF **rational + irrational is irrational (contradiction)**
1. Suppose not: some rational r and irrational s have $r + s$ rational.
1.1. Then $r = a/b$ and $r + s = c/d$ with $a, b, c, d \in \mathbb{Z}$, $b, d \neq 0$ *(by definition of rational)*.
1.2. So $s = c/d - a/b = (bc - ad)/(bd)$, with $bc - ad, bd \in \mathbb{Z}$ and $bd \neq 0$ *(closure; zero product property)*.
1.3. Hence s is rational *(by definition of rational)*, contradicting that s is irrational. ■

PROOF **The set of primes is infinite (contradiction)**
1. Suppose not: the primes are exactly $p_1, p_2, ..., p_n$. Let $N = p_1 p_2 \cdots p_n + 1$.
1.1. $N > 1$, so some prime p divides N *(existence of a prime divisor, proved below)* $p = p_i$ for some i *(the list is complete)*.
1.2. p_i divides $p_1 \cdots p_n = N - 1$ as well. But a prime dividing a cannot divide $a + 1$ *(consecutive-integer divisibility result, proved below)*. Contradiction.
2. Hence there are infinitely many primes. ■

PROOF **There exist irrational p, q with p^q rational (Lecture 1, Ex. 7; non-constructive proof by cases)**
1. $\sqrt{2}$ is irrational *(Theorem 4.7.1)*. Consider $\sqrt{2}^{\sqrt{2}}$: it is rational or irrational.
1.1. **Case 1:** rational. Take $p = q = \sqrt{2}$.
1.2. **Case 2:** irrational. Take $p = \sqrt{2}^{\sqrt{2}}$, $q = \sqrt{2}$: then $p^q = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2$, rational.
2. In either case the required p and q exist. ■

PROOF **Any two consecutive integers have opposite parity**
1. Let m and $m + 1$ be two arbitrary consecutive integers. m is even or odd *(Assumption 1)*.
1.1. **Case 1:** $m = 2k$ for some $k \in \mathbb{Z}$. Then $m + 1 = 2k + 1$ is odd *(by definition of odd)*.
1.2. **Case 2:** $m = 2k + 1$. Then $m + 1 = 2(k + 1)$ with $k + 1 \in \mathbb{Z}$ *(closure)*, so $m + 1$ is even.
2. In both cases m and $m + 1$ have opposite parity. ■

PROOF **The square of any odd integer has the form $8m + 1$ (division into cases via QR)**
1. Let n be a arbitrary odd integer. By the QR theorem with $d = 4$, $n = 4q$, $4q + 1$, $4q + 2$ or $4q + 3$ for some $q \in \mathbb{Z}$; as n is odd, $n = 4q + 1$ or $n = 4q + 3$.
1.1. **Case 1:** $n^2 = 16q^2 + 8q + 1 = 8(2q^2 + q) + 1$.
1.2. **Case 2:** $n^2 = 16q^2 + 24q + 9 = 8(2q^2 + 3q + 1) + 1$.
2. In both cases $n^2 = 8m + 1$ for an integer m *(closure)*. ■

PROOF **Theorem 4.4.6 · Triangle inequality $|x + y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$**
1. By Lemma 4.4.4, $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$; adding, $-(|x| + |y|) \leq x + y \leq |x| + |y|$.
1.1. **Case 1:** $x + y \geq 0$. Then $|x + y| = x + y \leq |x| + |y|$ *(definition of absolute value)*.
1.2. **Case 2:** $x + y < 0$. Then $|x + y| = -(x + y) \leq |x| + |y|$ *(from the left inequality)*.
2. So $|x + y| \leq |x| + |y|$ in all cases. ■

PROOF **For any integer a and prime p: if $p \mid a$ then $p \nmid (a + 1)$ (contradiction)**
1. Suppose not: $p \mid a$ and $p \mid (a + 1)$ for some integer a and prime p.
1.1. Then $a = pr$ and $a + 1 = ps$ for some $r, s \in \mathbb{Z}$ *(definition of divisibility)*, so $1 = ps - pr = p(s - r)$.

1.2. So $p \mid 1$ *(as $s - r \in \mathbb{Z}$)*, hence $p = 1$ or $p = -1$ *(Theorem 4.3.2)*. This is a contradiction, since a prime is greater than 1. ■

PROOF **If $a \mid b$ and $a \mid c$ then $a \mid (mb + nc)$ for all integers m, n**
1. Suppose $a \mid b$ and $a \mid c$: $b = ar$ and $c = as$ for some $r, s \in \mathbb{Z}$ *(definition of divisibility)*.
2. Then $mb + nc = a(mr + ns)$ *(basic algebra)*, and $mr + ns \in \mathbb{Z}$ *(closure)*, so $a \mid (mb + nc)$. ■

PROOF **Lecture 1 · If x and y are "colorful" (n colorful $\leftrightarrow \exists k \in \mathbb{Z}\,(n = 3k)$), so is $x + 2y$**
Work backwards first: the goal "$x + 2y$ is colorful" means we need an integer c with $x + 2y = 3c$.
1. Let x, y be colorful integers. Then $x = 3a$ and $y = 3b$ for some $a, b \in \mathbb{Z}$ *(by definition of colorful)*.
2. $x + 2y = 3a + 6b = 3(a + 2b)$ *(basic algebra)*. Let $c = a + 2b$, an integer *(closure)*.
3. So $x + 2y = 3c$, and $x + 2y$ is colorful *(by definition of colorful)*. ■

WORKED EXAMPLE **Disproof by counterexample**
- "$\forall a, b, c \in \mathbb{Z}\,(a \mid bc \to a \mid b \vee a \mid c)$" is false: $4 \mid 2 \cdot 2$ but $4 \nmid 2$.
- "$\forall a, b \in \mathbb{Z}\,(a \mid b \wedge b \mid a \to a = b)$" is false: take $a = 1$, $b = -1$. (True on $\mathbb{Z}^+$: Theorem 4.3.1.)
- "The product of two irrational numbers is irrational" is false: $\sqrt{2} \cdot \sqrt{8} = 4$ (Lecture 1).
- "$\forall a, b \in \mathbb{R}\,(a^2 = b^2 \to a = b)$" is false: take $a = 1$, $b = -1$ (Lecture 4).
- "There is no integer that is both even and odd" is **true**: if $n = 2a = 2b + 1$ then $1 = 2(a - b)$, so $1/2 \in \mathbb{Z}$, a contradiction.

PROOF **Theorem 4.2.1 · Every integer is a rational number**
1. Let a be a arbitrary integer. Then $a = a/1$, a quotient of integers with non-zero denominator.
2. Hence a is rational *(by definition of rational)*. ■

PROOF **Proof by exhaustion (Lecture 4, Ex. 3) · every even n with $4 \le n \le 26$ is a sum of two primes**
$4 = 2 + 2$, $6 = 3 + 3$, $8 = 3 + 5$, $10 = 5 + 5$, $12 = 5 + 7$, $14 = 3 + 11$, $16 = 5 + 11$, $18 = 7 + 11$, $20 = 7 + 13$, $22 = 5 + 17$, $24 = 5 + 19$, $26 = 7 + 19$. Every case is checked. ■

PROOF **For every integer n, $3 \mid (n^3 - n)$ (division into cases, QR with $d = 3$)**
1. Let $n \in \mathbb{Z}$. By the QR theorem, $n = 3q$, $3q + 1$ or $3q + 2$ for some $q \in \mathbb{Z}$. Note $n^3 - n = (n - 1)n(n + 1)$ *(basic algebra)*.
   1.1. **Case $n = 3q$:** the factor $n = 3q$, so $n^3 - n = 3 \cdot q(n - 1)(n + 1)$.
   1.2. **Case $n = 3q + 1$:** the factor $n - 1 = 3q$, so $n^3 - n = 3 \cdot qn(n + 1)$.
   1.3. **Case $n = 3q + 2$:** the factor $n + 1 = 3(q + 1)$, so $n^3 - n = 3 \cdot (q + 1)(n - 1)n$.
2. In each case the cofactor is an integer *(closure)*, so $3 \mid (n^3 - n)$ *(definition of divisibility)*. ■

PROOF **Every prime $p > 3$ has the form $6k + 1$ or $6k + 5$**
1. Let $p > 3$ be prime. By the QR theorem, $p = 6k + r$ with $r \in \{0, 1, 2, 3, 4, 5\}$.
   1.1. If $r \in \{0, 2, 4\}$, then $p = 2(3k + r/2)$ is even and > 2, so not prime.
   1.2. If $r = 3$, then $p = 3(2k + 1)$ with $p > 3$, so not prime.
2. Hence $r = 1$ or $r = 5$. ■

PROOF **Every integer $n > 1$ is divisible by a prime (via well-ordering)**
1. Let $n > 1$ be an integer, and let $D = \{d \in \mathbb{Z} : d > 1 \wedge d \mid n\}$. $D \neq \varnothing$, since $n \in D$.
2. D has a smallest element p *($(\mathbb{N}, \le)$ is well-ordered)*.
3. p is prime: suppose instead $p = rs$ with $1 < r < p$. Then $r \mid p$ and $p \mid n$, so $r \mid n$ *(Theorem 4.3.3)*, so $r \in D$ with $r < p$, contradicting the minimality of p.
4. Hence the prime p divides n. ■

PROOF **Lemma 4.4.4 · For every real r: $-|r| \le r \le |r|$ (division into cases)**
1. **Case $r \ge 0$:** $|r| = r$ *(definition of absolute value)*, and $-|r| = -r \le 0 \le r$. So $-|r| \le r = |r|$.
2. **Case $r < 0$:** $|r| = -r > 0$, so $-|r| = r$ and $r < 0 < |r|$. So $-|r| = r \le |r|$. ■

PAST PAPER **Using unique factorization · AY25 Q6(iii) in full**
Unique factorization: every integer $n > 1$ is a product of primes, unique up to order.
Claim: if $n = m^2$ is a perfect square and a prime p divides n, then $p^2 \mid n$.
If $m = 0$, then $n = 0$ and $p^2 \mid 0$. Otherwise write $|m| = p_1^{e_1} \cdots p_k^{e_k}$ *(unique factorization)* then $n = p_1^{2e_1} \cdots p_k^{2e_k}$. A prime dividing n is some p_i (uniqueness of the factorization), and its exponent $2e_i \ge 2$, so $p_i^2 \mid n$. ■
In a perfect square every prime appears to an even power.

WORKED EXAMPLE **Division into cases**

Prove: for every integer n, $n^2 + n$ is even.

1. Let n be a arbitrary integer. Then n is even or odd *(Assumption 1)*.
   1.1. **Case 1:** $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, and $2k^2 + k \in \mathbb{Z}$ *(closure)*.
   1.2. **Case 2:** $n = 2k + 1$. Then $n^2 + n = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1)$, and $(2k + 1)(k + 1) \in \mathbb{Z}$ *(closure)*.
2. In both cases $n^2 + n$ is even *(by definition of even)*. ■

WORKED EXAMPLE **Contradiction**

Prove: there is no smallest positive rational number.

1. Suppose not: let r be the smallest positive rational number.
   1.1. $r = a/b$ for some integers a, b with $b \neq 0$ *(definition of rational)*. Then $r/2 = a/(2b)$, where $a, 2b \in \mathbb{Z}$ and $2b \neq 0$, so $r/2$ is rational.
   1.2. $r/2 > 0$ and $r/2 < r$ *(as $r > 0$)*, contradicting the choice of r as the smallest.
2. Hence there is no smallest positive rational number. ■

Part IV · Set Theory

13 · Describing sets

DEFINITION **Set · Membership · Cardinality**

A **set** is an unordered collection of objects, its **members** or **elements**; order and repetition do not matter: $\{9, 8, 7\} = \{7, 8, 7, 9, 9\}$. $x \in S$ means x is an element of S; $x \notin S$ means it is not. The **cardinality** $|S|$ is the size of S, the number of elements of S.

DEFINITION **Set-roster, set-builder and replacement notation**

Set-roster: list all the elements between braces, e.g. $\{1, 2, 3\}$, $\{1, 2, ..., 100\}$, $\{1, 2, 3, ...\}$ (the ellipsis "..." reads "and so forth").

Set-builder: $\{x \in U : P(x)\}$ is the set of all $x \in U$ for which $P(x)$ is true. z is a member iff $z \in U$ **and** $P(z)$.

Replacement: $\{t(x) : x \in A\}$ is the set of all objects $t(x)$ as x ranges over A. z is a member iff $t(x) = z$ for **some** $x \in A$.

E.g. $\{x + 1 : x \in \mathbb{Z}_{\geq 0}\} = \mathbb{Z}^+$; $\{x \in \mathbb{Z} : -2 < x < 5\} = \{-1, 0, 1, 2, 3, 4\}$.

DEFINITION **Interval notation (Lecture 5)**

For real numbers $a \leq b$: $(a, b) = \{x \in \mathbb{R} : a < x < b\}$, $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$, $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$, $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$.

Unbounded: $(a, \infty) = \{x \in \mathbb{R} : x > a\}$, $[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$, $(-\infty, b) = \{x \in \mathbb{R} : x < b\}$, $(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$.

The interval (a, b) and the ordered pair (a, b) look the same; context tells them apart.

Symbol	Set
$\mathbb{N}$	natural numbers $\{0, 1, 2, ...\}$, **including 0**
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}$	integers, rationals, reals
$\mathbb{Z}^+, \mathbb{Z}^-, \mathbb{Z}_{\geq 0}$	positive, negative, non-negative integers (0 is neither positive nor negative)
$(a, b), [a, b]$	open / closed real intervals; also $(a, b], [a, \infty)$ ...

PAST PAPER **AY25 Q17(b) · Name these subsets of $\mathbb{N}$**

$A = \{0, 1, 3, 7, 15, 31, ...\}$ in replacement notation: $A = \{2^n - 1 : n \in \mathbb{N}\}$.

$B = \{3, 9, 15, 21, 27, ...\}$ in set-builder notation: $B = \{n \in \mathbb{N} : 3 \mid n \wedge n \text{ is odd}\}$.

(Many correct answers; check the first few terms.)

14 · Subsets, equality and the empty set

DEFINITION **Subset, superset · Proper subset · Set equality · Empty set, singleton**
$A \subseteq B \Leftrightarrow \forall x(x \in A \to x \in B)$: every element of A is in B ("A is contained in B"). $A \not\subseteq B \Leftrightarrow \exists x(x \in A \wedge x \notin B)$.
If $A \subseteq B$ we may write $B \supseteq A$: "B contains / includes A", "B is a **superset** of A".
$A \subsetneq B$ (proper, or strict, inclusion) iff $A \subseteq B$ and $A \neq B$. $A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A \Leftrightarrow \forall x(x \in A \leftrightarrow x \in B)$.
The **empty set** $\varnothing = \{\}$ is the set with no elements. A set with exactly one element is a **singleton**.

THEOREM **Theorem 6.2.4 · The empty set**
$\varnothing \subseteq A$ for every set A (and $\varnothing$ is unique). **Proof:** $\forall x(x \in \varnothing \to x \in A)$ is vacuously true, as $x \in \varnothing$ is always false. **Uniqueness:** the empty set is unique: two empty sets are subsets of one another by Theorem 6.2.4, hence equal.

Claim		Why
$\varnothing \in \{1,2,3\}$	F	$\varnothing$ is not listed
$\varnothing \subseteq \{1,2,3\}$	T	Thm 6.2.4
$\{2\} \in \{1,2,3\}$	F	elements are numbers
$\{2\} \subseteq \{1,2,3\}$	T	2 is an element
$\{2\} \in \{\{1\},\{2\},\{3\}\}$	T	$\{2\}$ is listed
$\{2\} \subseteq \{\{1\},\{2\},\{3\}\}$	F	2 is not listed
$\{9\} = \{\{9\}\}$	F	different elements

The $\varnothing$ ladder	$\lvert\cdot\rvert$
$\varnothing = \{\}$	0
$\{\varnothing\} = \mathcal{P}(\varnothing)$	1
$\{\varnothing, \{\varnothing\}\} = \mathcal{P}(\mathcal{P}(\varnothing))$	2
$\mathcal{P}(\mathcal{P}(\mathcal{P}(\varnothing)))$	4
$\mathcal{P}^4(\varnothing)$	16

$\varnothing \neq \{\varnothing\}$: the first is empty, the second has one element. $\varnothing \in \{\varnothing\}$ and $\varnothing \subseteq \{\varnothing\}$ are both true.

TRAP **$\subseteq$ is transitive but $\in$ is not · AY23 Q8**
Which hold for all sets? (i) $A \in B \wedge B \subseteq C \Rightarrow A \in C$ is **true**, since A is an element of B and hence of C.
(ii) $A \in B \wedge B \subseteq C \Rightarrow A \subseteq C$ is false: take $A = \{1\}$, $B = C = \{\{1\}\}$, but $1 \notin C$.
(iii), (iv) $A \subseteq B \wedge B \in C \Rightarrow A \in C$ or $A \subseteq C$ are false: take $A = \{x\}$, $B = \{x, y\}$, $C = \{\{x, y\}\}$. Only (i): **A**.

15 · Ordered pairs and Cartesian products

DEFINITION **Ordered pair, n-tuple · Cartesian product**
$(a,b) = (c,d) \Leftrightarrow a = c \wedge b = d$; likewise for n-tuples componentwise. $(1,2) \neq (2,1)$ though $\{1,2\} = \{2,1\}$.
$A \times B = \{(a,b) : a \in A \wedge b \in B\}$; $A_1 \times \cdots \times A_n = \{(a_1, ..., a_n) : a_i \in A_i\}$; $A^n = A \times \cdots \times A$.

Facts. $|A \times B| = |A| \cdot |B|$. $A \times \varnothing = \varnothing \times A = \varnothing$. $A \times B \neq B \times A$ in general. $(A \times B) \times C \neq A \times B \times C$: elements $((a,b),c)$ versus (a,b,c). $A \times (B \cup C) = (A \times B) \cup (A \times C)$, and likewise for $\cap$ and $\setminus$.

PAST PAPER **AY24 Q6 · Which hold for all sets A, B, C?**
(i) $\mathcal{P}(A) \subseteq \mathcal{P}(B) \to A \subseteq B$ is **true** (proof in §17). (ii) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ is **true** (proof in §17).
(iii) $A \times B \subseteq B \times C \to A \subseteq C$ is **false**: take $B = \varnothing$; then $A \times \varnothing = \varnothing$ is a subset of every set, whatever A is.
Answer **A**.

TRAP **T3 Q3(c) · $A \times \mathcal{P}(B) \neq \mathcal{P}(A \times B)$**
For every A, B, $\varnothing \in \mathcal{P}(A \times B)$ *($\varnothing \subseteq A \times B$)*, but $\varnothing \notin A \times \mathcal{P}(B)$ *(every member of a Cartesian product is an ordered pair; $\varnothing$ is not one)*. This named witness proves the sets differ. A cardinality "proof" ($n \cdot 2^k \neq 2^{nk}$ for $|A| = n$, $|B| = k$) is wrong: equality holds when $n = 1$, or $n = 2$ and $k = 1$.

16 · Operations on sets

DEFINITION **Universal set · Union, intersection, difference, complement**
A **universal set** (universe of discourse) U is the set containing every object under discussion, e.g. $\mathbb{R}$ when all sets considered are sets of reals.
For $A, B \subseteq U$: $A \cup B = \{x \in U : x \in A \vee x \in B\}$, $A \cap B = \{x \in U : x \in A \wedge x \in B\}$, $B \setminus A = \{x \in U : x \in B \wedge x \notin A\}$ (the **relative complement** of A in B), $\overline{A} = \{x \in U : x \notin A\} = U \setminus A$.
Procedural versions (the steps in element proofs): $a \in X \cup Y \Leftrightarrow a \in X \vee a \in Y$; $a \in X \cap Y \Leftrightarrow a \in X \wedge a \in Y$; $a \in X \setminus Y \Leftrightarrow a \in X \wedge a \notin Y$; $a \in \overline{X} \Leftrightarrow a \notin X$; $(a, b) \in X \times Y \Leftrightarrow a \in X \wedge b \in Y$.

DEFINITION **Symmetric difference (Tutorial 3) · Disjoint**
$A \oplus B = (A \setminus B) \cup (B \setminus A)$, the set of elements in exactly one of A and B. A and B are **disjoint** iff $A \cap B = \varnothing$; $A_1, A_2, \ldots$ are **mutually disjoint** iff $A_i \cap A_j = \varnothing$ whenever $i \neq j$. $\bigcup_{i=0}^{n} A_i$, $\bigcap_{i=0}^{n} A_i$ extend $\cup, \cap$.

THEOREM **Theorem 6.2.1 · Subset relations**
$A \cap B \subseteq A$, $A \cap B \subseteq B$ (inclusion of intersection); $A \subseteq A \cup B$, $B \subseteq A \cup B$ (inclusion in union); $A \subseteq B \wedge B \subseteq C \Rightarrow A \subseteq C$ (transitivity). The **set identities** (Theorem 6.2.2) are the right column of the table in §2.

PAST PAPER **AY23 Q7 · For which can no non-empty A, B exist?**
(i) $A \setminus B = A \cap B$: these two sets are disjoint, so both must be $\varnothing$; then $A = (A \setminus B) \cup (A \cap B) = \varnothing$. **Impossible.**
(ii) $A \setminus B = A \cup B$: then $B \subseteq A \cup B = A \setminus B$, so every element of B is not in B; $B = \varnothing$. **Impossible.**
(iii) $A \cup B = A \cap B$: take $A = B$. (iv) $A \oplus B = A \cup B$: take $A \cap B = \varnothing$. Answer **B** (only (i), (ii)).

17 · Proving and disproving set statements

METHOD **Three methods**
Element method. To show $X \subseteq Y$: "Let $z \in X$ … so $z \in Y$". To show $X = Y$: prove both inclusions, or one chain of $\Leftrightarrow$ steps each justified by a definition or a logic law.
Algebraic method. Rewrite using the set identities (§2), one named law per line.
Disproof. Give one concrete counterexample with every set written out. A Venn diagram helps in finding one.

PROOF $\{x \in \mathbb{Z} : x^2 = 1\} = \{1, -1\}$ **(two inclusions, Lecture 5)**
1. ($\subseteq$) Take any $z \in \{x \in \mathbb{Z} : x^2 = 1\}$.
   1.1. Then $z \in \mathbb{Z}$ and $z^2 = 1$, so $(z-1)(z+1) = 0$ *(by basic algebra)*.
   1.2. So $z = 1$ or $z = -1$, i.e. $z \in \{1, -1\}$.
2. ($\supseteq$) Take any $z \in \{1, -1\}$. Then $z \in \mathbb{Z}$ and $z^2 = 1$, so $z \in \{x \in \mathbb{Z} : x^2 = 1\}$.
3. Therefore the sets are equal *(by definition of set equality, from 1 and 2)*. ■

PROOF **De Morgan: $\overline{A \cup B} = \overline{A} \cap \overline{B}$ (chain of $\Leftrightarrow$)**
1. Let $z \in U$. Then
   1.1. $z \in \overline{A \cup B} \Leftrightarrow \sim(z \in A \cup B)$ *(by definition of complement)*
   1.2. $\Leftrightarrow \sim(z \in A \vee z \in B)$ *(by definition of $\cup$)*
   1.3. $\Leftrightarrow z \notin A \wedge z \notin B$ *(by De Morgan's law for logic)*
   1.4. $\Leftrightarrow z \in \overline{A} \wedge z \in \overline{B} \Leftrightarrow z \in \overline{A} \cap \overline{B}$ *(by definitions of complement, $\cap$)*. ■

PROOF $(A \cap B) \cup (A \setminus B) = A$ **(algebraic, Lecture 5)**
$(A \cap B) \cup (A \setminus B) = (A \cap B) \cup \left(A \cap \overline{B}\right)$ *(set difference law)*
$= A \cap \left(B \cup \overline{B}\right)$ *(distributive law)*
$= A \cap U = A$ *(complement law; identity law)*
■

PROOF $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ **(algebraic)**
$A \setminus (B \cap C)$
$= A \cap \overline{B \cap C}$ *(set difference)*

$= A \cap \left(\overline{B} \cup \overline{C}\right)$ *(De Morgan)*
$= \left(A \cap \overline{B}\right) \cup \left(A \cap \overline{C}\right)$ *(distributive)*
$= (A \setminus B) \cup (A \setminus C)$ *(set difference ×2)*. ■

PROOF **For all sets A, B: $A \subseteq B \Leftrightarrow \mathcal{P}(A) \subseteq \mathcal{P}(B)$**
1. ($\Rightarrow$) Suppose $A \subseteq B$. Let $X \in \mathcal{P}(A)$; then $X \subseteq A$ *(definition of power set)*, so $X \subseteq B$ *(transitivity of $\subseteq$)*, so $X \in \mathcal{P}(B)$.
2. ($\Leftarrow$) Suppose $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. $A \subseteq A$, so $A \in \mathcal{P}(A)$, hence $A \in \mathcal{P}(B)$, i.e. $A \subseteq B$ *(definition of power set)*. ■

WORKED EXAMPLE **T3 Q4 · Set equality by double inclusion**
Let $A = \{2n + 1 : n \in \mathbb{Z}\}$ and $B = \{2n - 5 : n \in \mathbb{Z}\}$. Prove $A = B$.
1. If $a \in A$, write $a = 2n + 1 = 2(n + 3) - 5$ for $n \in \mathbb{Z}$. Since $n + 3 \in \mathbb{Z}$ *(closure)*, $a \in B$ *(definition of B)*.
2. If $b \in B$, write $b = 2n - 5 = 2(n - 3) + 1$ for $n \in \mathbb{Z}$. Since $n - 3 \in \mathbb{Z}$, $b \in A$.
3. Thus $A = B$ *(double inclusion)*. ■

A substitution $k = n + 3$ must range over **all** integers: given k, use $n = k - 3$. Closure alone does not show this.

WORKED EXAMPLE **T3 Q5 · Element method as a set-builder chain**
Prove that for all sets A, B, C: $A \cap (B \setminus C) = (A \cap B) \setminus C$.
$A \cap (B \setminus C) = \{x : x \in A \wedge x \in (B \setminus C)\}$ *(definition of $\cap$)*
$= \{x : x \in A \wedge (x \in B \wedge x \notin C)\}$ *(definition of $\setminus$)*
$= \{x : (x \in A \wedge x \in B) \wedge x \notin C\}$ *(associativity of $\wedge$)*
$= \{x : x \in A \cap B \wedge x \notin C\}$ *(definition of $\cap$)*
$= (A \cap B) \setminus C$ *(definition of $\setminus$)*. ■

WORKED EXAMPLE **T3 Q6 · Symmetric difference by set identities only**
(a) $A = \{1, 4, 9, 16\}$, $B = \{2, 4, 6, ..., 16\}$: find $A \oplus B$. (b) Using Theorem 6.2.2, prove $A \oplus B = (A \cup B) \setminus (A \cap B)$.
(a) $A \setminus B = \{1, 9\}$, $B \setminus A = \{2, 6, 8, 10, 12, 14\}$, so $A \oplus B = \{1, 2, 6, 8, 9, 10, 12, 14\}$.
(b) $A \oplus B = (A \setminus B) \cup (B \setminus A)$ *(definition of $\oplus$)*
1. $= \left(A \cap \overline{B}\right) \cup \left(B \cap \overline{A}\right)$ *(set difference law)*
2. $= \left(\left(A \cap \overline{B}\right) \cup B\right) \cap \left(\left(A \cap \overline{B}\right) \cup \overline{A}\right)$ *(distributive law)*
3. $= \left((A \cup B) \cap \left(\overline{B} \cup B\right)\right) \cap \left(\left(A \cup \overline{A}\right) \cap \left(\overline{B} \cup \overline{A}\right)\right)$ *(distributive, commutative laws)*
4. $= ((A \cup B) \cap U) \cap \left(U \cap \left(\overline{A} \cup \overline{B}\right)\right)$ *(commutative, complement laws)*
5. $= (A \cup B) \cap \left(\overline{A} \cup \overline{B}\right)$ *(commutative, identity laws)*
6. $= (A \cup B) \cap \overline{A \cap B}$ *(De Morgan's law)*
7. $= (A \cup B) \setminus (A \cap B)$ *(set difference law)*. ■

WORKED EXAMPLE **Set identity**
Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
$A \setminus (B \cup C)$
$= A \cap \overline{B \cup C}$ *(set difference law)*
$= A \cap \left(\overline{B} \cap \overline{C}\right)$ *(De Morgan's law)*
$= (A \cap A) \cap \left(\overline{B} \cap \overline{C}\right)$ *(idempotent law)*
$= \left(A \cap \overline{B}\right) \cap \left(A \cap \overline{C}\right)$ *(associative, commutative laws)*
$= (A \setminus B) \cap (A \setminus C)$ *(set difference law)*. ■

WORKED EXAMPLE **Prove or disprove**
(a) $A \cup (B \setminus C) = (A \cup B) \setminus C$; (b) $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$; (c) $A \subseteq C \wedge B \subseteq C \rightarrow A \cup B \subseteq C$.
(a) **False:** $A = \{1\}$, $B = \varnothing$, $C = \{1\}$: left side $\{1\}$, right side $\varnothing$.
(b) **False:** $A = \{1\}$, $B = \{2\}$: $\{1, 2\} \in \mathcal{P}(A \cup B)$ but $\{1, 2\} \notin \mathcal{P}(A) \cup \mathcal{P}(B)$. (Only $\supseteq$ holds.)
(c) **True:** let $x \in A \cup B$; then $x \in A$ or $x \in B$ *(definition of $\cup$)* in either case $x \in C$ *(as $A \subseteq C$, $B \subseteq C$)*. ■

18 · Partitions and power sets

DEFINITION **Partition (Lecture 6)**
$\mathcal{C}$ is a **partition** of A iff (1) every element of $\mathcal{C}$ is a **non-empty** subset of A, and (2) every element of A lies in **exactly one** element of $\mathcal{C}$: $\forall x \in A \, \exists! S \in \mathcal{C} \, (x \in S)$. The elements of $\mathcal{C}$ are its **components**. (Equivalently: non-empty, mutually disjoint, union is A.)

DEFINITION **Power set · Theorem 6.3.1**
$\mathcal{P}(A)$ is the set of all subsets of A. If A is finite and $|A| = n$, then $|\mathcal{P}(A)| = 2^n$. Always $\varnothing \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$.
$\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$, but only $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.

PAST PAPER **AY24 Q7 and Q8 · Partitions, power sets**
Q7. (i) $\{\{2,3\},\{4\},\{3,2\},\{1\}\} = \{\{2,3\},\{4\},\{1\}\}$ **is** a partition of $\{1,2,3,4\}$, because repeated elements count once. (ii) $\{\mathbb{Z}\}$ **is** a partition of $\mathbb{Z}$ (one component). (iii) $\mathbb{Z}$ is **not**: its elements are numbers, not subsets. **C.**
Q8. (i) $|\mathcal{P}(S) \setminus \mathcal{P}(\varnothing)| = 2^{|S|} - 1$ is true, as $\mathcal{P}(\varnothing) = \{\varnothing\}$. (ii) $|S \cap \mathcal{P}(S)| = 0$ is false: $S = \{a, \{a\}\}$ gives $S \cap \mathcal{P}(S) = \{\{a\}\}$. (iii) $|S| \cdot |\mathcal{P}(S)| = |S \times \mathcal{P}(S)|$ is true. **D.**

PAST PAPER **AY25 Q7, Q8 · Counting**
Q7. A, B, C pairwise share exactly one element; $D = \mathcal{P}(A) \cap \mathcal{P}(B) \cap \mathcal{P}(C) = \mathcal{P}(A \cap B \cap C)$. The triple intersection lies inside each one-element pairwise intersection, so it has 0 or 1 elements: $|D| \in \{1, 2\}$. **B.**
Q8. $|A| = 2|B|$, $|B| = |C| + 2$, and $\{B, C, D\}$ partitions A, so $|A| = |B| + |C| + |D|$: $2|B| = |B| + (|B| - 2) + |D|$, giving $|D| = 2$. **C.**

PAST PAPER **AY24 Q16 (5 marks) · $A = \{\varnothing, x\}$**
$\mathcal{P}(A) = \{\varnothing, \{\varnothing\}, \{x\}, \{\varnothing, x\}\}$ and $\mathcal{P}(\varnothing) = \{\varnothing\}$, $\mathcal{P}(\mathcal{P}(\varnothing)) = \{\varnothing, \{\varnothing\}\}$.
- (a) $\mathcal{P}(A) \setminus \mathcal{P}(\mathcal{P}(\varnothing)) = \{\{x\}, \{\varnothing, x\}\}$.
- (b) $(\mathcal{P}(A) \setminus \mathcal{P}(\varnothing)) \setminus \{A\} = \{\{\varnothing\}, \{x\}\}$, so its power set is $\{\varnothing, \{\{\varnothing\}\}, \{\{x\}\}, \{\{\varnothing\}, \{x\}\}\}$.
- (c) $\mathcal{P}(\varnothing) \times \mathcal{P}(A) = \{(\varnothing, S) : S \in \mathcal{P}(A)\}$ and $\mathcal{P}(A) \times \mathcal{P}(\varnothing) = \{(S, \varnothing) : S \in \mathcal{P}(A)\}$ share only $(\varnothing, \varnothing)$, so the answer is $\{(\varnothing, \{\varnothing\}), (\varnothing, \{x\}), (\varnothing, \{\varnothing, x\})\}$.

PAST PAPER **AY23 Q6 and Q17 · $A_1 = \{\varnothing\}$, $A_2 = \{a\}$, $A_3 = \{a, \{a\}\}$**
Q6. Target $\varnothing \cup \mathcal{P}(\varnothing) = \{\varnothing\}$. $\varnothing \times \{\varnothing\} = \varnothing$ ✗; $\mathcal{P}(\varnothing \cup \mathcal{P}(\varnothing)) = \{\varnothing, \{\varnothing\}\}$ ✗; $\mathcal{P}(\varnothing) \setminus \varnothing = \{\varnothing\}$ ✓; $(\varnothing \cap \{\varnothing\}) \cup (\varnothing \cup \{\varnothing\}) = \{\varnothing\}$ ✓. **D.**
Q17. (a) $(A_1 \cap A_2) \setminus A_3 = \varnothing$. (b) $(A_1 \cup A_2) \setminus A_3 = \{\varnothing\}$. (c) $(A_1 \times A_2) \cup A_3 = \{(\varnothing, a), a, \{a\}\}$. (d) $(A_1 \cup A_2) \times A_3 = \{(\varnothing, a), (\varnothing, \{a\}), (a, a), (a, \{a\})\}$.

PROOF **Distributive law $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, by the element method with cases**
1. ($\subseteq$) Let $x \in A \cup (B \cap C)$. Then $x \in A$ or $x \in B \cap C$ *(definition of $\cup$)*.
   1.1. **Case $x \in A$:** then $x \in A \cup B$ and $x \in A \cup C$ *(definition of $\cup$)*, so $x \in (A \cup B) \cap (A \cup C)$.
   1.2. **Case $x \in B \cap C$:** then $x \in B$ and $x \in C$, so again $x \in A \cup B$ and $x \in A \cup C$.
2. ($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$, so $x \in A \cup B$ and $x \in A \cup C$.
   2.1. **Case $x \in A$:** then $x \in A \cup (B \cap C)$.
   2.2. **Case $x \notin A$:** then $x \in B$ and $x \in C$ *(as $x \in A \cup B$, $x \in A \cup C$)*, so $x \in B \cap C \subseteq A \cup (B \cap C)$.
3. Hence the two sets are equal *(definition of set equality)*. ■

PROOF **For all sets A, B: $A \subseteq B \leftrightarrow A \cup B = B$**
1. ($\rightarrow$) Suppose $A \subseteq B$. $B \subseteq A \cup B$ *(inclusion in union)*. Conversely let $x \in A \cup B$: if $x \in A$ then $x \in B$ *(as $A \subseteq B$)* if $x \in B$, done. So $A \cup B = B$.
2. ($\leftarrow$) Suppose $A \cup B = B$. Let $x \in A$; then $x \in A \cup B$ *(inclusion in union)* $= B$. So $A \subseteq B$. ■

PROOF **For every set A, $A \times \varnothing = \varnothing$ (proof by contradiction)**
1. Suppose not: some $z \in A \times \varnothing$. Then $z = (a, b)$ with $a \in A$ and $b \in \varnothing$ *(definition of $\times$)*.
2. But $\varnothing$ has no elements, a contradiction. So $A \times \varnothing$ has no elements: $A \times \varnothing = \varnothing$. ■

PROOF **$A \times (B \cup C) = (A \times B) \cup (A \times C)$ (chain of $\leftrightarrow$)**
$(x, y) \in A \times (B \cup C)$
$\leftrightarrow x \in A \wedge (y \in B \vee y \in C)$ *(definitions of $\times$, $\cup$)*
$\leftrightarrow (x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)$ *(distributive law of logic)*
$\leftrightarrow (x, y) \in (A \times B) \cup (A \times C)$ *(definitions of $\times$, $\cup$)*. ■

PROOF **$(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$ (algebraic)**
$(A \cup B) \setminus C$
$= (A \cup B) \cap \overline{C}$ *(set difference law)*
$= \left(A \cap \overline{C}\right) \cup \left(B \cap \overline{C}\right)$ *(distributive, commutative)*
$= (A \setminus C) \cup (B \setminus C)$ *(set difference law)*. ■

WORKED EXAMPLE **Disproving a set identity · "$(A \setminus B) \cup (B \setminus C) = A \setminus C$ for all sets"**
False. Counterexample: $A = \varnothing$, $B = \{1\}$, $C = \varnothing$. Then $(A \setminus B) \cup (B \setminus C) = \varnothing \cup \{1\} = \{1\}$, while $A \setminus C = \varnothing$. Write out every set and compute both sides.

DRILL **Nested sets and power sets (answers after the arrow)**
- $|\{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}| \to$ **3**
- $\mathcal{P}(\{\varnothing\}) \to \{\varnothing, \{\varnothing\}\}$
- $\{\varnothing\} \times \{\varnothing\} \to \{(\varnothing, \varnothing)\}$
- $\varnothing \times \{\varnothing\} \to \varnothing$
- $\mathcal{P}(\{a, b\}) \setminus \{\varnothing\} \to \{\{a\}, \{b\}, \{a, b\}\}$
- $\{1, \{1\}\} \cap \mathcal{P}(\{1\}) \to \{\{1\}\}$
- $|\mathcal{P}(A \times B)|$ with $|A| = 2$, $|B| = 3 \to 2^6 = 64$
- $\{x \in \mathbb{Z} : x^2 < 5\} \to \{-2, -1, 0, 1, 2\}$
- $\{2n + 1 : n \in \{0, 1, 2\}\} \to \{1, 3, 5\}$
- $\{\{1, 2\}, \{3\}\}$ a partition of $\{1, 2, 3\}$? $\to$ **yes**; $\{\{1, 2\}, \{2, 3\}\} \to$ **no** (2 twice); $\{\{1, 2\}, \{3\}, \varnothing\} \to$ **no** ($\varnothing$ component)

PROOF **If $A \subseteq B$ then $A \cap C \subseteq B \cap C$ and $A \cup C \subseteq B \cup C$**
1. Let $x \in A \cap C$: $x \in A$ and $x \in C$ *(definition of $\cap$)* $x \in B$ *(as $A \subseteq B$)* so $x \in B \cap C$.
2. Let $x \in A \cup C$: if $x \in A$ then $x \in B \subseteq B \cup C$; if $x \in C$ then $x \in B \cup C$ *(inclusion in union)*. ■

Reading only (Lecture 5.3). Unrestricted comprehension breaks: Russell's $R = \{x : x \notin x\}$ gives $R \in R \leftrightarrow R \notin R$; a set of "everything" $\mathcal{D}$ contradicts $|\mathcal{P}(\mathcal{D})| > |\mathcal{D}|$ (Cantor). Zermelo (1908), later Fraenkel and Skolem, fixed this with the ZF axioms. Cantor, Russell, Zermelo and Fraenkel contributed to modern set theory; Gauss did not (AY25 Q1).

WORKED EXAMPLE **Power set of a set that contains $\varnothing$**
Let $A = \{\varnothing, \{\varnothing\}\}$. Find $\mathcal{P}(A)$ and $A \cap \mathcal{P}(A)$.
$\mathcal{P}(A) = \{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\}$, with four subsets since $|A| = 2$. For $A \cap \mathcal{P}(A)$ test each element of A: $\varnothing \in \mathcal{P}(A)$ (as $\varnothing \subseteq A$), and $\{\varnothing\} \in \mathcal{P}(A)$ (as $\{\varnothing\} \subseteq A$, since $\varnothing \in A$). So $A \cap \mathcal{P}(A) = \{\varnothing, \{\varnothing\}\} = A$, i.e. $A \subseteq \mathcal{P}(A)$.

Part V · Relations

19 · Relations and inverses

DEFINITION **Relation · Domain, co-domain, range**
A (binary) **relation from A to B** is a subset $R \subseteq A \times B$; xRy ("x is R-related to y") means $(x, y) \in R$, and $x \not R y$ means $(x, y) \notin R$. A **relation on A** is a relation from A to A, i.e. a subset of $A \times A = A^2$. Its **directed graph** draws each element of A once, with an arrow $x \to y$ whenever xRy and a loop at x when xRx.
$Dom(R) = \{a \in A : aRb \text{ for some } b \in B\}$, $coDom(R) = B$, $Range(R) = \{b \in B : aRb \text{ for some } a \in A\}$.
An ***n*-ary relation** on $A_1 \times \cdots \times A_n$ is a subset of it (binary, ternary, quaternary for $n = 2, 3, 4$). Relational databases are built on n-ary relations.

DEFINITION **Inverse relation**
$R^{-1} = \{(y, x) \in B \times A : (x, y) \in R\}$, i.e. $\forall x \in A, y \in B\,((y, x) \in R^{-1} \leftrightarrow (x, y) \in R)$. Reverse every arrow.

PAST PAPER **AY25 Q9, Q10 · Relations are sets of pairs**
Q9. $A = \{1, 2, 3\}$, $B = \{2, 5\}$. $S_2 = \mathcal{P}(A \times B)$ is **exactly** the set S_3 of relations from A to B; $S_1 = \mathcal{P}(A) \times \mathcal{P}(B)$ holds **pairs of sets**, a different kind of object. So $S_4 \subseteq S_3 \subseteq S_2$. Answer **A**.
Q10. The relation "$p \to q$" on $\{T, F\}$ is $R = \{(T, T), (F, T), (F, F)\}$. $R^{-1} = \{(T, T), (T, F), (F, F)\}$ is "$q \to p$", the **converse**. The converse is equivalent to the inverse, so the answer key accepted **B or D**.

20 · Composition of relations

DEFINITION **Composition (Lecture 6.1.4)**
For $R \subseteq A \times B$ and $S \subseteq B \times C$, the **composition** $S \circ R$ is the relation from A to C with
$$\forall x \in A, \forall z \in C\,(x(S \circ R)z \leftrightarrow \exists y \in B\,(xRy \wedge ySz)).$$
x and z are related iff there is a path $x \overset{R}{\to} y \overset{S}{\to} z$. ***S* ∘ *R* applies *R* first.**

METHOD **Computing $S \circ R$ by hand**
For each pair $(x, y) \in R$, list every $(y, z) \in S$ that **starts where it ends**, and record (x, z). For $R \circ R \circ R$, compute $R_2 = R \circ R$ first, then $R \circ R_2$. Stop and re-check pairs that could join through two different middles.

PAST PAPER **AY25 Q17(a) (4 marks) · $S = \{(1,3),(3,6),(6,9),(9,12)\}$, $R = \{(6,6),(9,9)\}$**
$R \circ S$ (apply S then R): $(3,6)(6,6) \to (3,6)$; $(6,9)(9,9) \to (6,9)$. So $R \circ S = \{(3,6),(6,9)\}$.
$S \circ R$ (apply R then S): $(6,6)(6,9) \to (6,9)$; $(9,9)(9,12) \to (9,12)$. So $S \circ R = \{(6,9),(9,12)\}$.
(i) $(R \circ S) \setminus (S \circ R) = \{(\mathbf{3},\mathbf{6})\}$.
(ii) $(R^{-1} \circ S^{-1}) \circ (S \circ R) = (S \circ R)^{-1} \circ (S \circ R)$; with $(S \circ R)^{-1} = \{(9,6),(12,9)\}$ this is $\{(\mathbf{6},\mathbf{6}),(\mathbf{9},\mathbf{9})\}$.

PAST PAPER **AY24 Q11, Q17(e), Q12 · Powers and formula relations**
Q11. $R = \{(a,a),(a,b),(b,c),(c,a),(c,b)\}$ on $\{a,b,c\}$. $R \circ R$ is all of $A \times A$ except (b,c); then bRc and $c(R \circ R)c$ put (b,c) into $R \circ R \circ R$, which is $\boldsymbol{A \times A}$. Answer B.
Q17(e). $xR_1y \leftrightarrow y = 2x$, $xR_2y \leftrightarrow x = y^2$ on $\mathbb{Z}^+$. $x(R_1 \circ R_2)z \leftrightarrow \exists y(x = y^2 \wedge z = 2y)$, so $\boldsymbol{x = z^2/4}$ (with z even).
Q12. $|R \circ R|$ has no fixed relation to $|R|$: on $\{a,b\}$, $R = \{(a,a),(a,b),(b,a)\}$ gives $|R \circ R| = 4 > 3$; $\{(a,a),(a,b)\}$ gives $2 = 2$ but $\neq 2^2$; $\{(a,a)\}$ gives $1 = 1^2$. None of (i)–(iii) always holds. Answer **E**.

PROOF **$(S \circ R)^{-1} = R^{-1} \circ S^{-1}$ (Lecture 6, slide 18)**
Let $R \subseteq A \times B$ and $S \subseteq B \times C$. For all $z \in C$, $x \in A$: $(z,x) \in (S \circ R)^{-1} \leftrightarrow (x,z) \in S \circ R$ *(definition of inverse)*
$\leftrightarrow \exists y \in B\,(xRy \wedge ySz)$ *(definition of $\circ$)*
$\leftrightarrow \exists y \in B\,(zS^{-1}y \wedge yR^{-1}x)$ *(definition of inverse; commutative law)*
$\leftrightarrow (z,x) \in R^{-1} \circ S^{-1}$ *(definition of $\circ$)*. ■

PROOF **Proposition (Lecture 6, slide 18): $(T \circ S) \circ R = T \circ (S \circ R)$**
Let $R \subseteq A \times B$, $S \subseteq B \times C$, $T \subseteq C \times D$, $x \in A$ and $w \in D$.
$x((T \circ S) \circ R)w$
$\leftrightarrow \exists y(xRy \wedge \exists z(ySz \wedge zTw))$
$\leftrightarrow \exists z \exists y(xRy \wedge ySz \wedge zTw)$ *(z is not free in xRy; same-kind quantifiers commute; associative law)*
$\leftrightarrow \exists z(\exists y(xRy \wedge ySz) \wedge zTw)$ *(y is not free in zTw)*
$\leftrightarrow \exists z(x(S \circ R)z \wedge zTw) \leftrightarrow x(T \circ (S \circ R))w$. ■

PAST PAPER **T4 Q6 (AY23/24 exam) · Seven copies of an equivalence relation**
$R = \{(1,1),(1,2),(2,1),(2,2),(3,3)\}$ on $\{1,2,3\}$; find $R \circ R \circ \cdots \circ R$ (7 copies). R is an equivalence relation, so $R \circ R = R$ (AY23 Q12, §23). By associativity, $R^7 = ((R \circ R) \circ (R \circ R)) \circ ((R \circ R) \circ R) = (R \circ R) \circ (R \circ R)$
$= R \circ R = R$.

WORKED EXAMPLE **T4 Q9 · Inverse and composition by the element method**
$S = \{(m,n) \in \mathbb{Z}^2 : m^2 + n^2 \text{ is even}\}$. Determine (a) S^{-1}, (b) $S \circ S$, (c) $S \circ S^{-1}$.
(a) $S^{-1} = \{(x,y) \in \mathbb{Z}^2 : (y,x) \in S\}$ *(definition of inverse)*
$= \{(x,y) \in \mathbb{Z}^2 : y^2 + x^2 \text{ is even}\}$ *(definition of S)*
$= \{(x,y) \in \mathbb{Z}^2 : x^2 + y^2 \text{ is even}\}$ *(commutativity of $+$)*
$= S$.
(b) $S \circ S = S$:
1. ($\subseteq$) Let $(x,z) \in S \circ S$. Then $(x,y) \in S$ and $(y,z) \in S$ for some $y \in \mathbb{Z}$ *(definition of composition)*.
   1.1. So $x^2 + y^2$ and $y^2 + z^2$ are even *(definition of S)*, hence $x^2 + 2y^2 + z^2$ is even *(sum of two even integers)*.
   1.2. As $2y^2$ is even *(definition of even, closure)*, $x^2 + z^2$ is even *(difference of two even integers)*, so $(x,z) \in S$ *(definition of S)*.
2. ($\supseteq$) Let $(x,z) \in S$. Note $(x,x) \in S$, as $x^2 + x^2$ is even. Since $(x,x) \in S$ and $(x,z) \in S$, $(x,z) \in S \circ S$ *(definition of composition)*.
3. Therefore $S \circ S = S$. ■

(c) $S \circ S^{-1} = S \circ S = S$ by (a) and (b).

WORKED EXAMPLE **Lecture 6 · Composition as "going through" a middle set**
"takes" $\subseteq$ Students $\times$ Modules and "held in" $\subseteq$ Modules $\times$ Venues. Ann takes CS1010, CS1231, MA1101; Bryan takes CS1010, IS1103; Candy takes CS1231, IS1103; Danny takes nothing. CS1010 is held in LT15; CS1231 in ICube and SR1; IS1103 in SR1; CS2100 in LT15; MA1101 nowhere.
"held in $\circ$ takes" = "goes to": Ann $\to$ {LT15, ICube, SR1}; Bryan $\to$ {LT15, SR1}; Candy $\to$ {ICube, SR1}; Danny $\to$ nothing. CS2100 contributes no pair, since nobody takes it.

21 · Properties of relations on a set A

DEFINITION **Reflexive, symmetric, transitive (Lecture 6.2) · Antisymmetric (6.4) · Asymmetric (Tutorial 5)**

	R has the property iff	R fails it iff
Reflexive	$\forall x \in A\,(xRx)$	$\exists x \in A\,(x \not R x)$
Symmetric	$\forall x, y \in A\,(xRy \to yRx)$	$\exists x, y\,(xRy \wedge y \not R x)$
Transitive	$\forall x, y, z \in A\,(xRy \wedge yRz \to xRz)$	$\exists x, y, z\,(xRy \wedge yRz \wedge x \not R z)$
Antisymmetric	$\forall x, y \in A\,(xRy \wedge yRx \to x = y)$	$\exists x, y\,(xRy \wedge yRx \wedge x \neq y)$
Asymmetric	$\forall x, y \in A\,(xRy \to y \not R x)$	$\exists x, y\,(xRy \wedge yRx)$

THEOREM **Useful facts**

- R symmetric $\Leftrightarrow R = R^{-1}$ (Tutorial 4).
- R transitive $\Leftrightarrow R \circ R \subseteq R$.
- R reflexive $\Rightarrow R \subseteq R \circ R$.
- Asymmetric $\Rightarrow$ antisymmetric (and irreflexive).
- **Antisymmetric is not "not symmetric":** any $R \subseteq \{(x, x) : x \in A\}$ is both.
- "=" is an equivalence relation **and** a partial order (AY23 Q10, AY24 Q9).
- The empty relation on $A \neq \varnothing$ is symmetric, transitive, antisymmetric and asymmetric, but not reflexive.
- If R, S **both** have the property: reflexivity survives $\cup, \cap, \circ$; symmetry survives $\cup, \cap$ but not $\circ$; transitivity and antisymmetry survive only $\cap$.
- Write "0 is related to itself", not "0 is reflexive".

METHOD **Proof skeletons for the properties**

Reflexive: "Let $x \in A$. … so xRx." **Symmetric:** "Let $x, y \in A$ with xRy. … so yRx."
Transitive: "Let $x, y, z \in A$ with xRy and yRz. … so xRz." **Antisymmetric:** "Let xRy and yRx. … so $x = y$."
Disproof: one explicit counterexample, named elements, each membership checked.

PAST PAPER **AY23 Q9 and Q10 · Counterexamples on small sets**

Q9 (R reflexive, S symmetric). On $\{a, b, c\}$ take $R = \{(a, a), (b, b), (c, c), (a, c)\}$, $S = \{(a, b), (b, a)\}$: $R \circ S = \{(a, b), (b, a), (b, c)\}$ is not symmetric; $R \cup S$ is not symmetric; $(R \circ S)^{-1}$ is not transitive. Only "$R \cup S$ is reflexive" survives: **A**.
Q10. Asymmetric $\Rightarrow$ antisymmetric: true. "Not reflexive $\Rightarrow$ not asymmetric": false ($R = \{(a, b)\}$). "Not reflexive $\Rightarrow$ asymmetric": false ($R = \{(a, a)\}$ on $\{a, b\}$). "No relation is both an equivalence relation and a partial order": false (=). Only (i): **A**.

PAST PAPER **AY24 Q10, AY25 Q11, AY25 Q14**

AY24 Q10. $pRq \leftrightarrow (p \to q)$ on $\{T, F\}$: $R = \{(T, T), (F, F), (F, T)\}$ is reflexive, antisymmetric, transitive, not symmetric: **D**.
AY25 Q11. $ARB \leftrightarrow |A \cap B| = 1$ on $\mathcal{P}(\mathbb{Z})$: symmetric; not reflexive ($|\varnothing \cap \varnothing| = 0$); not transitive. **Symmetric but not an equivalence relation.** Answer B.
AY25 Q14. Reflexive R on $\{1, 2, 3, 4\}$ with $(xRy \wedge xRz) \to y = z$: reflexivity forces xRx, so each x is related to **nothing else**. Only the identity relation qualifies, so $|\mathcal{F}| = 1$. Answer B.

PROOF **R is symmetric $\leftrightarrow R = R^{-1}$ (Tutorial 4)**

1. ($\to$) Suppose R symmetric. $(x, y) \in R \to (y, x) \in R$ *(symmetry)* $\to (x, y) \in R^{-1}$ *(definition of inverse)*, so $R \subseteq R^{-1}$. Conversely $(x, y) \in R^{-1} \to (y, x) \in R \to (x, y) \in R$ *(symmetry)*, so $R^{-1} \subseteq R$.
2. ($\leftarrow$) Suppose $R = R^{-1}$ and xRy. Then $(y, x) \in R^{-1} = R$, i.e. yRx. ■

PROOF **R is transitive $\leftrightarrow R \circ R \subseteq R$ (AY23 Q12 model answer, extended)**

1. ($\to$) Let $(x, z) \in R \circ R$. There is some y with xRy and yRz *(definition of composition)* so xRz *(transitivity)*, i.e. $(x, z) \in R$.
2. ($\leftarrow$) Let xRy and yRz. Then $(x, z) \in R \circ R$ *(definition of composition)* $\subseteq R$, so xRz. ■

Similarly, **reflexive $\Rightarrow R \subseteq R \circ R$**: given $(x, z) \in R$, use xRx and xRz with middle $y = x$.

WORKED EXAMPLE **T5 Q7 · Asymmetric vs antisymmetric**

On $A = \{a, b, c\}$ find a relation that is (a) asymmetric and antisymmetric; (b) antisymmetric but not asymmetric; (c) asymmetric but not antisymmetric; (d) neither.
(a) $\{(a, b), (a, c)\}$. (b) $\{(a, a), (b, c)\}$. (c) **No solution:** every asymmetric relation is antisymmetric (below). (d) $\{(a, b), (b, a), (a, c)\}$.

1. Let R be an asymmetric relation on A: $\forall x, y \in A\,(xRy \to y\not{R}x)$ *(definition of asymmetry)*.
   1.1. $\equiv \forall x, y \in A\,(x\not{R}y \vee y\not{R}x)$ *(implication law)*.
   1.2. $\Rightarrow \forall x, y \in A\,((x\not{R}y \vee y\not{R}x) \vee x = y)$ *(generalization)*.
   1.3. $\equiv \forall x, y \in A\,(\sim(xRy \wedge yRx) \vee x = y)$ *(De Morgan's law)*.
   1.4. $\equiv \forall x, y \in A\,(xRy \wedge yRx \to x = y)$ *(implication law)*.
2. The final formula is the definition of antisymmetry, so R is antisymmetric: asymmetry makes it vacuously true. ■

22 · Closures

DEFINITION **Transitive closure (Lecture 6.2.2); reflexive and symmetric closures**
The **transitive closure** of R on A is the relation R^t such that (1) R^t is transitive, (2) $R \subseteq R^t$, (3) $R^t \subseteq S$ for every transitive $S \supseteq R$. It is the **smallest** transitive relation containing R. Likewise the **reflexive closure** $R^r = R \cup \{(x, x) : x \in A\}$ and the **symmetric closure** $R^s = R \cup R^{-1}$.

METHOD **Computing R^t**
$R^t = R \cup R^2 \cup R^3 \cup \cdots$ where $R^2 = R \circ R$; on a finite set stop once a power adds nothing new (at most $|A|$ powers). On the digraph: draw $x \to z$ whenever some path leads from x to z. A cycle back to the starting point creates a loop.

PAST PAPER **AY24 Q17(a), (b) (5 marks)**
(a) $R = \{(a, b), (a, c)\}$ on $\{a, b, c\}$: $|R^r| = 2 + 3 = \mathbf{5}$; $|R^s| = |\{(a, b), (a, c), (b, a), (c, a)\}| = \mathbf{4}$; $|R^t| = \mathbf{2}$, since no pair ends where another begins, so R is already transitive.
(b) $S = \{(a, b), (b, a), (b, d), (d, c), (b, b), (d, d)\}$; $T = S^t \setminus S$ consists of the new pairs obtained from paths: $a \to b \to a$ gives (a, a); $a \to b \to d$ gives (a, d); $a \to b \to d \to c$ gives (a, c); $b \to d \to c$ gives (b, c). $\boldsymbol{T = \{(a, a), (a, c), (a, d), (b, c)\}}$.

WORKED EXAMPLE **T5 Q6 · Reflexive closure (the closure-proof template)**
Let R be a relation on a non-empty set A, and define S on A by $xSy \leftrightarrow ((x = y) \vee (xRy))$. Show: (a) S is reflexive; (b) $R \subseteq S$; (c) if S' is a reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$. What is S called?
(a) Let $x \in A$. $x = x$, so xSx *(definition of S)*. Therefore S is reflexive.
(b) Suppose $(x, y) \in R$. Then xSy *(definition of S)*, so $(x, y) \in S$. Therefore $R \subseteq S$ *(definition of $\subseteq$)*.
(c) Let $(x, y) \in S$: $x = y$ or xRy *(definition of S)*.
1. **Case 1 ($x = y$):** $xS'y$ *(S' is reflexive)*, so $(x, y) \in S'$.
2. **Case 2 (xRy):** $(x, y) \in R \subseteq S'$, so $(x, y) \in S'$.

In all cases $(x, y) \in S'$. Therefore $S \subseteq S'$. ■
S is the **reflexive closure** of R: the smallest reflexive relation on A containing R. Parts (a)–(c) mirror the three conditions defining a closure.

WORKED EXAMPLE **Composition and closure**
$A = \{1, 2, 3, 4\}$, $R = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$. Find $R \circ R$, $R^{-1} \circ R$ and $|R^t|$.
$R \circ R = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$ (two steps round the 4-cycle). $R^{-1} \circ R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$: follow an arrow of R, then return along the same arrow. From any vertex a path reaches every vertex, itself included, so $R^t = A \times A$ and $|R^t| = 16$.

WORKED EXAMPLE **All the closures**
$A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 3)\}$. Find R^r, R^s, R^t and the smallest equivalence relation containing R.
$R^r = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\}$ (5 pairs); $R^s = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$ (4); $R^t = \{(1, 2), (2, 3), (1, 3)\}$ (3).
The smallest equivalence relation containing R is $A \times A$ (9 pairs): ignoring directions, 1–2–3 is one connected piece, so there is a single class.

THEOREM **Closure facts**
The transitive closure of a **symmetric** relation is symmetric, and of a **reflexive** relation is reflexive (reverse or extend the paths). Hence the **smallest equivalence relation containing R** is $\left(R \cup R^{-1} \cup \{(x, x) : x \in A\}\right)^t$. Its classes are the connected components of the digraph of R when arrow directions are ignored.

23 · Equivalence relations

DEFINITION **Relation induced by a partition · Equivalence relation**
Given a partition $\mathcal{C}$ of A, the **induced relation** is $xRy \leftrightarrow x, y$ lie in the same component; it is reflexive, symmetric and transitive (Theorem 8.3.1).
R on A is an **equivalence relation** iff it is **reflexive, symmetric and transitive**. Notation: $\sim$.

DEFINITION **Equivalence class · Quotient $A/\sim$**
For an equivalence relation $\sim$ on A and $a \in A$, the **equivalence class** of a (the **class of a**) is $[a]_\sim = \{x \in A : a \sim x\}$; procedurally, $\forall x \in A\,(x \in [a]_\sim \leftrightarrow a \sim x)$.
$A/\sim = \{[x]_\sim : x \in A\}$, the set of all equivalence classes, read "the quotient of A by $\sim$". Its **elements** are the equivalence classes.

DEFINITION **Congruence modulo n (Lecture 6.3.4)**
Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$. a is **congruent to b modulo n**, written $a \equiv b \pmod n$, iff $a - b = nk$ for some $k \in \mathbb{Z}$; in other words, $n \mid (a-b)$.
E.g. $7 \equiv 1 \pmod 2$; $-3 \equiv 12 \pmod 5$; $-4 \not\equiv 5 \pmod 7$. The classes are $[x] = \{x + nk : k \in \mathbb{Z}\}$, and $\mathbb{Z}/\sim_n = \{[0], [1], ..., [n-1]\}$.

PROOF **Lemma Rel.1 (Lecture 6)**
Let $\sim$ be an equivalence relation on A. For $x, y \in A$, these are equivalent: (i) $x \sim y$; (ii) $[x] = [y]$; (iii) $[x] \cap [y] \neq \varnothing$.
1. ((i) $\Rightarrow$ (ii)) Suppose $x \sim y$; then $y \sim x$ *(symmetry)*.
   1.1. For every $z \in [x]$: $x \sim z$ *(definition of $[x]$)*, so $y \sim z$ *(transitivity, as $y \sim x$)*, so $z \in [y]$. Thus $[x] \subseteq [y]$.
   1.2. Symmetrically $[y] \subseteq [x]$, so $[x] = [y]$.
2. ((ii) $\Rightarrow$ (iii)) Suppose $[x] = [y]$. $x \sim x$ *(reflexivity)*, so $x \in [x] = [x] \cap [y]$ *(idempotent law)* hence $[x] \cap [y] \neq \varnothing$.
3. ((iii) $\Rightarrow$ (i)) Take $z \in [x] \cap [y]$. Then $x \sim z$ and $y \sim z$ *(definitions of $[x], [y]$)* so $z \sim y$ *(symmetry)* and $x \sim y$ *(transitivity)*. ■

PROOF **Proposition (Lecture 6, slide 54) · Congruence mod n is an equivalence relation on $\mathbb{Z}$ ($a \equiv b \pmod n \leftrightarrow n \mid (a-b)$)**
1. (Reflexive) For all $a \in \mathbb{Z}$: $a - a = 0 = n \cdot 0$, so $a \equiv a \pmod n$ *(definition of congruence)*.
2. (Symmetric) Let $a \equiv b \pmod n$: $a - b = nk$ for some $k \in \mathbb{Z}$. Then $b - a = n(-k)$ with $-k \in \mathbb{Z}$ *(closure)*, so $b \equiv a \pmod n$.
3. (Transitive) Let $a \equiv b$ and $b \equiv c \pmod n$: $a - b = nk$, $b - c = nl$. Then $a - c = n(k+l)$ with $k + l \in \mathbb{Z}$, so $a \equiv c \pmod n$. ■

PAST PAPER **AY23 Q19 (6 marks) · $aSb \leftrightarrow \exists k \in \mathbb{Z}\,(a - b = k/2)$ on $\mathbb{Q}$**
(a) Prove S transitive (official model answer):
1. Let $a, b, c \in \mathbb{Q}$ such that aSb and bSc.
2. There exist $m, n \in \mathbb{Z}$ such that $a - b = m/2$ and $b - c = n/2$ *(by definition of S)*.
3. Now $a - c = (a-b) + (b-c) = m/2 + n/2 = (m+n)/2$ *(by basic algebra)*.
4. $m + n \in \mathbb{Z}$ *(by closure of integers under addition)*.
5. Therefore aSc *(by definition of S)*. ■

(b) Two distinct classes: $[0] = \{k/2 : k \in \mathbb{Z}\}$ and $[0.1]$. **(c)** $\mathbb{Q}/S = \{[x] : x \in \mathbb{Q} \wedge 0 \leq x < 1/2\}$. Each class has exactly one representative in $[0, 1/2)$, so there are no duplicates.

PAST PAPER **AY25 Q17(c), (d), Q15 · AY24 Q9 · AY23 Q12**
AY25 Q17(c). "Same biological father **and** mother": **yes**, an equivalence relation. "Share **a** biological parent": **no**, because it is not transitive (half-siblings).
AY25 Q17(d). $x \sim y \leftrightarrow \exists d \in \mathbb{Z}^+(d \mid x \wedge d \mid y)$: $d = 1$ always works, so everything is related: $\mathbb{Z}^+/\sim = \{\mathbb{Z}^+\} = \{[1]\}$.
AY25 Q15. "$\forall x \in A\,([x] \subseteq A/\sim)$" is **false**: classes are **elements** of $A/\sim$, not subsets.
"$\exists x \in A/\sim \forall y \in x\,(y \in A)$" and "$|A| = \sum_{x \in A/\sim} |x|$" are true: **E**.
AY24 Q9. Divisibility on 10 primes is the identity relation: 10 classes **and** 10 minimal elements. Answer **C**.
AY23 Q12. For an equivalence relation R, all four hold: $R = R^{-1}$ (symmetry), so $R^{-1} \circ R = R \circ R^{-1} = R \circ R$; $R \circ R \subseteq R$ (transitivity: $xRy \wedge yRz \Rightarrow xRz$); $R \subseteq R \circ R$ (reflexivity: take the middle $y = x$). Hence $R \circ R^{-1} = R$: **D**.

PROOF **Theorem Rel.2 · Theorem 8.3.4 · If $\sim$ is an equivalence relation on A, then $A/\sim$ is a partition of A (Lecture 6)**

1. $A/\sim$ is a set *(by definition)*.
2. Every element is a non-empty subset of A: $[x] = \{y \in A : x \sim y\} \subseteq A$, and $x \in [x]$ *(reflexivity)*.
3. Every $x \in A$ is in exactly one class:
   3.1. $x \in [x]$, so x is in at least one class.
   3.2. If $x \in [y]$ and $x \in [z]$, then $[y] \cap [z] \neq \varnothing$, so $[y] = [z]$ *(Lemma Rel.1, (iii) ⇒ (ii))*. ■

PROOF **Theorem 8.3.1 · The relation induced by a partition $\mathcal{C}$ of A is an equivalence relation**

1. (Reflexive) Let $x \in A$. x lies in some component S *(definition of partition)*, so $x, x \in S$ and xRx.
2. (Symmetric) If xRy, then x, y lie in one component S, so $y, x \in S$ and yRx.
3. (Transitive) Let xRy and yRz: $x, y \in S_1$ and $y, z \in S_2$ for components S_1, S_2. Then $y \in S_1 \cap S_2$, so $S_1 = S_2$ *(each element lies in exactly one component)*, hence $x, z \in S_1$ and xRz. ■

Together with Theorem Rel.2, this gives a one-to-one correspondence between partitions and equivalence relations on A. Counts: §55.

PROOF **On $A = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, $(a,b) \sim (c,d) \leftrightarrow ad = bc$ is an equivalence relation**

1. (Reflexive) $ab = ba$ *(commutativity)*, so $(a,b) \sim (a,b)$.
2. (Symmetric) If $ad = bc$ then $cb = da$ *(commutativity)*, so $(c,d) \sim (a,b)$.
3. (Transitive) Let $ad = bc$ and $cf = de$, with $b, d, f \neq 0$.
   3.1. Multiply: $adf = bcf = bde$, so $d(af - be) = 0$ *(basic algebra)*.
   3.2. $d \neq 0$, so $af = be$ *(zero product property)*, i.e. $(a,b) \sim (e,f)$. ■

Each class is one rational number: $[(1,2)] = \{(1,2), (2,4), (-3,-6), ...\}$; $A/\sim$ "is" $\mathbb{Q}$.

METHOD **Writing $A/\sim$ without duplicates**

Pick **one representative per class** from a region that meets every class exactly once, then write $\{[x] : x \in$ that region$\}$.

- Congruence mod 3: $\mathbb{Z}/\equiv_3 = \{[0], [1], [2]\}$, with $[r] = \{3k + r : k \in \mathbb{Z}\}$.
- AY23 Q19: $\mathbb{Q}/S = \{[x] : x \in \mathbb{Q} \wedge 0 \leq x < 1/2\}$.
- Lecture Ex. 15: nonempty $A \subseteq \{1,2,3\}$, related iff they have the same least element: $[\{1\}]$ $= \{\{1\}, \{1,2\}, \{1,3\}, \{1,2,3\}\}$, $[\{2\}] = \{\{2\}, \{2,3\}\}$, $[\{3\}] = \{\{3\}\}$, three classes.

PAST PAPER **T4 Q8 (AY20/21 exam) · Classes of ordered pairs**

On $\mathbb{Z}^+ \times \mathbb{Z}^+$, $(a,b) \sim (c,d) \leftrightarrow ab = cd$. Write the set-builder line first: $[(1,1)]$
$= \{(x,y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : (1,1) \sim (x,y)\}$ *(definition of class)*
$= \{(x,y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 1\}$ *(definition of $\sim$)*
$= \{(1,1)\}$. Likewise $[(4,3)] = \{(x,y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 12\} = \{(1,12), (2,6), (3,4), (4,3), (6,2), (12,1)\}$.

WORKED EXAMPLE **T4 Q10 · Equivalence relation using Appendix A**

On $\mathbb{Z} \setminus \{0\}$ let $a \sim b \leftrightarrow ab > 0$. (a) Prove $\sim$ is an equivalence relation. (b) Determine all distinct equivalence classes.

1. (Reflexivity) Let $a \in \mathbb{Z} \setminus \{0\}$. Since $a \neq 0$, $a^2 > 0$ *(T21)*. Thus $a \sim a$ *(definition of $\sim$)*.
2. (Symmetry) Let $a, b \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$. Then $ab > 0$ *(definition of $\sim$)*, so $ba > 0$ *(commutative law of $\times$)*, so $b \sim a$.
3. (Transitivity) Let $a, b, c \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$ and $b \sim c$. Then $ab > 0$ and $bc > 0$ *(definition of $\sim$)*.
   3.1. Multiplying, $ab^2c > 0$ *(Ord1)*, so $(ac)b^2 > 0$ *(associative, commutative laws of $\times$)*.
   3.2. So ac and b^2 are both positive or both negative *(T25)*.
   3.3. Since $b^2 > 0$ *(T21, as $b \neq 0$)*, $ac > 0$. Thus $a \sim c$.
4. Therefore $\sim$ is an equivalence relation. ■

(b) T25 rules out pairs of opposite signs; the product of two positive or two negative integers is positive. The classes are $[1] = \mathbb{Z}^+$ and $[-1] = \mathbb{Z}^-$, so $(\mathbb{Z} \setminus \{0\})/\sim = \{\mathbb{Z}^+, \mathbb{Z}^-\}$.

DEFINITION **Representative · Well-defined operation on $A/\sim$**

Any $b \in [a]$ is a **representative** of $[a]$, since then $[b] = [a]$: mod 5, $[2] = [7] = [12]$.

Let $\sim$ be an equivalence relation on A and $*$ a binary operation on A (so $a * b \in A$ for all $a, b \in A$). The induced rule $[a] * [b] = [a * b]$ is **well defined** iff the answer depends only on the classes, not on the representatives chosen: $a \sim a' \wedge b \sim b' \Rightarrow a * b \sim a' * b'$.

PROOF **$+$ and $\cdot$ are well defined on the classes of congruence mod n**

1. Fix $n \in \mathbb{Z}^+$ and let $a, a', b, b' \in \mathbb{Z}$ with $a \equiv a' \pmod n$ and $b \equiv b' \pmod n$: $a - a' = nk$ and $b - b' = nm$ for some $k, m \in \mathbb{Z}$ *(definition of congruence)*.
2. (Addition) $(a+b) - (a'+b') = n(k+m)$ with $k+m \in \mathbb{Z}$ *(closure)*, so $a+b \equiv a'+b' \pmod n$.
3. (Multiplication) $ab - a'b' = (a-a')b + a'(b-b') = n(kb + a'm)$ with $kb + a'm \in \mathbb{Z}$ *(closure)*, so $ab \equiv a'b' \pmod n$. ■

E.g. mod 5: $[2] \cdot [4] = [8] = [3]$ and $[7] \cdot [9] = [63] = [3]$.

TRAP **Check reflexive, symmetric, transitive first · then check well-definedness**

- $a \sim b \leftrightarrow ab \geq 0$ is reflexive and symmetric, but **not** transitive on either $\mathbb{Z}$ or $\mathbb{R}$ ($1 \sim 0$ and $0 \sim -1$, but $1 \nsim -1$), so it does not give a partition into equivalence classes. On $\mathbb{Z} \setminus \{0\}$ its classes are $\mathbb{Z}^+$ and $\mathbb{Z}^-$ (§23, T4 Q10). On $\mathbb{R} \setminus \{0\}$ it means "same sign": $(\mathbb{R} \setminus \{0\})/\sim = \{[1], [-1]\} = \{(0, \infty), (-\infty, 0)\}$.
- $a \sim b \leftrightarrow a - b = 2k$ for some $k \in \mathbb{Z}$, on $\mathbb{R}$: $[a] = \{a + 2k : k \in \mathbb{Z}\}$ and $\mathbb{R}/\sim = \{[r] : 0 \leq r < 2\}$ (infinitely many classes). $+$ is well defined, but $\cdot$ is **not**: $[0.5] = [2.5]$, yet $[0.5] \cdot [0.5] = [0.25] \neq [1.25] = [2.5] \cdot [0.5]$, since $1.25 - 0.25 = 1$ is not of the form $2k$.

WORKED EXAMPLE **Equivalence relation and quotient**

On $\mathbb{Z}$ let $xRy \leftrightarrow 3 \mid (x^2 - y^2)$. Prove R is an equivalence relation and find $\mathbb{Z}/R$.

1. (Reflexive) $x^2 - x^2 = 0 = 3 \cdot 0$, so xRx *(definition of divisibility)*.
2. (Symmetric) If $x^2 - y^2 = 3k$ then $y^2 - x^2 = 3(-k)$ with $-k \in \mathbb{Z}$, so yRx.
3. (Transitive) If $x^2 - y^2 = 3k$ and $y^2 - z^2 = 3l$, then $x^2 - z^2 = 3(k+l)$, so xRz.
4. (Classes) By the QR theorem $x = 3q + r$ with $r \in \{0, 1, 2\}$; $x^2 = 9q^2 + 6qr + r^2$, so $x^2 \bmod 3 = 0$ if $r = 0$ and 1 if $r = 1, 2$. Hence $\mathbb{Z}/R = \{[0], [1]\} = \{\{3k : k \in \mathbb{Z}\}, \{n \in \mathbb{Z} : 3 \nmid n\}\}$. ■

WORKED EXAMPLE **Classes of a relation on pairs**

On $\mathbb{Z} \times \mathbb{Z}$ let $(a, b) \sim (c, d) \leftrightarrow a + d = b + c$. Show $\sim$ is an equivalence relation and describe $(\mathbb{Z} \times \mathbb{Z})/\sim$.

1. Rewrite: $(a, b) \sim (c, d) \leftrightarrow a - b = c - d$ *(basic algebra)*. Then reflexivity, symmetry and transitivity are those of $=$ on $\mathbb{Z}$ (e.g. $a - b = c - d$ and $c - d = e - f$ give $a - b = e - f$).
2. So $[(a, b)] = \{(c, d) : c - d = a - b\}$, a diagonal line of lattice points, and each class contains exactly one point $(k, 0)$ with $k = a - b$.
3. $(\mathbb{Z} \times \mathbb{Z})/\sim = \{[(k, 0)] : k \in \mathbb{Z}\}$, with one class for each integer and no duplicates. ■

24 · Partial orders

DEFINITION **Partial order (Lecture 6.4.2) · Poset**

R on A is a **partial order** iff it is **reflexive, antisymmetric and transitive**; then (A, R) is a **partially ordered set** (poset). A general partial order is written $\preccurlyeq$ ("curly less than or equal to"). Examples: $\leq$ on $\mathbb{R}$; $\subseteq$ on any set of sets; $\mid$ on $\mathbb{Z}^+$. Non-examples: $<$ (not reflexive); congruence mod n (not antisymmetric); $\mid$ on $\mathbb{Z}$ ($2 \mid -2$ and $-2 \mid 2$, but $2 \neq -2$).

TRAP **T5 Q1 · Disprove with one named counterexample**

On strings over $\{s, u\}$, $aRb \leftrightarrow \text{len}(a) \leq \text{len}(b)$ is **not** a partial order: sRu and uRs but $s \neq u$, so R is not antisymmetric. One counterexample suffices; there is no need to check reflexivity or transitivity.

PROOF **$\mid$ is a partial order on any set $A \subseteq \mathbb{Z}^+$ (Lecture 6, Ex. 20)**

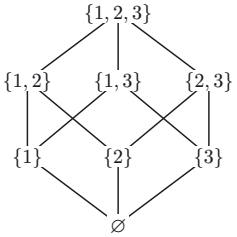
1. (Reflexive) Let $a \in A$. $a = 1 \cdot a$, so $a \mid a$ *(by definition of divisibility)*.
2. (Antisymmetric) Let $a, b \in A$ with $a \mid b$ and $b \mid a$. Then $a \leq b$ and $b \leq a$ *(by Theorem 4.3.1, as $a, b \in \mathbb{Z}^+$)*, so $a = b$.
3. (Transitive) Let $a \mid b$ and $b \mid c$. Then $a \mid c$ *(by Theorem 4.3.3)*. ■

DEFINITION **Hasse diagram (Lecture 6.4.3)**

For a partial order $\preccurlyeq$ on a finite A and distinct x, y: **if $x \preccurlyeq y$ and no $m \in A$ has $x \preccurlyeq m \preccurlyeq y$ (with $m \neq x, y$), draw x below y joined by a line; otherwise no line.**

From the digraph: place arrows pointing up; delete (1) loops, (2) arrows implied by transitivity, (3) arrowheads.

Reading it: $x \preccurlyeq y$ iff you can climb from x to y along lines.



DEFINITION **Comparable · Compatible (AY24 paper) · Chain, antichain (AY24 paper) · Total order**

a, b are **comparable** iff $a \preccurlyeq b$ or $b \preccurlyeq a$; otherwise **noncomparable**. a, b are **compatible** iff $\exists c \in A\,(a \preccurlyeq c \wedge b \preccurlyeq c)$.

A **chain** is a subset C in which every two elements are comparable; a **maximal chain** M has no $t \notin M$ with $M \cup \{t\}$ a chain; the **length** of a chain is one less than its number of elements. An **antichain** is a subset in which no two distinct elements are comparable.

$\preccurlyeq$ is a **total (linear) order** iff it is a partial order and $\forall x, y \in A\,(x \preccurlyeq y \vee y \preccurlyeq x)$; its Hasse diagram is a single chain. $\mid$ on $\mathbb{Z}^+$ is partial, not total ($3 \nmid 5$, $5 \nmid 3$); $\leq$ on $\mathbb{Q}$ is total.

DEFINITION **Maximal, minimal, largest, smallest (Lecture 6.4.5)**

For a poset $(A, \preccurlyeq)$ and $c \in A$:

- c **maximal** $\Leftrightarrow \forall x \in A\,(c \preccurlyeq x \to c = x)$: nothing lies strictly above c. Equivalently, every $x \in A$ has $x \preccurlyeq c$ or is not comparable with c.
- c **minimal** $\Leftrightarrow \forall x \in A\,(x \preccurlyeq c \to c = x)$: nothing lies strictly below c. Equivalently, every $x \in A$ has $c \preccurlyeq x$ or is not comparable with c.
- c **largest** $\Leftrightarrow \forall x \in A\,(x \preccurlyeq c)$: every element lies below c.
- c **smallest** $\Leftrightarrow \forall x \in A\,(c \preccurlyeq x)$: every element lies above c.

Largest = greatest = maximum; smallest = least = minimum. On a Hasse diagram: maximal = no line going up; minimal = no line going down.

DEFINITION **Well-ordered set (Lecture 6.4.7)**

Let $\preccurlyeq$ be a total order on a set A. A is **well-ordered** iff every non-empty subset of A contains a smallest element. Symbolically,

$$\forall S \in \mathcal{P}(A)\,(S \neq \varnothing \to \exists x \in S\,\forall y \in S\,(x \preccurlyeq y)).$$

$(\mathbb{N}, \leq)$ is well-ordered. $(\mathbb{Z}, \leq)$ is not: $\mathbb{Z}$ itself (or $\mathbb{Z}^-$) is a non-empty subset with no smallest element.

THEOREM **Extremal-element facts**

- **Proposition (Lecture 6, slide 83):** smallest ⇒ minimal; largest ⇒ maximal. If c is smallest and $x \preccurlyeq c$, then also $c \preccurlyeq x$, so $x = c$ by antisymmetry.
- A largest (smallest) element, if it exists, is **unique** (two largest elements precede each other, hence are equal by antisymmetry), and is then the **only** maximal (minimal) element.
- Conversely, in a **finite** poset a unique maximal element is largest. This fails for infinite posets.
- A finite non-empty poset has at least one maximal and one minimal element.
- Distinct maximal elements are noncomparable (so are distinct minimal ones).
- An element can be both maximal and minimal: an isolated point.
- Infinite posets may have none: $(\mathbb{Z}, \leq)$.
- Every total order on a finite set is well-ordered: a non-empty subset has a minimal element, and in a total order a minimal element is smallest.

PAST PAPER **AY23 Q13, Q14 · Facts about partial orders**

(i) "No element is both smallest and largest" is false: $(\{a\}, \{(a,a)\})$. (ii) "An element can be maximal and minimal but neither largest nor smallest" is true: $(\{a,b\}, \{(a,a),(b,b)\})$. (iii) "Distinct maximal elements are comparable" is false, as it would contradict maximality. (iv) "There may be no maximal or minimal element" is true: $(\mathbb{Z}, \leq)$. **C**.

Q14. On $\{1,2,3\}$: $P = \{(1,1),(1,2),(2,2)\}$ is not reflexive ($(3,3)$ missing); $Q = \{(1,1),(1,2),(1,3),(2,2),(3,2),(3,3)\}$ **is** a partial order; $R = \{(1,1),(1,2),(2,2),(2,3),(3,3),(3,1)\}$ is not transitive ($1R2$, $2R3$, not $1R3$), and $R^t = A \times A$ is not antisymmetric. Only Q: **B**.

WORKED EXAMPLE **Partial order and extremal elements**

On $\mathbb{Z}^+$ let $a \preccurlyeq b \leftrightarrow b = a \cdot 2^k$ for some $k \in \mathbb{N}$. Prove $\preccurlyeq$ is a partial order; find its minimal and maximal elements.

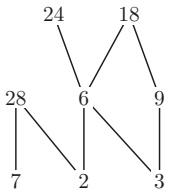
1. (Reflexive) $a = a \cdot 2^0$ and $0 \in \mathbb{N}$.

2. (Antisymmetric) If $b = a2^j$ and $a = b2^k$, then $a = a2^{j+k}$, so $2^{j+k} = 1$, $j = k = 0$, and $a = b$.
3. (Transitive) If $b = a2^j$ and $c = b2^k$ then $c = a2^{j+k}$ with $j + k \in \mathbb{N}$ *(closure)*.
4. (Minimal) $b \preccurlyeq a$ with $b \neq a$ means $a = b2^k$, $k \geq 1$, which is possible iff a is even. So the minimal elements are exactly the **odd** numbers.
5. (Maximal) None: $a \preccurlyeq 2a$ and $2a \neq a$. Hence no largest or smallest element either. ■

25 · Linearization and Kahn's algorithm

DEFINITION Linearization (Lecture 6.4.7)
A **linearization** of a partial order $\preccurlyeq$ on A is a **total order** $\preccurlyeq^*$ on A with $\forall x, y \in A\,(x \preccurlyeq y \to x \preccurlyeq^* y)$. It lists the elements one at a time without breaking any precedence.

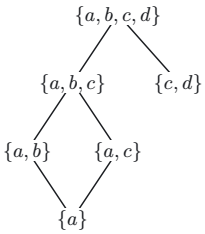
METHOD Kahn's algorithm (1962) · Testing a proposed line-up
Kahn: set $A_0 := A$, $i := 0$; while $A_i \neq \varnothing$: pick a **minimal** element c_i of A_i, set $A_{i+1} := A_i \setminus \{c_i\}$, $i := i + 1$. Output $c_0 \preccurlyeq^* c_1 \preccurlyeq^* \cdots$. Different choices of minimal element give different linearizations.
Test: a sequence is a linearization iff **for every line of the Hasse diagram, the lower element comes earlier**. A single reversed pair means it is not a linearization.



PAST PAPER AY24 Q13–Q15 · $(\{2, 3, 6, 7, 9, 18, 24, 28\}, |)$
Covers: $2 \mid 6$, $2 \mid 28$, $3 \mid 6$, $3 \mid 9$, $7 \mid 28$, $6 \mid 18$, $6 \mid 24$, $9 \mid 18$.
Q13 Longest maximal chains have length 2:
$\{2, 6, 18\}, \{2, 6, 24\}, \{3, 6, 18\}, \{3, 6, 24\}, \{3, 9, 18\}$. There are **5**; answer B.
Q14 $\{2, 3, 28\}$ is not an antichain ($2 \mid 28$); $\{7, 9, 24\}$ is; $\{6\}$ is, vacuously. **D**.
Q15 $2, 3, 6, 7, 9, 18, 24, 28$ ✓; $7, 3, 2, 28, 9, 6, 18, 24$ ✓; $7, 2, 3, 6, 18, 9, 24, 28$ ✗, since $9 \mid 18$ but 18 comes first. **B**.

PAST PAPER AY24 Q17(c), (d) · $(E = \{2, 4, 6, 8, 10, 12\}, |)$
Covers: $2 \mid 4$, $2 \mid 6$, $2 \mid 10$, $4 \mid 8$, $4 \mid 12$, $6 \mid 12$. **Minimal:** 2. **Maximal:** 8, 10, 12. **Smallest:** 2. **Largest:** none.
Two maximal chains of different lengths: $\{2, 4, 8\}$ (length 2) and $\{2, 10\}$ (length 1); $\{2, 4, 12\}$, $\{2, 6, 12\}$ also work for the first.

PAST PAPER AY25 Q17(e) · $(\{\{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}, \{a, b, c, d\}, \{c, d\}\}, \subseteq)$
Covers: $\{a\} \subset \{a, b\}$, $\{a, c\} \subset \{a, b, c\} \subset \{a, b, c, d\}$, and $\{c, d\} \subset \{a, b, c, d\}$. **Smallest element:** no, since $\{a\}$ and $\{c, d\}$ are both minimal and noncomparable. **Maximal:** yes, $\{a, b, c, d\}$ (which is also largest).



PAST PAPER AY23 Q15, Q18 · $x \preccurlyeq z \leftrightarrow F_x = F_z \vee |F_x| < |F_z|$ on $\{11, ..., 16\}$, F_x = positive divisors
Divisor counts: $11 \to 2$, $13 \to 2$, $14 \to 4$, $15 \to 4$, $16 \to 5$, $12 \to 6$. The order compares counts only, so 11, 13 are incomparable, as are 14, 15. **Minimal:** 11, 13. **Smallest:** none. **Maximal:** 12. **Largest:** 12.
Q15: a linearization must list counts in non-decreasing order: $13, 11, 15, 14, 16, 12$ ✓ and $11, 13, 14, 15, 16, 12$ ✓; the other two put 12 too early. **B**.

PROOF AY24 Q17(f), (g) · $(a, b)W(c, d) \leftrightarrow a \leq c \wedge b \geq d$ on $\mathbb{Z} \times \mathbb{Z}$
(f) "Every two elements are comparable" is false. Counterexample: $(1, 3)$ and $(2, 4)$, since neither $1 \leq 2 \wedge 3 \geq 4$ nor $2 \leq 1 \wedge 4 \geq 3$ holds.
(g) "Every two elements are compatible" is true.
1. Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be arbitrary elements of $\mathbb{Z} \times \mathbb{Z}$.
2. Let $c = (\max(x_1, y_1), \min(x_2, y_2)) \in \mathbb{Z} \times \mathbb{Z}$.
3. Then $x_1 \leq c_1$, $y_1 \leq c_1$ *(definition of max)* and $x_2 \geq c_2$, $y_2 \geq c_2$ *(definition of min)*.
4. So xWc and yWc *(definition of W)*: x and y are compatible. ■

Relation	Refl.	Sym.	Trans.	Antisym.	Verdict
$=$ on any set	✓	✓	✓	✓	equivalence **and** partial order
$\leq$ on $\mathbb{R}$	✓	✗	✓	✓	total order
$<$ on $\mathbb{R}$	✗	✗	✓	✓	neither (asymmetric)
$\mid$ on $\mathbb{Z}^+$	✓	✗	✓	✓	partial order, not total
$\mid$ on $\mathbb{Z}$	✓	✗	✓	✗	neither ($2 \mid -2 \mid 2$)
$\equiv \pmod n$ on $\mathbb{Z}$	✓	✓	✓	✗	equivalence, n classes
$\subseteq$ on $\mathcal{P}(A)$	✓	✗	✓	✓	partial order
$\lvert X\rvert = \lvert Y\rvert$ on $\mathcal{P}(A)$	✓	✓	✓	✗	equivalence, classes by size
$xy \geq 0$ on $\mathbb{Z}$	✓	✓	✗	✗	neither: $-1, 0, 1$
$\lvert x-y\rvert \leq 1$ on $\mathbb{Z}$	✓	✓	✗	✗	neither: $0, 1, 2$
$\varnothing$ on $A \neq \varnothing$	✗	✓	✓	✓	neither
$A \times A$	✓	✓	✓	✗ if $\lvert A\rvert \geq 2$	equivalence, one class

WORKED EXAMPLE **Lecture 6, Ex. 26 · Kahn's algorithm on $\{d \in \mathbb{Z}^+ : d \mid 30\}$**

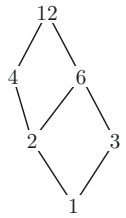
Step	Minimal elements of A_i	Choose
A_0	1	$c_0 = 1$
A_1	2, 3, 5	$c_1 = 3$
A_2	2, 5	$c_2 = 2$
A_3	5, 6	$c_3 = 6$
A_4	5	$c_4 = 5$
A_5	10, 15	$c_5 = 15$
A_6	10	$c_6 = 10$
A_7	30	$c_7 = 30$

Result: $1 \preccurlyeq^* 3 \preccurlyeq^* 2 \preccurlyeq^* 6 \preccurlyeq^* 5 \preccurlyeq^* 15 \preccurlyeq^* 10 \preccurlyeq^* 30$. Any other choice at steps 1–3 or 5 gives another valid linearization.

WORKED EXAMPLE **Hasse diagram and Kahn**

For $(\{1, 2, 3, 4, 6, 12\}, \mid)$: draw the Hasse diagram; give the minimal, maximal, smallest, largest elements; produce a linearization with Kahn's algorithm.

Covers: $1 \mid 2$, $1 \mid 3$, $2 \mid 4$, $2 \mid 6$, $3 \mid 6$, $4 \mid 12$, $6 \mid 12$ (diagram right). Minimal and smallest: **1**. Maximal and largest: **12**. Kahn: A_0 has the single minimal element 1; then 2 and 3 are minimal and we take 2; then 3 and 4, and we take 3; then 4 and 6, and we take 4; then 6; then 12. Linearization: $1 \preccurlyeq^* 2 \preccurlyeq^* 3 \preccurlyeq^* 4 \preccurlyeq^* 6 \preccurlyeq^* 12$ (also valid: $1, 3, 2, 6, 4, 12$).



Part VI · Functions

26 · Functions, images and preimages

DEFINITION **Function (L7 s7)**

A **function** f from a set X to a set Y, written $f : X \to Y$, is a relation $f \subseteq X \times Y$ satisfying

(F1) $\forall x \in X\, \exists y \in Y\, ((x, y) \in f)$; every input has an output.

(F2) $\forall x \in X\, \forall y_1, y_2 \in Y\, (((x, y_1) \in f \wedge (x, y_2) \in f) \to y_1 = y_2)$; the output is unique.

Equivalently, $\forall x \in X\, \exists! y \in Y\, ((x, y) \in f)$. On an arrow diagram: every element of X has exactly one arrow leaving it.

DEFINITION **Image, preimage (L7 s11) · Setwise image and preimage (L7 s12) · Domain, co-domain, range (L7 s16)**

Write $f(x) = y$ iff $(x, y) \in f$; f **maps** x to y ($f : x \mapsto y$); x is the **argument**, $f(x)$ the **image** of x, and x is **a preimage** of y.

For $A \subseteq X$ and $B \subseteq Y$:

$$f(A) = \{f(x) : x \in A\} \quad \text{(setwise image)}, \qquad f^{-1}(B) = \{x \in X : f(x) \in B\} \quad \text{(setwise preimage)}.$$

X is the **domain**, Y the **co-domain**, and the **range** is $f(X) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$. Always range $\subseteq$ co-domain.

TRAP **Two meanings of f^{-1} (L7 s12, s15)**
- $f^{-1}(B)$ with B a **subset** of the co-domain is the setwise preimage. It exists for every function and is a set.
- $f^{-1}(y)$ with y an **element** of the co-domain is the inverse function (§28). It exists only when f is a bijection.
- For the preimage of a single element write $f^{-1}(\{b\})$, never $f^{-1}(b)$, unless f is known to be bijective.

WORKED EXAMPLE **L7 Ex. 1, 4, 6 · T6 D1, Q1, Q2**
- **Not a function.** "$f(x)$ is the real y with $x^2+y^2=1$" on $\mathbb{R}$: $x=2$ has no y (F1 fails); $x=0$ has $y=\pm1$ (F2 fails).
- **T6 D1.** $f(z)=1$ if $2 \mid z$, 2 if $3 \mid z$: not a function ($f(6)$ has two values, $f(1)$ has none). $f:\mathbb{R}\to\mathbb{Z}$, $f(x)=2x$: not a function ($f(0.3)=0.6\notin\mathbb{Z}$). $f:\mathbb{Z}\to\mathbb{R}$, $f(x)=2x$: a function.
- **T6 Q1** on $\mathbb{N}$. $xR_1y \leftrightarrow x^2=y^2$: a function (F1: take $y=x$; F2: $y_1^2=y_2^2$ with $y_1,y_2\geq 0$ gives $y_1=y_2$). $xR_2y \leftrightarrow y \mid x$: not a function ($6R_22$ and $6R_23$). $xR_3y \leftrightarrow y=x+1$: a function.
- **Ex. 4.** $g:\mathbb{Z}\to\mathbb{Z}$, $g(x)=x^2$: $g(\{-1,0,1\})=\{0,1\}$ and $g^{-1}(\{0,1,2\})=\{-1,0,1\}$.
- **Ex. 6.** $f:\mathbb{Z}\to\mathbb{Z}$, $f(x)=2x+1$: domain $\mathbb{Z}$, co-domain $\mathbb{Z}$, range = the odd integers.
- **T6 Q2.** $\text{len}:\{s,u\}^*\to\mathbb{Z}_{\geq 0}$: $\text{len}(\{\varepsilon,ss,uu,ssss\})=\{0,2,4\}$; $\text{len}^{-1}(\{3\})$ is the set of all 8 strings of length 3; the inverse function len^{-1} does not exist, since $\text{len}(s)=\text{len}(u)$.

PROOF **T6 Q9 · $X\subseteq f^{-1}(f(X))$ and $f(f^{-1}(Y))\subseteq Y$; the reverse inclusions can fail**
Let $f:A\to B$, $X\subseteq A$, $Y\subseteq B$.
1. ($X\subseteq f^{-1}(f(X))$) Let $x\in X$. Then $f(x)\in f(X)$ *(definition of $f(X)$)*, so $x\in f^{-1}(f(X))$ *(definition of setwise preimage)*.
2. ($f(f^{-1}(Y))\subseteq Y$) Let $y\in f(f^{-1}(Y))$. Then $y=f(x)$ for some $x\in f^{-1}(Y)$ *(definition of setwise image)*, and $f(x)\in Y$ *(definition of setwise preimage)*. So $y\in Y$. ■

Counterexamples for the reverse. $f:\{-1,1\}\to\{0\}$, $X=\{1\}$: $f^{-1}(f(X))=\{-1,1\}\not\subseteq X$.
$f:\{0\}\to\{-1,1\}$ with $f(0)=1$, $Y=\{-1\}$: $f(f^{-1}(Y))=\varnothing$, so $Y\not\subseteq f(f^{-1}(Y))$.
Equality holds in the first when f is injective and in the second when f is surjective.

27 · Sequences, strings and function equality

DEFINITION **Sequence (L7 s18) · Fibonacci sequence (L7 s19) · String (L7 s20)**
A **sequence** $a_0,a_1,a_2,\ldots$ is represented by a function a with domain $\mathbb{Z}_{\geq 0}$ and $a(n)=a_n$; any function with domain $\mathbb{Z}_{\geq m}$ represents a sequence.
Fibonacci: $F_0=0$, $F_1=1$, $F_{n+2}=F_{n+1}+F_n$: $\quad 0,1,1,2,3,5,8,13,21,34,55,\ldots$
A **string** (word) over a set A is an expression $a_0a_1\cdots a_{l-1}$ with $l\in\mathbb{Z}_{\geq 0}$ and each $a_i\in A$; l is its **length**; the **empty string** ε has length 0; A^* is the set of all strings over A. Two sequences (strings of the same length) are equal iff they agree at every index.

THEOREM **Theorem 7.1.1 · Function equality (L7 s22)**
$f:A\to B$ and $g:C\to D$ are equal iff (i) $A=C$ and $B=D$, and (ii) $f(x)=g(x)$ for all $x\in A$.

WORKED EXAMPLE **L7 Ex. 11–13 · T6 D3**
- On $X=\{0,1,2\}$, $f(x)=(x^2+x+1) \bmod 3$ and $g(x)=(x+2)^2 \bmod 3$ both give $1,0,1$, so $f=g$.
- $f,g:\{0,2\}\to\mathbb{Z}$ with $f(x)=2x$, $g(x)=x^2$: equal (same domain, co-domain, values $0,4$).
- $f:\mathbb{Z}\to\mathbb{Z}$ and $g:\mathbb{Z}\to\mathbb{Q}$ with $f(x)=x^3=g(x)$: **not** equal, since the co-domains differ.
- **T6 D3.** $f,g:\mathbb{Q}\to\mathbb{Q}$, $f(x)=\lfloor x\rfloor+1$, $g(x)=\lceil x\rceil$: both have range $\mathbb{Z}$, but $f(0)=1\neq 0=g(0)$, so $f\neq g$ (they agree exactly off the integers).

28 · Injections, surjections, bijections, inverses

DEFINITION **Injection (L7 s25) · Surjection (L7 s27) · Bijection (L7 s29)**
Let $f:X\to Y$.
- f is **injective** (one-to-one) iff $\forall x_1,x_2\in X\,(f(x_1)=f(x_2)\Rightarrow x_1=x_2)$. Contrapositive: $x_1\neq x_2 \Rightarrow f(x_1)\neq f(x_2)$.
- f is **surjective** (onto) iff $\forall y\in Y\,\exists x\in X\,(y=f(x))$: range = co-domain.
- f is **bijective** (a one-to-one correspondence) iff it is injective and surjective: $\forall y\in Y\,\exists!x\in X\,(y=f(x))$.

Arrows **into** each element of the co-domain: at most one (injective), at least one (surjective), exactly one (bijective).

METHOD **Negations (to disprove)**
- **Not injective:** $\exists x_1, x_2 \in X\,(f(x_1) = f(x_2) \wedge x_1 \neq x_2)$. Name two inputs.
- **Not surjective:** $\exists y \in Y\,\forall x \in X\,(y \neq f(x))$. Name one y and show no x reaches it.

METHOD **Proof templates**
- **Injective:** "Let $x_1, x_2 \in X$ with $f(x_1) = f(x_2)$" ... "so $x_1 = x_2$."
- **Surjective:** "Take any $y \in Y$. Let $x = ...$" (solve $y = f(x)$ on scratch paper). Check $x \in X$ and $f(x) = y$.
- **Bijective:** both, or exhibit an inverse (Theorem 7.2.3).

PROOF **L7 Ex. 14, 16 · $f : \mathbb{Q} \to \mathbb{Q}$, $f(x) = 3x + 1$ is a bijection** **Ex. 15, 17 · $g : \mathbb{Z} \to \mathbb{Z}$, $g(x) = x^2$ is neither**

1. (Injective) Let $x_1, x_2 \in \mathbb{Q}$ with $f(x_1) = f(x_2)$. Then $3x_1 + 1 = 3x_2 + 1$, so $x_1 = x_2$ *(basic algebra)*.
2. (Surjective) Take any $y \in \mathbb{Q}$. Let $x = (y-1)/3$. Then $x \in \mathbb{Q}$ *(closure of $\mathbb{Q}$)* and $f(x) = 3 \cdot \frac{y-1}{3} + 1 = y$. ■

g is not injective: $g(1) = 1 = g(-1)$ but $1 \neq -1$. g is not surjective: $g(x) = x^2 \geq 0 > -1$ for all $x \in \mathbb{Z}$, so -1 has no preimage.

The domain matters: $x \mapsto 3x + 1$ on $\mathbb{Z}$ is injective but not surjective ($y = 0$ would need $x = -1/3 \notin \mathbb{Z}$).

WORKED EXAMPLE **T6 Q4 · Classify four functions**

(a) $f : \mathbb{Q} \to \mathbb{Q}$, $x \mapsto 12x + 31$: bijective, with $f^{-1}(y) = (y - 31)/12$, because $y = 12x + 31 \Leftrightarrow x = (y - 31)/12$ for all $x, y \in \mathbb{Q}$.

(b) $g : \text{Bool}^2 \to \text{Bool}$, $(p, q) \mapsto p \wedge \sim q$: not injective ($g(F, T) = F = g(F, F)$); surjective ($g(T, F) = T$).

(c) $h : \text{Bool}^2 \to \text{Bool}^2$, $(p, q) \mapsto (p \wedge q, p \vee q)$: not injective ($h(T, F) = (F, T) = h(F, T)$); not surjective: if $p \wedge q = T$ then $p \vee q = T$, so (T, F) is not in the range.

(d) $k : \mathbb{Z} \to \mathbb{Z}$, $k(x) = x$ if x is even, $2x - 1$ if x is odd.

1. k preserves parity: x even gives $k(x) = x$ even; x odd gives $k(x) = 2(x-1) + 1$ odd. So x is even iff $k(x)$ is even.
2. (Injective) Let $k(x_1) = k(x_2)$. If this value is even, both x_i are even and $x_1 = k(x_1) = k(x_2) = x_2$. If it is odd, both are odd and $2x_1 - 1 = 2x_2 - 1$, so $x_1 = x_2$.
3. (Not surjective) Suppose $k(x) = 3$. Then x is odd *(line 1)*, so $2x - 1 = 3$ and $x = 2$, which is even: contradiction. So 3 has no preimage. ■

DEFINITION **Inverse function (L7 s31)**

Let $f : X \to Y$. A function $g : Y \to X$ is an **inverse** of f iff

$$\forall x \in X\,\forall y \in Y\,(y = f(x) \Leftrightarrow x = g(y)).$$

The inverse, when it exists, is unique and is written f^{-1}. Example: $f(x) = 3x + 1$ on $\mathbb{Q}$ has $f^{-1}(y) = (y - 1)/3$.

PROOF **Proposition · Uniqueness of inverses (L7 s34)** **Theorem 7.2.3 · f is bijective iff f has an inverse; then f^{-1} is a bijection (L7 s35–36)**

Uniqueness. If $g_1, g_2 : Y \to X$ are inverses of f, then for all $x \in X$, $y \in Y$: $x = g_1(y) \Leftrightarrow y = f(x) \Leftrightarrow x = g_2(y)$. So $g_1 = g_2$ *(Theorem 7.1.1)*.

Theorem 7.2.3.

1. ("if") Suppose f has an inverse $g : Y \to X$.
   1.1. (Injective) Let $f(x_1) = f(x_2) = y$. Then $x_1 = g(y)$ and $x_2 = g(y)$ *(g is an inverse)*, so $x_1 = x_2$.
   1.2. (Surjective) Let $y \in Y$ and $x = g(y)$. Then $y = f(x)$ *(g is an inverse)*.
2. ("only if") Suppose f is bijective. Then $\forall y \in Y\,\exists! x \in X\,(y = f(x))$ *(definition of bijection)*. Define $g(y)$ to be that unique x. Then g is well defined and $y = f(x) \Leftrightarrow x = g(y)$, so g is an inverse of f.
3. (f^{-1} is a bijection) f is an inverse of f^{-1}, since the defining condition is symmetric in f and g. So f^{-1} is bijective by line 1. ■

THEOREM **T6 Q5 (AY22/23 S2 exam) · Left and right inverses**

$g : B \to A$ is a **left inverse** of $f : A \to B$ iff $g(f(a)) = a$ for all $a \in A$; $h : B \to A$ is a **right inverse** iff $f(h(b)) = b$ for all $b \in B$.

***f* is injective iff it has a left inverse; *f* is surjective iff it has a right inverse** (for $A \neq \varnothing$). Neither kind of inverse implies the other. In T6 Q4: left inverses for f, k; right inverses for f, g.

29 · Composition of functions

DEFINITION Composition (L7 s39) · Identity function (L7 s41)
For $f : X \to Y$ and $g : Y \to Z$, the **composition** $g \circ f : X \to Z$ is defined by $(g \circ f)(x) = g(f(x))$ for all $x \in X$. **f is applied first.**
The **identity function** on X is $\mathrm{id}_X : X \to X$ with $\mathrm{id}_X(x) = x$.

THEOREM Theorems on composition (L7 s44–52)
- **7.3.1.** For $f : X \to Y$: $f \circ \mathrm{id}_X = f$ and $\mathrm{id}_Y \circ f = f$.
- **7.3.2.** If $f : X \to Y$ is a bijection: $f^{-1} \circ f = \mathrm{id}_X$ and $f \circ f^{-1} = \mathrm{id}_Y$.
- **Associativity.** For $f : A \to B$, $g : B \to C$, $h : C \to D$: $(h \circ g) \circ f = h \circ (g \circ f)$.
- **Not commutative.** $f(x) = 3x$, $g(x) = x + 1$ on $\mathbb{Z}$: $(g \circ f)(0) = 1 \neq 3 = (f \circ g)(0)$.
- **7.3.3.** f, g injective $\Rightarrow g \circ f$ injective. **7.3.4.** f, g surjective $\Rightarrow g \circ f$ surjective. Hence bijections compose to a bijection.

PROOF Associativity (L7 s47) · Theorem 7.3.3 (L7 s50) · Theorem 7.3.4 (L7 s52)
Associativity. Both sides have domain A and co-domain D, and for every $x \in A$: $((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x))) = h((g \circ f)(x)) = (h \circ (g \circ f))(x)$. Equal by Theorem 7.1.1.
7.3.3.
1. Let $f : X \to Y$ and $g : Y \to Z$ be injective, and let $x_1, x_2 \in X$ with $(g \circ f)(x_1) = (g \circ f)(x_2)$.
2. Then $g(f(x_1)) = g(f(x_2))$ *(definition of composition)*.
3. So $f(x_1) = f(x_2)$ *(g is injective)*, and then $x_1 = x_2$ *(f is injective)*. ■

7.3.4.
1. Let $f : X \to Y$ and $g : Y \to Z$ be surjective, and let $z \in Z$.
2. There is $y \in Y$ with $g(y) = z$ *(g is surjective)*, and $x \in X$ with $f(x) = y$ *(f is surjective)*.
3. Then $(g \circ f)(x) = g(f(x)) = g(y) = z$. ■

TRAP T6 D4 · A wrong surjectivity proof
"$\forall y \exists x\, f(x) = y$, and $\forall z \exists y\, g(y) = z$, so $(g \circ f)(x) = g(y) = z$." The x, y, z in the last line were never fixed: the proof must **start** with an arbitrary $z \in Z$, **then** obtain y from g, **then** obtain x from f for that particular y.

PROOF T6 Q3 · For bijections $f : A \to B$, $g : B \to C$: $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$
1. f^{-1}, g^{-1} are bijections *(Theorem 7.2.3)* $g \circ f$ and $f^{-1} \circ g^{-1}$ are bijections *(Theorems 7.3.3, 7.3.4)* so $(g \circ f)^{-1}$ exists.
2. Both $(g \circ f)^{-1}$ and $f^{-1} \circ g^{-1}$ have domain C and co-domain A.
3. Let $z \in C$. Let $y = g^{-1}(z)$, so $z = g(y)$; let $x = f^{-1}(y)$, so $y = f(x)$ *(definition of inverse)*.
4. Then $(g \circ f)(x) = g(f(x)) = g(y) = z$, so $(g \circ f)^{-1}(z) = x$ *(definition of inverse)*.
5. Also $(f^{-1} \circ g^{-1})(z) = f^{-1}(g^{-1}(z)) = f^{-1}(y) = x$.
6. The two functions agree at every $z \in C$, so they are equal *(Theorem 7.1.1)*. ■

PROOF T6 Q6 · If $g \circ f$ is injective, then f is injective · If $g \circ f$ is surjective, then g is surjective
1. Let $f : B \to C$ and let g have domain C with $g \circ f$ injective. Let $x_1, x_2 \in B$ with $f(x_1) = f(x_2)$.
2. Then $(g \circ f)(x_1) = g(f(x_1)) = g(f(x_2)) = (g \circ f)(x_2)$, so $x_1 = x_2$ *($g \circ f$ is injective)*. ■

For the second: given z, surjectivity of $g \circ f$ gives x with $g(f(x)) = z$, and $y = f(x)$ is a preimage of z under g.
The other halves fail: $X = Z = \{0\}$, $Y = \{0, 1\}$, $f(0) = 0$, $g(0) = g(1) = 0$. Then $g \circ f = \mathrm{id}_{\{0\}}$ is bijective, yet g is not injective and f is not surjective.

PAST PAPER Order of a bijection · T6 Q7, Q8 · T8 Q10(c) · AY25 final Q11, Q12, Q26(b)
The **order** of a bijection $f : A \to A$ is the least $n \in \mathbb{Z}^+$ with $f \circ f \circ \cdots \circ f = \mathrm{id}_A$ (n copies). The **order of an input** x is the least $m \in \mathbb{Z}^+$ with $f^{(m)}(x) = x$.
T6 Q7. On $\{1, 2, 3\}$, g swaps 1, 2 and h swaps 2, 3: orders 2 and 2. $g \circ h$ sends $1 \mapsto 2 \mapsto 3 \mapsto 1$: order 3; likewise $h \circ g$.
T6 Q8. $g(x) = 3x \bmod 5$: $g^{(3)}(21) = g^{(2)}(3) = g(4) = 2$. Order of 3: $3 \mapsto 4 \mapsto 2 \mapsto 1 \mapsto 3$, so 4. Orders of $0, 1, 2, 3, 4$ are $1, 4, 4, 4, 4$: classes $\{0\}$ and $\{1, 2, 3, 4\}$.
T8 Q10(c). $f = \{(1, 2), (2, 4), (3, 1), (4, 3)\}$, $g = \{(1, 4), (2, 1), (3, 2), (4, 3)\}$: orders 4 and 4; $f^{-1} \circ g$ sends $1 \mapsto 2 \mapsto 3 \mapsto 1$, $4 \mapsto 4$: order 3.
AY25 Q11, Q12. $f(p, q) = (p, p \oplus q)$ and $g(p, q) = (p \oplus q, q)$ on "Bool"^{2}. $f \circ f = \mathrm{id}$, so $f = f^{-1}$; but $f^{-1} \circ f = \mathrm{id} \neq f$. $(g \circ f)(p, q) = (q, p \oplus q)$ has order 3.
AY25 Q26(b). $z : \mathbb{Z}_5 \to \mathbb{Z}_5$, $z(x) = 2x + 1$: $z^{-1}([3]) = [1]$; $z^{(2)}(x) = 4x + 3$; $[3] \mapsto [2] \mapsto [0] \mapsto [1] \mapsto [3]$, so the order of $[3]$ is 4.

30 · Addition and multiplication on $\mathbb{Z}_n$

DEFINITION $\mathbb{Z}_n$ (L7 s54–55) · Addition and multiplication on $\mathbb{Z}_n$ (L7 s56)
$\mathbb{Z}_n$ denotes the quotient $\mathbb{Z}/\underset{n}{\sim}$, where $\underset{n}{\sim}$ is congruence mod n: $\mathbb{Z}_n = \{[0],[1],...,[n-1]\}$ with $[r] = \{nk + r : k \in \mathbb{Z}\}$.
For $[x],[y] \in \mathbb{Z}_n$: $[x]+[y]=[x+y]$ and $[x]\cdot[y]=[x\cdot y]$. In $\mathbb{Z}_3$: $[0]+[1]=[1]$, $[2]\cdot[2]=[4]=[1]$.

TRAP $\mathbb{Z}_n$ has two meanings in the slides
Lecture 7 uses $\mathbb{Z}_n = \mathbb{Z}/\underset{n}{\sim}$ (a set of n classes). Lecture 9, slide 9 writes $\mathbb{Z}_n = \{1,2,...,n\}$ when defining finite sets. Read the definition in the question.

PROOF Propositions · + and · on $\mathbb{Z}_n$ are well defined (L7 s57–58)
Claim: for all $[x_1],[y_1],[x_2],[y_2] \in \mathbb{Z}_n$, if $[x_1]=[x_2]$ and $[y_1]=[y_2]$ then $[x_1]+[y_1]=[x_2]+[y_2]$ and $[x_1]\cdot[y_1] = [x_2]\cdot[y_2]$.
1. Let $[x_1]=[x_2]$ and $[y_1]=[y_2]$. Then $x_1 \equiv x_2$ and $y_1 \equiv y_2 \pmod n$ *(Lemma Rel.1)*.
2. So $x_1 - x_2 = nk$ and $y_1 - y_2 = nl$ for some $k,l \in \mathbb{Z}$ *(definition of congruence)*.
3. $(x_1+y_1)-(x_2+y_2) = n(k+l)$, so $x_1+y_1 \equiv x_2+y_2 \pmod n$, hence $[x_1+y_1]=[x_2+y_2]$ *(Lemma Rel.1)*.
4. $x_1y_1 - x_2y_2 = (nk+x_2)(nl+y_2) - x_2y_2 = n(nkl+ky_2+lx_2)$ with $nkl+ky_2+lx_2 \in \mathbb{Z}$ *(closure)*, hence $[x_1y_1]=[x_2y_2]$. ■

The representative-level version of this proof is in §23.

METHOD Is a rule on classes well defined?
A rule $F([x]) = ...$ defines a function on $A/\sim$ only if the output does not depend on the representative: **show $x \sim x' \Rightarrow$ same output**. To disprove, name two representatives of one class with different outputs.
- $[x]_3 \mapsto [x]_2$ from $\mathbb{Z}_3$ to $\mathbb{Z}_2$: **not** well defined ($[0]_3 = [3]_3$ but $[0]_2 \neq [3]_2$). From $\mathbb{Z}_4$ to $\mathbb{Z}_2$ it is well defined, as $4 \mid (x-x') \Rightarrow 2 \mid (x-x')$.
- **T6 Q10.** On $\mathbb{Q}/\sim$ with $x \sim y \leftrightarrow x-y \in \mathbb{Z}$: + is well defined ($(x_1+y_1)-(x_2+y_2) = k+l \in \mathbb{Z}$); · is **not**: $[1/2]=[-1/2]$, but $[1/2]\cdot[1/2] = [1/4] \neq [-1/4] = [1/2]\cdot[-1/2]$, as $1/4-(-1/4) = 1/2 \notin \mathbb{Z}$.
- **T8 Q10 (AY22/23 S2).** If $g : X \to Y$ satisfies $g(a)=g(b) \Leftrightarrow a \sim b$, then $f([x]) = g(x)$ is well defined ($[x]=[x'] \Rightarrow x \sim x' \Rightarrow g(x)=g(x')$) **and** injective ($f([x])=f([x']) \Rightarrow g(x)=g(x') \Rightarrow x \sim x' \Rightarrow [x]=[x']$).

Part VII · Induction and Recursion

31 · Sequences, sums and products

DEFINITION Sequence, term, explicit formula (L8 s5–6) · Summation (L8 s7) · Product (L8 s11)
A **sequence** is an ordered list $a_m, a_{m+1}, ..., a_n$ (or infinite) of **terms**; k in a_k is the **index**. An **explicit formula** gives a_k in terms of k. For integers $m \le n$:

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + \cdots + a_n, \qquad \prod_{k=m}^{n} a_k = a_m \cdot a_{m+1} \cdot \cdots \cdot a_n.$$

k is the **index**, m the **lower limit**, n the **upper limit**. **Recursive form:** $\sum_{k=m}^{m} a_k = a_m$ and $\sum_{k=m}^{n} a_k = \left(\sum_{k=m}^{n-1} a_k\right) + a_n$ for $n > m$; likewise for $\prod$. An **empty sum** is 0; an **empty product** is 1.

THEOREM Theorem 5.1.1 (L8 s14) · Standard sequences (L8 s18–20)
For real sequences a_k, b_k, real c and $n \ge m$:

$$\sum_{k=m}^{n} a_k + \sum_{k=m}^{n} b_k = \sum_{k=m}^{n} (a_k + b_k), \quad c \cdot \sum_{k=m}^{n} a_k = \sum_{k=m}^{n} c \cdot a_k, \quad \left(\prod_{k=m}^{n} a_k\right) \cdot \left(\prod_{k=m}^{n} b_k\right) = \prod_{k=m}^{n} (a_k \cdot b_k).$$

Arithmetic: $a_k = a_{k-1} + d$, so $a_n = a_0 + dn$ and $\sum_{k=0}^{n-1} a_k = \frac{n}{2}(2a_0 + (n-1)d)$.
Geometric: $a_k = ra_{k-1}$, so $a_n = a_0 r^n$ and $\sum_{k=0}^{n-1} a_k = a_0 \frac{1-r^n}{1-r}$ for $r \neq 1$.
Squares $1,4,9,16,...$; triangle numbers $1,3,6,10,...$ $(n(n+1)/2)$; lazy caterer $1,2,4,7,11,16,...$ $(n(n+1)/2+1)$.

WORKED EXAMPLE L8 Ex. 4, 5, 7 · Telescoping, a product, change of variable
- **Telescoping.** $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, so $\sum_{k=1}^{n} \frac{1}{k(k+1)} = \left(1-\frac{1}{2}\right) + \left(\frac{1}{2}-\frac{1}{3}\right) + \cdots + \left(\frac{1}{n}-\frac{1}{n+1}\right) = 1 - \frac{1}{n+1}$.
- $\prod_{k=1}^{5}(k+2) = 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 = 2520$.

- **Change of variable.** The index is a dummy: $\sum_{k=1}^{3} k^2 = \sum_{i=1}^{3} i^2 = \sum_{k=3}^{5} (k-2)^2$. Put $j = k-1$: $\sum_{k=1}^{n+1} \frac{k}{n+k} = \sum_{j=0}^{n} \frac{j+1}{n+j+1}$. Change the limits and every occurrence of the index together.

32 · Mathematical induction

DEFINITION **Principle of Mathematical Induction, PMI (L8 s24) · Closed form (L8 s27)**

Let $P(n)$ be a property defined for integers n and let a be a fixed integer. Suppose

1. $P(a)$ is true (**basis step**)
2. for all integers $k \geq a$, if $P(k)$ is true then $P(k+1)$ is true. (**inductive step**; $P(k)$ is the **inductive hypothesis**)

Then $P(n)$ is true for all integers $n \geq a$. PMI is taken as an axiom.

A sum is in **closed form** when it is written without an ellipsis or a summation symbol.

METHOD **The format used in class (L8 s26) · Mistakes to avoid (T7 Q1, D2, D3)**

1. Let $P(n) \equiv (...)$ for $n \geq a$. (set up the predicate)
2. Basis step: verify $P(a)$.
3. Assume $P(k)$ is true for an arbitrary $k \geq a$. (inductive hypothesis)
4. Inductive step: derive $P(k+1)$, marking where $P(k)$ is used.
5. Therefore $P(n)$ is true for all $n \geq a$ by MI.

- $P(k)$ is a **statement**, not a number: never write "$(k+1) + P(k)$".
- Do not start from $P(k+1)$ and reduce it to a true identity; start from one side of $P(k+1)$ and reach the other.
- Do not assume "$P(k)$ for all k"; that is the conclusion.
- Check the **basis value matches the claim** (T7 D2: the claim is for $n \in \mathbb{N}$, so the basis must be $n = 0$, and the step must be shown, not asserted).
- The inductive step must cover an **arbitrary** object of size $k+1$ (T7 D3: start from any group of $k+1$ people and remove one, rather than building up from a group of k).

PROOF **Theorem 5.2.2 (5th ed. 5.2.1) · $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ for all $n \geq 1$ (L8 s26)**

1. Let $P(n) \equiv (1 + 2 + \cdots + n = n(n+1)/2)$ for $n \in \mathbb{Z}^+$.
2. Basis step: $1 = 1(1+1)/2$, so $P(1)$ is true.
3. Assume $P(k)$ is true for an arbitrary $k \geq 1$: $1 + 2 + \cdots + k = k(k+1)/2$.
4. Inductive step: $1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1)$ *(inductive hypothesis)* $= \frac{(k+1)(k+2)}{2} = \frac{(k+1)((k+1)+1)}{2}$. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for all $n \in \mathbb{Z}^+$. ■

PROOF **Theorem 5.2.3 (5th ed. 5.2.2) · $\sum_{i=0}^{n} r^i = \frac{r^{n+1}-1}{r-1}$ for real $r \neq 1$ and $n \geq 0$ (L8 s28)**

1. Let $P(n) \equiv \left(\sum_{i=0}^{n} r^i = (r^{n+1} - 1)/(r-1)\right)$ for $n \geq 0$.
2. Basis step: $r^0 = 1 = (r^1 - 1)/(r-1)$, so $P(0)$ is true.
3. Assume $P(k)$ for an arbitrary $k \geq 0$.
4. Inductive step: $\sum_{i=0}^{k+1} r^i = \sum_{i=0}^{k} r^i + r^{k+1} = \frac{r^{k+1}-1}{r-1} + r^{k+1}$ *(inductive hypothesis)* $= \frac{r^{k+1}-1+r^{k+1}(r-1)}{r-1} = \frac{r^{(k+1)+1}-1}{r-1}$.
5. Therefore $P(n)$ is true for all $n \geq 0$. ■

PROOF **Proposition 5.3.1 (5th ed. 5.3.2) · $3 \mid (2^{2n} - 1)$ for all $n \geq 0$ (L8 s29)**

1. Let $P(n) \equiv (3 \mid (2^{2n} - 1))$ for $n \geq 0$.
2. Basis step: $2^0 - 1 = 0 = 3 \cdot 0$, so $P(0)$ is true.
3. Assume $P(k)$ for an arbitrary $k \geq 0$: $2^{2k} - 1 = 3r$ for some $r \in \mathbb{Z}$ *(definition of divisibility)*.
4. Inductive step: $2^{2(k+1)} - 1 = 2^{2k} \cdot 4 - 1 = 2^{2k} \cdot 3 + \left(2^{2k} - 1\right) = 2^{2k} \cdot 3 + 3r = 3\left(2^{2k} + r\right)$, and $2^{2k} + r \in \mathbb{Z}$ *(closure)*. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for all $n \geq 0$. ■

PROOF **Proposition 5.3.2 (5th ed. 5.3.3) · $2n + 1 < 2^n$ for all $n \geq 3$ (L8 s30)**

1. Let $P(n) \equiv (2n + 1 < 2^n)$ for $n \geq 3$.
2. Basis step: $2 \cdot 3 + 1 = 7 < 8 = 2^3$, so $P(3)$ is true.
3. Assume $P(k)$ for an arbitrary $k \geq 3$.
4. Inductive step: $2(k+1) + 1 = (2k+1) + 2 < 2^k + 2$ *(inductive hypothesis)* $< 2^k + 2^k$ *($2 < 2^k$ for $k \geq 2$)* $= 2^{k+1}$.
5. Therefore $P(n)$ is true for all $n \geq 3$. ■

The basis cannot be lowered: $n = 1$ gives $3 < 2$ and $n = 2$ gives $5 < 4$, both false.

TRAP **L8 Ex. 12 · "All cows have the same colour"**
The step "cows 1 to k share a colour, cows 2 to $k+1$ share a colour, and cows 2 to k are common to both" fails for $k=1$: the overlap $\{2,\dots,k\}$ is **empty**, so nothing links cow 1 to cow 2. $P(1)\Rightarrow P(2)$ is false, and the chain breaks at its first link. The inductive step must work for **every** $k\geq a$.

WORKED EXAMPLE **L8 Ex. 13 · Tiling by L-trominoes**
$P(n)$: every $2^n\times 2^n$ board with one square removed can be tiled by L-trominoes. $P(1)$: the board is one L-tromino. Step: split a $2^{k+1}\times 2^{k+1}$ board into four $2^k\times 2^k$ quadrants; place one tromino at the centre covering one square of each quadrant not containing the removed square. Each quadrant now has one square missing and is tiled by $P(k)$.

PROOF **T7 Q2 · $1^2+2^2+\cdots+n^2=n(n+1)\frac{2n+1}{6}$** **T7 Q3 · Bernoulli: $1+nx\leq(1+x)^n$ for real $x\geq -1$**
Q2. $P(1)$: $1=(1\cdot 2\cdot 3)/6$. Step: $1^2+\cdots+k^2+(k+1)^2=\frac{k(k+1)(2k+1)}{6}+(k+1)^2$ *(IH)* $=\frac{(k+1)(2k^2+7k+6)}{6}$ $=\frac{(k+1)(k+2)(2k+3)}{6}$, which is $P(k+1)$.
Q3. $P(1)$: $1+x=(1+x)^1$. Step: $(1+x)^{k+1}=(1+x)^k(1+x)\geq(1+kx)(1+x)$ *(IH, and $1+x\geq 0$)* $=1+(k+1)x+kx^2\geq 1+(k+1)x$ *($kx^2\geq 0$).* ■

PROOF **T7 Q4 · For odd a: $2^{n+2}\mid(a^{2^n}-1)$ for all $n\in\mathbb{Z}^+$**
1. Let $P(n)\equiv(2^{n+2}\mid(a^{2^n}-1))$.
2. Basis step: $a=2p+1$ for some $p\in\mathbb{Z}$, so $a^2-1=(a-1)(a+1)=4p(p+1)=8m$ for some $m\in\mathbb{Z}$ *(a product of two consecutive integers is even)*. So $P(1)$ is true.
3. Assume $P(k)$: $a^{2^k}-1=2^{k+2}m$ for some $m\in\mathbb{Z}$.
4. Inductive step: $a^{2^{k+1}}-1=\left(a^{2^k}\right)^2-1=\left(a^{2^k}-1\right)\left(a^{2^k}+1\right)=2^{k+2}m\left(2^{k+2}m+2\right)=2^{k+3}m\left(2^{k+1}m+1\right)$. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for all $n\in\mathbb{Z}^+$ by MI. ■

33 · Strong induction and well-ordering

DEFINITION **1PI and 2PI (L8 s37)**
Let $P(n)$ be a property of integers $n\geq a$.
- **Weak (regular) induction, 1PI:** if $P(a)$ holds, and $P(k)\Rightarrow P(k+1)$ for every $k\geq a$, then $P(n)$ holds for all $n\geq a$.
- **Strong induction, 2PI:** if $P(a)$ holds, and $(P(a)\wedge P(a+1)\wedge\cdots\wedge P(k))\Rightarrow P(k+1)$ for every $k\geq a$, then $P(n)$ holds for all $n\geq a$.
- **2PI, variant with several basis cases:** if $P(a),P(a+1),\dots,P(b)$ hold, and $P(k)\Rightarrow P(k+b-a+1)$ for every $k\geq a$, then $P(n)$ holds for all $n\geq a$.

1PI, 2PI and the well-ordering principle are equivalent in power.

METHOD **Choosing the form**
Use 2PI when $P(k+1)$ depends on an earlier case that is not $P(k)$: a factor of $k+1$, the case $k-3$, or two or three previous terms of a recurrence. **Prove as many basis cases as the step reaches back**: a step using $P(k-3)$ needs four consecutive basis cases; a recurrence $a_{n+3}=a_{n+2}+a_{n+1}+a_n$ needs three.

PROOF **L8 Ex. 14 · Every integer greater than 1 is divisible by a prime (2PI, L8 s39)**
1. Let $P(n)\equiv(n$ is divisible by a prime) for $n>1$.
2. Basis step: $2\mid 2$ and 2 is prime, so $P(2)$ is true.
3. Inductive step: let $k\geq 2$ and assume $P(i)$ for all integers i with $2\leq i\leq k$.
   3.1. Case 1 ($k+1$ is prime): $k+1$ is divisible by the prime $k+1$.
   3.2. Case 2 ($k+1$ is not prime): $k+1=ab$ with $1<a<k+1$ and $1<b<k+1$. Then $2\leq a\leq k$, so a prime p divides a *(inductive hypothesis)* and $a\mid(k+1)$. So $p\mid(k+1)$ *(Theorem 4.3.3)*.
4. Therefore every integer greater than 1 is divisible by a prime. ■

The well-ordering proof of the same result is in §12.

PROOF **L8 Ex. 15, 16 · Every amount $n\geq 12$ is $4a+5b$ with $a,b\in\mathbb{N}$ (1PI and 2PI, L8 s40–41)** **T7 Q5 · $n\geq 8$ is $3x+5y$**
By 1PI. $P(12)$: $12=4\cdot 3$. Step: suppose $k=4a+5b$. Case 1 ($a\geq 1$): replace one 4 by a 5: $k+1$ $=4(a-1)+5(b+1)$. Case 2 ($a=0$): then $k=5b\geq 12$ forces $b\geq 3$; replace three 5s by four 4s: $k+1$ $=4\cdot 4+5(b-3)$.
By 2PI.

1. Let $P(n) \equiv (n = 4a + 5b \text{ for some } a, b \in \mathbb{N})$ for $n \geq 12$.
2. Basis step: $12 = 4 \cdot 3 + 5 \cdot 0$; $13 = 4 \cdot 2 + 5 \cdot 1$; $14 = 4 \cdot 1 + 5 \cdot 2$; $15 = 4 \cdot 0 + 5 \cdot 3$.
3. Assume $P(i)$ for $12 \leq i \leq k$, for an arbitrary $k \geq 15$.
4. Inductive step: $12 \leq k - 3 \leq k$, so $k - 3 = 4a + 5b$ for some $a, b \in \mathbb{N}$ *(inductive hypothesis)* then $k + 1 = 4(a + 1) + 5b$.
5. Therefore $P(n)$ is true for all $n \geq 12$. ■

T7 Q5. Same pattern: basis $8 = 3 + 5$, $9 = 3 \cdot 3$, $10 = 5 \cdot 2$; step $k + 3 = 3(x + 1) + 5y$. By 1PI instead: if $y > 0$, $k + 1 = 3(x + 2) + 5(y - 1)$; if $y = 0$, then $x \geq 3$ and $k + 1 = 3(x - 3) + 5 \cdot 2$. (11 is not $4a + 5b$ and 7 is not $3x + 5y$, so the bounds are sharp.)

PROOF **T7 Q6 · Every positive integer is a sum of distinct powers of 2 (2PI)** **T7 Q7 · $a_n < 3^n$**

Q6. $P(1)$: $1 = 2^0$. Let $k \geq 1$ and assume $P(1), ..., P(k)$. Write $k + 1 = 2m$ or $2m + 1$ with $m \in \mathbb{Z}$; then $1 \leq m \leq k$, so $m = 2^{i_1} + \cdots + 2^{i_l}$ with $i_1 < \cdots < i_l$ *(inductive hypothesis)*. If $k + 1 = 2m$, then $k + 1 = 2^{i_1+1} + \cdots + 2^{i_l+1}$. If $k + 1 = 2m + 1$, then $k + 1 = 2^0 + 2^{i_1+1} + \cdots + 2^{i_l+1}$, and $0 < i_1 + 1$. Exponents stay distinct.

Q7. $a_0 = 0, a_1 = 2, a_2 = 7$, $a_{n+3} = a_{n+2} + a_{n+1} + a_n$. Basis: $0 < 1$, $2 < 3$, $7 < 9$. Step: if $P(k), P(k + 1), P(k + 2)$ hold, then $a_{k+3} < 3^{k+2} + 3^{k+1} + 3^k < 3 \cdot 3^{k+2} = 3^{k+3}$. ■

PAST PAPER **T7 Q8 (AY23/24 S1 exam) · T8 Q9 (AY22/23 S2 exam) · T7 Q9 · Fibonacci and Hanoi**

T7 Q8. $P(a, b) \equiv F(a + b) = F(a + 1)F(b) + F(a)F(b - 1)$. Basis: $P(0, b)$: $F(b) = 1 \cdot F(b) + 0 \cdot F(b - 1)$; $P(1, b)$: $F(b + 1) = F(b) + F(b - 1) = F(2)F(b) + F(1)F(b - 1)$. Step: $F(k + 1 + b) = F(k + b) + F(k - 1 + b)$ *(definition of F)* $= (F(k + 1)F(b) + F(k)F(b - 1)) + (F(k)F(b) + F(k - 1)F(b - 1))$ *($P(k, b)$, $P(k - 1, b)$)* $= (F(k + 1) + F(k))F(b) + (F(k) + F(k - 1))F(b - 1)$ *(distributive law, twice)* $= F(k + 2)F(b) + F(k + 1)F(b - 1)$.

T8 Q9. $\text{Even}(F_n) \Leftrightarrow \text{Even}(F_{n+3})$. Basis $P(0)$: $F_0 = 0$, $F_3 = 2$; $P(1)$: $F_1 = 1$, $F_4 = 3$. Step ($k \geq 1$): $\text{Even}(F_{k+1}) \Leftrightarrow (\text{Even}(F_k) \Leftrightarrow \text{Even}(F_{k-1}))$ *(Fact 1)* $\Leftrightarrow (\text{Even}(F_{k+3}) \Leftrightarrow \text{Even}(F_{k+2}))$ *($P(k)$, $P(k - 1)$)* $\Leftrightarrow \text{Even}(F_{k+4})$ *(Fact 1)*.

T7 Q9, Tower of Hanoi. $T(1) = 1$. To move $k + 1$ disks: move the top k to the spare peg ($2^k - 1$ moves, IH), move the largest (1 move), move the k back on top ($2^k - 1$ moves). $T(k + 1) = 2(2^k - 1) + 1 = 2^{k+1} - 1$.

DEFINITION **Well-Ordering Principle for the integers (L8 s43)**

Every non-empty subset of $\mathbb{Z}_{\geq 0}$ has a smallest element. (Epp's version allows any set of integers bounded below; the course uses the statement above.)

PROOF **Well-Ordering Principle, proved by 2PI (L8 s44) · L8 Ex. 17**

1. Suppose not: let $S \subseteq \mathbb{Z}_{\geq 0}$ be non-empty with no smallest element.
2. For $n \in \mathbb{Z}_{\geq 0}$ let $P(n) \equiv (n \notin S)$.
3. Let $k \in \mathbb{Z}_{\geq 0}$ with $P(0), ..., P(k - 1)$ true, i.e. $0, 1, ..., k - 1 \notin S$. If $k \in S$, then k is the smallest element of S, contradicting line 1. So $k \notin S$ and $P(k)$ is true. (For $k = 0$ the hypothesis is empty, so this also gives the basis.)
4. Hence $P(n)$ is true for all $n \in \mathbb{Z}_{\geq 0}$ *(2PI)*, so $S = \varnothing$, contradicting line 1. ■

Ex. 17. (a) The positive reals have no least element ($x/2 < x$); no violation, as WOP concerns sets of integers. (b) $\{n \in \mathbb{Z}_{\geq 0} : n^2 < n\} = \varnothing$; no violation, as WOP concerns non-empty sets. (c) Non-negative integers of the form $46 - 7k$: $..., 18, 11, 4$; least element 4.

34 · Recursion and structural induction

DEFINITION **Recurrence relation, initial conditions (L8 s48)**

A **recurrence relation** for $a_0, a_1, a_2, ...$ is a formula relating each term a_k to certain predecessors $a_{k-1}, a_{k-2}, ..., a_{k-i}$. The **initial conditions** give the first terms $a_0, ..., a_{i-1}$.

Examples: $F_n = F_{n-1} + F_{n-2}$ with $F_0 = 0, F_1 = 1$; $0! = 1$, $n! = n \cdot (n - 1)!$; $a^0 = 1$, $a^n = a^{n-1} \cdot a$.

DEFINITION **Recursively defined set (L8 s56, s58) · Structural induction (L8 s58)**
A recursive definition of a set S has three clauses:
- **(base clause)** certain elements, the **founders**, are in S;
- **(recursion clause)** S is closed under certain functions, the **constructors**: if f is a constructor and $x \in S$, then $f(x) \in S$;
- **(minimality clause)** membership of S can always be demonstrated by finitely many successive applications of the clauses above.

Structural induction over S. To prove $\forall x \in S\, P(x)$, it suffices to show
(basis step) $P(c)$ for every founder c; and **(induction step)** $\forall x \in S\,(P(x) \Rightarrow P(f(x)))$ for every constructor f.

WORKED EXAMPLE **L8 Ex. 21–24 · Recursively defined sets**
- $\mathbb{Z}_{\geq 0}$: founder 0; constructor $x \mapsto x+1$. $2\mathbb{Z}$: founder 0; constructors $x \mapsto x-2$ and $x \mapsto x+2$.
- **Legal parentheses P:** base $() \in P$; if $E \in P$ then $(E) \in P$; if $E, F \in P$ then $EF \in P$. Derivation of $(())()$: $()$ by the base; $(())$ by rule (a) with $E = ()$; $(())()$ by rule (b) with $E = (())$, $F = ()$.
- **Ex. 24.** $1 \in S$; $x \in S \Rightarrow 2x \in S \wedge 3x \in S$. Then $S = \{2^a 3^b : a, b \in \mathbb{N}\}$: $9, 12 \in S$ and $10, 11, 13 \notin S$ (each has a prime factor other than 2 and 3).

METHOD **Membership questions on a recursive set**
To show $x \in S$: write a derivation, one clause per line.
To show $x \notin S$: find an **invariant** P true of every founder and preserved by every constructor, with $P(x)$ false; prove $\forall y \in S\, P(y)$ by structural induction.

PROOF **T7 Q10 · $2 \in S$; $x \in S \Rightarrow 3x \in S \wedge x^2 \in S$. Which of $0, 6, 15, 16, 36$ are in S?**
$6 = 3 \cdot 2 \in S$; $36 = 6^2 \in S$; $4 = 2^2 \in S$, so $16 = 4^2 \in S$.
$0 \notin S$: every $n \in S$ satisfies $n \geq 2$.
1. Let $P(n) \equiv (n \geq 2)$ for $n \in S$.
2. (Basis step) $2 \geq 2$, so $P(2)$ is true.
3. (Induction step) Let $x \in S$ with $x \geq 2$. Then $3x \geq 6 \geq 2$ and $x^2 \geq 4 \geq 2$, so $P(3x)$ and $P(x^2)$ are true.
4. Hence $\forall n \in S\, P(n)$ by structural induction over S. ■

$15 \notin S$: every element of S is even (2 is even; if x is even so are $3x$ and x^2), by the same argument.

PAST PAPER **T7 Q11 · AY25 final Q14–Q16 · Recursive sets of sets and strings**
T7 Q11. $A = \{1,2,3,4,5\}$, $B = \{1,3,5,7,9\}$ are founders; constructors $\cap$, $\cup$, $\setminus$. $C = \{2,4,7,9\}$ $= (A \setminus B) \cup (B \setminus A) \in S$. $D = \{2,3,4,5\} \notin S$: the invariant "$1 \in X \Leftrightarrow 3 \in X$" holds for A and B and is preserved by $\cap$, $\cup$, $\setminus$ (check the four cases of $1, 3 \in X$ or not, $1, 3 \in Y$ or not), but fails for D.
AY25 Q14, Q15. $a \in S$; $x \in S \Rightarrow xa \in S$; $x \in S \Rightarrow bxb \in S$. Elements of length at most 4: a; aa; aaa, bab; $aaaa$, $baba$, $baab$: **7**. A string of length n arises from one of length $n-1$ (append a) or $n-2$ (wrap in b), in exactly one way, so $G_n = G_{n-1} + G_{n-2}$ with $G_1 = G_2 = 1$: $G_n = F_n$.
AY25 Q16. $S_0 = \{1,2,3\}$, $S_{n+1} = (S_n \cup \{x+3 : x \in S_n\}) \setminus \{2,3\}$. $S_1 = \{1,4,5,6\}$, $S_2 = \{1,4,5,6,7,8,9\}$, and so on: the union of all S_n is $\mathbb{Z}^+$ (2 and 3 come from S_0).

Part VIII · Cardinality

35 · Pigeonhole principle and finite sets

THEOREM **Pigeonhole Principle · Dual Pigeonhole Principle (L9 s6)**
Let A and B be **finite** sets.
- **PHP.** If there is an injection $f : A \to B$, then $|A| \leq |B|$. Contrapositive: if $m > n$ pigeons are put into n pigeonholes, some pigeonhole holds at least two pigeons.
- **Dual PHP.** If there is a surjection $f : A \to B$, then $|A| \geq |B|$. Contrapositive: if $m < n$ pigeons are put into n pigeonholes, some pigeonhole is empty.

PROOF **PHP and Dual PHP (L9 s7–8)**
PHP.
1. Let $A = \{a_1, a_2, ..., a_m\}$ with $m = |A|$.
2. If $a_i \neq a_j$ then $f(a_i) \neq f(a_j)$ *(f is injective)*, so $f(a_1), ..., f(a_m)$ are m different elements of B.
3. Hence $|B| \geq m = |A|$. ■

Dual PHP.
1. Let $B = \{b_1, b_2, ..., b_n\}$ with $n = |B|$. For each b_i pick $a_i \in A$ with $f(a_i) = b_i$ *(f is surjective)*.
2. If $b_i \neq b_j$ then $f(a_i) \neq f(a_j)$, so $a_i \neq a_j$ *(f is a function)*.
3. So $a_1, ..., a_n$ are n different elements of A, and $|A| \geq n = |B|$. ■

DEFINITION **Finite, infinite (L9 s9) · Cardinality of a finite set (L9 s10)**
Let $\mathbb{Z}_n = \{1, 2, ..., n\}$ (in this lecture only). A set S is **finite** iff $S = \varnothing$ or there is a bijection from S to $\mathbb{Z}_n$ for some $n \in \mathbb{Z}^+$; otherwise S is **infinite**.
The **cardinality** of a finite set S is $|S| = 0$ if $S = \varnothing$, and $|S| = n$ if there is a bijection $f : S \to \mathbb{Z}_n$.

THEOREM **Finite sets (L9 s10–11)**
- **Equality of cardinality.** For finite A, B: $|A| = |B|$ iff there is a bijection $f : A \to B$. ($\Leftarrow$: PHP and Dual PHP. $\Rightarrow$: map $a_i \mapsto b_i$.)
- **Theorem Cardinality.1.** Any subset of a finite set is finite. Contrapositive: if $A \subseteq B$ and A is infinite, then B is infinite.

PAST PAPER **T8 Q8 (AY22/23 S1 exam) · $|A| \leq |R|$ for a relation R on a finite set A?**
(i) Reflexive: true. Define $f : A \to R$ by $f(x) = (x, x)$; this is a function into R because R is reflexive. If $f(x_1) = f(x_2)$ then $(x_1, x_1) = (x_2, x_2)$, so $x_1 = x_2$ *(equality of ordered pairs)*. f is an injection, so $|A| \leq |R|$ *(PHP)*.
(ii) Symmetric, (iii) transitive: false. The empty relation on a non-empty A is symmetric and transitive, with $|A| > 0 = |\varnothing|$.

36 · Same cardinality

DEFINITION **Same cardinality (Cantor, L9 s12)**
For **any** sets A and B: A has the **same cardinality** as B, written $|A| = |B|$, iff there is a bijection $f : A \to B$.
This defines "$|A| = |B|$" without defining $|A|$ for infinite sets.

PROOF **Theorem 7.4.1 · Same-cardinality is an equivalence relation (L9 s14–16)**
1. (Reflexive) $\mathrm{id}_A : A \to A$ is injective ($\mathrm{id}_A(x_1) = \mathrm{id}_A(x_2)$ gives $x_1 = x_2$) and surjective ($\mathrm{id}_A(x) = x$). So $|A| = |A|$.
2. (Symmetric) Let $|A| = |B|$, with a bijection $f : A \to B$ *(Cantor's definition)*. Then $f^{-1} : B \to A$ is a bijection *(Theorem 7.2.3)*. So $|B| = |A|$.
3. (Transitive) Let $|A| = |B|$ and $|B| = |C|$, with bijections $f : A \to B$ and $g : B \to C$. Then $g \circ f$ has the inverse $f^{-1} \circ g^{-1}$ *(T6 Q3, §29)*, so $g \circ f : A \to C$ is a bijection *(Theorem 7.2.3)*. So $|A| = |C|$. ■

PROOF $|2\mathbb{Z}| = |\mathbb{Z}|$ **(L9 s17–18) · L9 Ex. 1, 2**
Define $H : \mathbb{Z} \to 2\mathbb{Z}$ by $H(n) = 2n$.
1. (Injective) If $H(n_1) = H(n_2)$ then $2n_1 = 2n_2$, so $n_1 = n_2$.
2. (Surjective) Let $m \in 2\mathbb{Z}$. Then $m = 2k$ for some $k \in \mathbb{Z}$ *(definition of even)*, and $H(k) = 2k = m$.
3. So H is a bijection and $|\mathbb{Z}| = |2\mathbb{Z}|$ *(Cantor's definition)*. ■

Likewise $|\mathbb{N}| = |\mathbb{N} \setminus \{0\}|$ via $x \mapsto x + 1$, and $|\mathbb{N}| = |\mathbb{N} \setminus \{1, 3, 5, ...\}|$ via $x \mapsto 2x$.
An infinite set can have the same cardinality as a proper subset of itself. For a finite A and a proper subset B, always $|B| < |A|$.

37 · Countable sets

DEFINITION **$\aleph_0$ · Countably infinite · Countable, uncountable (L9 s23)**
$\aleph_0 = |\mathbb{Z}^+|$ (some authors use $\mathbb{N}$; either may be used). A set S is **countably infinite** iff $|S| = \aleph_0$, i.e. there is a bijection between $\mathbb{Z}^+$ (or $\mathbb{N}$) and S.
A set is **countable** iff it is finite or countably infinite; otherwise it is **uncountable**.

THEOREM **Proposition 9.1 · Lemma 9.2 · Countability via sequences (L9 s33–34)**
For an **infinite** set B:
- **Proposition 9.1.** B is countable iff there is a sequence $b_0, b_1, b_2, ... \in B$ in which every element of B appears **exactly once**.
- **Lemma 9.2.** B is countable iff there is a sequence $b_0, b_1, b_2, ...$ in which every element of B **appears**.

A bijection $f : \mathbb{Z}_{\geq 0} \to B$ is the sequence $f(0), f(1), ...$: surjective = every element appears at least once; injective = at most once. For Lemma 9.2 ("if"): delete the terms not in B and all but the first appearance of each element; what remains lists B exactly once.

WORKED EXAMPLE **Standard countable sets (L9 s24–30, s35) · T8 Q1, D1**
- $\mathbb{Z}$**:** list $0, 1, -1, 2, -2, ...$ Formula on $\mathbb{Z}^+$: $f(n) = n/2$ if n is even, $-(n-1)/2$ if n is odd. On $\mathbb{N}$ (T8 Q1): $g(n) = (-1)^n \left\lfloor \frac{n+1}{2} \right\rfloor$, giving $0, -1, 1, -2, 2, ...$
- $\mathbb{Z}^+ \times \mathbb{Z}^+$**:** count along the diagonals $(1,1), (1,2), (2,1), (1,3), (2,2), (3,1), ...$ The bijection is $f(x,y) = \frac{(x+y-2)(x+y-1)}{2} + x$.
- $\mathbb{Q}^+$**:** put p/q in a grid, walk the diagonals, and skip values already counted: $\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{3}{1}, \frac{1}{3}, ...$ Skipping gives injectivity; every grid point is reached, giving surjectivity.
- **Perfect squares** $\{0, 1, 4, 9, ...\}$ (T8 D1): $n \mapsto n^2$ is a bijection from $\mathbb{N}$; also a subset of $\mathbb{N}$ (Theorem 7.4.3).

THEOREM **Closure theorems (L9 s31, s41–45)**
- **Cartesian product.** If A and B are countably infinite, so is $A \times B$; likewise $A_1 \times \cdots \times A_n$ for $n \geq 2$.
- **Unions.** The union of countably many countable sets is countable: if $A_1, A_2, ...$ are countable, so is $\bigcup_{i=1}^{\infty} A_i$.
- **Theorem 7.4.3.** Any subset of a countable set is countable. **Corollary 7.4.4.** Any set with an uncountable subset is uncountable.
- **Proposition 9.3.** Every infinite set has a countably infinite subset.
- **Lemma 9.4.** If A and B are countably infinite, then $A \cup B$ is countable.

PROOF **Lemma 9.4 (L9 s45) · Proposition 9.3 (L9 s44) · Theorem 7.4.3 (L9 s41–42)**
Lemma 9.4. Take sequences $a_0, a_1, ...$ and $b_0, b_1, ...$ in which every element of A, respectively B, appears *(Lemma 9.2)*. Then $a_0, b_0, a_1, b_1, ...$ is a sequence in which every element of $A \cup B$ appears, so $A \cup B$ is countable *(Lemma 9.2)*.
Proposition 9.3. Let B be infinite. Choose $b_0, b_1, b_2, ...$ from B one at a time, each different from all earlier choices; this is always possible, since otherwise $B = \{b_0, ..., b_{n-1}\}$ would be finite. Then $\{b_0, b_1, ...\} \subseteq B$ is countably infinite.
Theorem 7.4.3. Let A be countable and $B \subseteq A$. If B is finite it is countable. Otherwise A is countably infinite; take a bijection $f : \mathbb{Z}^+ \to A$ and let $M = f^{-1}(B)$. Define $g(1) = f(i_1)$ with $i_1 = \min M$ and $g(k) = f(i_k)$ with $i_k = \min\{m \in M : m > i_{k-1}\}$. Then $g : \mathbb{Z}^+ \to B$ lists the elements of B in order of their indices, each exactly once, so it is a bijection. ■

PROOF **T8 Q2 · B countably infinite, C finite $\Rightarrow$ $B \cup C$ countable T8 Q5 · A countable union of countably infinite sets is countable**
Q2. Take a sequence $b_0, b_1, ...$ in which every element of B appears *(Lemma 9.2)* and write $C = \{c_0, ..., c_{n-1}\}$. Then $c_0, ..., c_{n-1}, b_0, b_1, ...$ contains every element of $B \cup C$, so $B \cup C$ is countable *(Lemma 9.2)*. (As a bijection: with $C \setminus B = \{c_0, ..., c_{k-1}\}$ and a bijection $f : \mathbb{N} \to B$, set $g(x) = c_x$ if $x < k$ and $f(x-k)$ otherwise.)
Q5.
1. $\mathbb{Z}^+ \times \mathbb{Z}^+$ is countable, so there is a bijection $f : \mathbb{Z}^+ \to \mathbb{Z}^+ \times \mathbb{Z}^+$.
2. For each $i \in \mathbb{Z}^+$ take a sequence $b_{i,1}, b_{i,2}, ...$ in which every element of S_i appears *(Lemma 9.2)*.
3. Define $c_k = b_{i,j}$ where $(i,j) = f(k)$.
4. Let $x \in \bigcup_{i \in \mathbb{Z}^+} S_i$. Then $x \in S_i$ for some i, so $x = b_{i,j}$ for some j; and $f(k) = (i,j)$ for some k *(f is surjective)*. So $x = c_k$.
5. Every element of the union appears in $c_1, c_2, ...$, so the union is countable *(Lemma 9.2)*. ■

TRAP **T8 Q3, Q4 · Induction reaches every finite n, never infinity**
- "$\bigcup_{i=1}^{n} A_i$ is finite (or countable) for every n" is proved by induction on n. It does **not** follow that $\bigcup_{i=1}^{\infty} A_i$ is finite: $A_i = \{i\}$ gives $\mathbb{Z}^+$. The infinite union of countable sets **is** countable, but that needs the argument of T8 Q5.
- In an inductive step the new set A_{k+1} must be **arbitrary**; choosing $A_{k+1} = \varnothing$ proves nothing.
- T8 D2: "uncountable and infinite" is redundant, since every uncountable set is infinite.

38 · Uncountable sets

PROOF **Theorem 7.4.2 (Cantor) · The interval $(0,1) = \{x \in \mathbb{R} : 0 < x < 1\}$ is uncountable (L9 s38–40)**

1. Suppose $(0,1)$ is countable. It is not finite, so it is countably infinite.
2. List its elements as a sequence $x_1, x_2, x_3, ...$ with decimal expansions $x_n = 0.a_{n1}a_{n2}a_{n3}...$, each $a_{ij} \in \{0, 1, ..., 9\}$ (use 0.5000... and not 0.4999...).
3. Construct $d = 0.d_1d_2d_3...$ with $d_n = 1$ if $a_{nn} \neq 1$, and $d_n = 2$ if $a_{nn} = 1$.
4. For every $n \in \mathbb{Z}^+$, $d_n \neq a_{nn}$, so $d \neq x_n$.
5. But $d \in (0,1)$, so d should be in the list: contradiction. Therefore $(0,1)$ is uncountable. ■

This is **Cantor's diagonalization argument**: build an object that differs from the n-th listed object at position n.

THEOREM **Consequences (L9 s43, s46–47)**

- $\mathbb{R}$ is uncountable, since $(0,1) \subseteq \mathbb{R}$ *(Corollary 7.4.4)*.
- $|\mathbb{R}| = |(0,1)|$: bend $(0,1)$ into a circle resting on the number line at 0 and project each point from the top of the circle onto the line; this projection is a bijection. (One formula: $x \mapsto \tan(\pi(x - 1/2))$.)
- The irrationals $\mathbb{R} \setminus \mathbb{Q}$ are uncountable: otherwise $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be countable.

PROOF **T8 Q7 · If A is countably infinite, then $\mathcal{P}(A)$ is uncountable**

1. Suppose $\mathcal{P}(A)$ is countable. It is infinite, since $\{a\} \in \mathcal{P}(A)$ for every $a \in A$.
2. There is a sequence $a_0, a_1, a_2, ...$ in which every element of A appears exactly once, and a sequence $B_0, B_1, B_2, ...$ in which every element of $\mathcal{P}(A)$ appears exactly once *(Proposition 9.1)*.
3. Define $B = \{a_i : a_i \notin B_i\}$. Then $B \in \mathcal{P}(A)$.
4. Let $i \in \mathbb{N}$. If $a_i \notin B_i$ then $a_i \in B$; if $a_i \in B_i$ then $a_i \notin B$ (each a_i appears once only). Either way $B \neq B_i$.
5. So B is not in the sequence, contradicting line 2. Therefore $\mathcal{P}(A)$ is uncountable. ■

PROOF **T8 Q6 · For X infinite and Y finite there is a bijection $X \cup Y \to X$**

1. Take a countably infinite $X_0 \subseteq X$ *(Proposition 9.3)* and let $Y_0 = Y \setminus X$, a finite set.
2. $X_0 \cup Y_0$ is countably infinite *(T8 Q2)*, so there is a bijection $f : X_0 \cup Y_0 \to X_0$.
3. Define $g : X \cup Y \to X$ by $g(z) = f(z)$ if $z \in X_0 \cup Y_0$, and $g(z) = z$ otherwise. Then g is a bijection. ■

METHOD **Countable or uncountable? (T8 D3 · AY25 final Q13, Q17, Q18)**

Show countable: exhibit a sequence containing every element (Lemma 9.2), or build the set from countable sets by subset, finite product, or countable union.

Show uncountable: diagonalize, or find an uncountable subset, or biject with $\mathcal{P}(\mathbb{N})$ or $\mathbb{R}$.

Set		Reason
$\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Z}^+ \times \mathbb{Z}^+$, $\mathbb{Q} \times \mathbb{Q}$ (points with rational coordinates)	countable	sequences above; product theorem
all functions $\{0,1\} \to \mathbb{N}$	countable	same as $\mathbb{N} \times \mathbb{N}$
all 2-element subsets of $\mathbb{N}$; all finite strings over a finite alphabet	countable	inject into $\mathbb{N} \times \mathbb{N}$; countable union of finite sets
any subset of $\mathbb{Q}$; any partition of $\mathbb{N}$ (as a set of components)	countable	Theorem 7.4.3; send each component to its least element
$(0,1)$, $\mathbb{R}$, $\mathbb{R} \setminus \mathbb{Q}$	uncountable	Theorem 7.4.2 and Corollary 7.4.4
$\mathcal{P}(\mathbb{N})$; all functions $\mathbb{N} \to \{0,1\}$; all infinite sequences of integers	uncountable	T8 Q7; diagonalization
the set of all partitions of $\mathbb{N}$	uncountable	$\{A, \mathbb{N} \setminus A\}$ for each A gives uncountably many

AY25 Q17. A countably infinite, B uncountable: $A \cup B$ is uncountable (it contains B) and $A \cap B$ is countable (a subset of A). Both true.

AY25 Q13. $(\mathbb{Z} \times \mathbb{Z}^+)/\sim$ with $(a,b) \sim (c,d) \leftrightarrow ad = bc$ is in bijection with $\mathbb{Q}$, hence with $\mathbb{Z}$ and with $\mathbb{Z} \times \mathbb{Z}^+$: all countably infinite.

Part IX · Counting and Probability

39 · Sample spaces, the multiplication rule, permutations

DEFINITION Sample space, event (L10 s8) · Equally likely probability (L10 s9)

A **sample space** is the set of all possible outcomes of a random process or experiment. An **event** is a subset of a sample space.

If S is a **finite** sample space in which **all outcomes are equally likely** and E is an event in S, then

$$P(E) = \frac{|E|}{|S|} = \text{the number of outcomes in } \frac{E}{\text{the total number of outcomes in}} S.$$

THEOREM Theorem 9.1.1 · Number of elements in a list (L10 s12) Theorem 9.2.1 · Multiplication rule (L10 s21)

- If $m \le n$ are integers, there are $n - m + 1$ integers from m to n inclusive.
- If an operation consists of k steps, where step i can be performed in n_i ways **regardless of how the preceding steps were performed**, then the whole operation can be performed in $n_1 n_2 \cdots n_k$ ways.
- **Theorem 5.2.4 (L10 s24).** A finite set A has $|\mathcal{P}(A)| = 2^{|A|}$ subsets: each element is picked or dropped.

WORKED EXAMPLE L10 s13, Ex. 1–5 · Lists, possibility trees, the multiplication rule

- **Multiples of 5 from 100 to 999:** $5 \cdot 20, \ldots, 5 \cdot 199$, so $199 - 20 + 1 = 180$; probability $180/900 = 1/5$.
- **Tournament** (play until one team wins two in a row or three in total): the possibility tree has **10** leaves; 4 of them need five games, so $P = 4/10 = 2/5$ if all are equally likely.
- **PINs** of 4 symbols from 36: $36^4 = 1\,679\,616$ with repetition; $36 \cdot 35 \cdot 34 \cdot 33 = 1\,413\,720$ without; $P(\text{no repeat}) \approx 0.8417$.
- **Officers** (president, treasurer, secretary from Ann, Bob, Cyd, Dan; Ann is not president; Cyd or Dan is secretary). $3 \cdot 3 \cdot 2 = 18$ is **wrong**: the number of choices for secretary depends on earlier choices. Reorder the steps: secretary (2 ways), then president (2), then treasurer (2): $2 \cdot 2 \cdot 2 = 8$.
- **ICANDOIT** from 26 tiles: without replacement the probability is 0 (the letter I is needed twice); with replacement it is $(1/26)^8$.

DEFINITION Permutation (L10 s32) · r-permutation, $P(n, r)$ (L10 s36)

A **permutation** of a set of objects is an ordering of the objects in a row. An ***r*-permutation** of a set of n elements is an ordered selection of r of its elements; their number is $P(n, r)$.

THEOREM Theorem 9.2.2 · Theorem 9.2.3 (L10 s34, s37) · Circular permutations (T9 Q5)

- A set with $n \ge 1$ elements has $n!$ permutations; $0! = 1$.
- For $1 \le r \le n$: $P(n, r) = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}$. $P(5, 2) = 20$, $P(7, 4) = 840$, $P(5, 5) = 120$.
- n distinct objects can be arranged around a circle in $(n-1)!$ ways (rotations are the same arrangement).

WORKED EXAMPLE L10 Ex. 6 · T9 Q1, Q2, Q5

- **COMPUTER:** $8! = 40\,320$ orderings; with CO kept together in order, treat CO as one unit: $7! = 5040$; probability $7!/8! = 1/8$.
- **T9 Q1.** 5-card sequences with at least one picture card (12 picture cards, 40 others): with replacement $52^5 - 40^5 = 277\,804\,032$; without replacement $P(52, 5) - P(40, 5) = 232\,914\,240$.
- **T9 Q2 (AY23/24 exam).** Five-digit cheque numbers readable upside down use digits $0, 1, 6, 8, 9$: 5^5. Those that read the same after rotation (the middle digit is 0, 1 or 8; the last two digits are forced): $5 \cdot 5 \cdot 3 = 75$. Confusable numbers: $5^5 - 75 = 3050$.
- **T9 Q5.** (a) 8 boys and 4 girls at a round table, no two girls adjacent: seat the boys (7!), then put the girls into 4 of the 8 gaps ($P(8, 4)$): $7! \cdot 1680 = 8\,467\,200$. (b) 6 people, Eric not next to Freddy: $5! - 2 \cdot 4! = 72$. (c) $n - 1$ people at n chairs: treat the empty chair as a person: $(n-1)!$.

40 · Addition, difference, inclusion and exclusion

THEOREM **Theorems 9.3.1–9.3.3 (L10 s40, s43, s50) · Complement (L10 s48)**

- **Addition rule.** If a finite set A is the union of mutually disjoint subsets $A_1, ..., A_k$, then $|A| = |A_1| + |A_2| + \cdots + |A_k|$.
- **Difference rule.** If A is finite and $B \subseteq A$, then $|A \setminus B| = |A| - |B|$. (Proof: B and $A \setminus B$ are disjoint with union A; apply the addition rule.)
- **Inclusion/exclusion.** For finite A, B, C: $|A \cup B| = |A| + |B| - |A \cap B|$ and $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.
- **Complement.** For an event A in a finite sample space S: $P(\overline{A}) = 1 - P(A)$.

WORKED EXAMPLE **L10 Ex. 7–9 · T9 Q3**

- **Passwords** of 1 to 3 letters: $26 + 26^2 + 26^3 = 18\,278$ (addition rule over the lengths).
- **PINs with a repeated symbol:** $36^4 - 36 \cdot 35 \cdot 34 \cdot 33 = 265\,896$; probability $\approx 0.158 = 1 - 0.842$.
- **Multiples of 3 or 5 from 1 to 1000:** $333 + 200 - 66 = 467$ (multiples of both are the multiples of 15). **Neither:** $1000 - 467 = 533$, using De Morgan: $\overline{A} \cap \overline{B} = \overline{A \cup B}$.
- **T9 Q3.** Permutations of $1, ..., n$ ($n \geq 3$) with 1, 2 or 3 in its correct position: $|P_k| = (n-1)!$, $|P_i \cap P_j| = (n-2)!$, $|P_1 \cap P_2 \cap P_3| = (n-3)!$, so the count is $3(n-1)! - 3(n-2)! + (n-3)! = (3n^2 - 12n + 13)(n-3)!$.

METHOD **"At least one" and "at most"**

"At least one" is usually shortest by the difference rule: total minus "none". "At most one" is "none" plus "exactly one" by the addition rule. Overlapping conditions joined by "or" need inclusion/exclusion. Before multiplying, check that each step's count is independent of earlier choices; if not, reorder the steps or split into cases.

41 · The pigeonhole principle in counting

THEOREM **Pigeonhole Principle · Generalized PHP and its contrapositive (L10 s55, s62, s65)**

- **PHP.** A function from one finite set to a smaller finite set cannot be one-to-one: at least two elements of the domain have the same image.
- **Generalized PHP.** For $f : X \to Y$ with $|X| = n$, $|Y| = m$ and a positive integer k: if $k < n/m$, then some $y \in Y$ is the image of at least $k + 1$ distinct elements of X.
- **Contrapositive form.** If for each $y \in Y$ the set $f^{-1}(\{y\})$ has at most k elements, then $n \leq km$.

Equivalently: among n pigeons in m holes, some hole holds at least $\lceil n/m \rceil$ pigeons.

METHOD **Writing a pigeonhole argument**

State (1) what the pigeons are, (2) what the holes are, (3) the rule sending each pigeon to a hole, (4) the two counts, then conclude with the principle by name. For "how many must be drawn to guarantee ...", find the **largest** selection that avoids the target and add 1.

WORKED EXAMPLE **L10 Ex. 10–12 · Decimal expansions (L10 s59–61)**

- 13 people: two share a birth month ($13 > 12$). Singapore: two people have the same number of hairs (5.47 million people, at most 300 001 possible counts).
- **85 people, 26 initials:** $3 < 85/26$, so at least 4 share a last initial. By the contrapositive: if at most 3 shared each initial there would be at most $3 \cdot 26 = 78$ people.
- **Rational numbers have terminating or repeating decimals.** In the long division of a by b every remainder lies in $\{0, ..., b-1\}$ *(Theorem 4.4.1)*. If no remainder is 0, then among $b + 1$ successive remainders two are equal, $r_j = r_k$ with $j < k$ *(PHP)*, and the digits repeat from there. $3/14 = 0.2\overline{142857}$: remainders $3, 2, 6, 4, 12, 8, 10, 2, ...$
- **42 students, 12 computers, at most 6 per computer: at least 5 computers are used by 3 or more students.** Let k be the number of such computers. They serve at most $6k$ students; the other $12 - k$ serve at most $2(12 - k)$. So $42 \leq 6k + 24 - 2k = 4k + 24$, giving $k \geq 4.5$, hence $k \geq 5$.

PROOF **T9 Q8 (AY21/22 exam) · Among any 5 distinct non-negative integers, two differ by a multiple of 4 T9 Q9 · Chess master**

Q8.

1. Let the pigeonholes be the classes $[0], [1], [2], [3]$ of congruence mod 4, and send each of the 5 integers to its class.
2. Two of the integers, x and y, lie in the same class $[i]$ *(PHP: $5 > 4$)*. So $x = 4k + i$ and $y = 4l + i$ for some $k, l \in \mathbb{Z}$.

3. Then $x-y=4(k-l)$ with $k-l\in\mathbb{Z}$ *(closure)*, so $4\mid(x-y)$. ■

Q9. 77 days, at least one game a day, at most 12 a week. Let P_i be the number of games played up to day i: $1\le P_1<P_2<\cdots<P_{77}\le 132$. Let $Q_i=P_i+21\le 153$. The 154 numbers $P_1,\ldots,P_{77},Q_1,\ldots,Q_{77}$ take values in $\{1,\ldots,153\}$, so two are equal *(PHP)*. The P_i are distinct and so are the Q_i, hence $P_j=Q_i=P_i+21$ for some i,j: exactly 21 games were played on days $i+1$ to j. ■

PAST PAPER **T10 Q10 · T10 Q12 · AY25 final Q24(f)**

T10 Q10. At a party of $n\ge 2$ people where the host shakes hands with everyone, two people shook the same number of hands: every degree lies in $\{1,\ldots,n-1\}$ (each person shook the host's hand), so n people share $n-1$ possible values *(PHP)*.

T10 Q12. Any simple graph G on 6 vertices, or its complement, contains a triangle. Colour the edges of K_6 black (in G) or red (in $\overline{G}$). Fix a vertex v: of its 5 edges, at least 3 have one colour *(generalized PHP, $2<5/2$)*, say red, to u_1,u_2,u_3. If some edge $\{u_x,u_y\}$ is red, then v,u_x,u_y form a red triangle; otherwise u_1,u_2,u_3 form a black triangle.

AY25 Q24(f). Socks: 8 red, 7 blue, 6 green, 5 black, 4 white. To avoid both "4 of one colour" and "3 each of two colours", take 3 of one colour and 2 of each of the other four: 11 socks. So **12** guarantee one of the two.

42 · Combinations and multisets

DEFINITION **r-combination, $\binom{n}{r}$ (L11 s4) · Multiset (L11 s18)**

For integers $0\le r\le n$, an ***r*-combination** of a set of n elements is a **subset** of r of them. $\binom{n}{r}$, read "n choose r", is the number of r-combinations (also written $C(n,r)$, ${}_nC_r$).

An ***r*-combination with repetition allowed**, or **multiset of size *r***, chosen from a set X of n elements is an unordered selection of r elements of X with repetition allowed, written $\left[x_{i_1},x_{i_2},\ldots,x_{i_r}\right]$.

THEOREM **Theorems 9.5.1, 9.5.2, 9.6.1 (L11 s10, s15, s21)**

- **9.5.1.** $\binom{n}{r}=\frac{P(n,r)}{r!}=\frac{n!}{r!(n-r)!}$, because an r-permutation is an r-subset followed by one of its $r!$ orderings: $P(n,r)=\binom{n}{r}\cdot r!$.
- **9.5.2 Permutations with indistinguishable objects.** If n objects consist of n_1 of type 1, ..., n_k of type k, with $n_1+\cdots+n_k=n$, the number of distinguishable orderings is
$$\binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\cdots-n_{k-1}}{n_k}=\frac{n!}{n_1!\,n_2!\cdots n_k!}.$$
- **9.6.1 Multisets.** The number of multisets of size r from n elements, equivalently the number of ways to select r objects from n categories with repetition, is $\binom{r+n-1}{r}$. (Stars and bars: a string of r crosses and $n-1$ bars.)

Choosing k from n (L11 s23)	**Order matters**	**Order does not matter**
Repetition allowed	n^k	$\binom{k+n-1}{k}$
Repetition not allowed	$P(n,k)=n!/(n-k)!$	$\binom{n}{k}=n!/(k!(n-k)!)$

WORKED EXAMPLE **L11 Ex. 1–7**

- **Committees.** 3-person committees from 4 people: $\binom{4}{3}=4$. $P(4,2)=12=\binom{4}{2}\cdot 2!$, so $\binom{4}{2}=6$.
- **Teams of 5 from 5 men and 7 women.** 3 men and 2 women: $\binom{5}{3}\binom{7}{2}=10\cdot 21=210$. At least one man: $\binom{12}{5}-\binom{7}{5}=792-21=771$. At most one man: $\binom{7}{5}+\binom{5}{1}\binom{7}{4}=21+175=196$.
- **MISSISSIPPI** (4 S, 4 I, 2 P, 1 M): $\binom{11}{4}\binom{7}{4}\binom{3}{2}\binom{1}{1}=\frac{11!}{4!\,4!\,2!\,1!}=34\,650$.
- **Multisets of size 3 from $\{1,2,3,4\}$:** $\binom{3+4-1}{3}=\binom{6}{3}=20$; e.g. $[1,3,4]$ is the string $\times\,|\,|\times|\times$.
- **Lunch for 20 guests from 6 stalls:** $\binom{20+6-1}{20}=\binom{25}{20}=53\,130$.
- **$x_1+x_2+x_3=20$:** non-negative integer solutions $\binom{22}{20}=231$. Positive solutions: put $y_i=x_i-1$, so $y_1+y_2+y_3=17$: $\binom{19}{17}=171$.

PAST PAPER **T9 Q4, Q6, Q7 · T10 Q2 · AY25 final Q24(a)–(e), (g)**

T9 Q6. 5 from 7 men and 6 women with at least 3 men: $\binom{7}{3}\binom{6}{2}+\binom{7}{4}\binom{6}{1}+\binom{7}{5}=525+210+21=756$; probability $756/\binom{13}{5}=756/1287\approx 0.5874$.

T9 Q7 (AY21/22). \$50 000 over 4 properties in \$1000 units: $x_1+\cdots+x_4=50$ has $\binom{53}{50}=23\,426$ solutions. Not necessarily all invested: add a slack variable x_5: $\binom{54}{50}=316\,251$.

T9 Q4, T10 Q2. n boxes, white balls in a non-empty consecutive block: $\sum_{k=1}^{n}(n-k+1)=n(n+1)/2$; or choose the two ends of the block among $n+1$ gaps: $\binom{n+1}{2}$.

AY25 Q24. (a) 3 red, 4 blue, 5 green balls in a row: $12!/(3!\,4!\,5!)=27\,720$. (b) Committee of 5 from 6 men and 5 women with at least 3 women: $\binom{5}{3}\binom{6}{2}+\binom{5}{4}\binom{6}{1}+\binom{5}{5}=150+30+1=181$. (c) 6 distinct letters with exactly 2 vowels: choose the 2 vowel positions, fill them, fill the rest: $\binom{6}{2}\cdot P(5,2)\cdot P(21,4)=15\cdot 20\cdot 143\,640$ $=43\,092\,000$. (d) n couples at a round table, men and women alternating: $(n-1)!\,n!$. (e) Every woman next to her husband: seat the n couples as units, then order each couple: $(n-1)!\,2^n$. (g) Three distinguishable dice with sum 10: **27**.

43 · Pascal's formula, the binomial theorem, combinatorial proofs

THEOREM **Theorem 9.7.1 Pascal's formula (L11 s26) · Theorem 9.7.2 Binomial theorem (L11 s34) · Identities (L11 s29–33)**

- **Pascal.** For positive integers $r\le n$: $\binom{n+1}{r}=\binom{n}{r-1}+\binom{n}{r}$.
- **Binomial theorem.** For real a,b and integer $n\ge 0$: $(a+b)^n=\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$. $\binom{n}{r}$ is a **binomial coefficient**.
- $\binom{n}{r}=\binom{n}{n-r}$; $k\binom{n}{k}=n\binom{n-1}{k-1}$; $\binom{n}{0}+\binom{n}{1}+\cdots+\binom{n}{n}=2^n$.

DEFINITION **Combinatorial proof (L11 s27)**

A **combinatorial proof** uses counting as the basis of the proof. Two kinds: a **bijective proof** (a bijection between two sets shows they have the same size) and a **proof by double counting** (count one set in two ways; the two expressions are equal).

PROOF **Pascal's formula, algebraic and combinatorial (L11 s28) · Three double-counting proofs (L11 s30–33)**

Algebraic. $\binom{n}{r-1}+\binom{n}{r}=\frac{n!}{(n-r+1)!(r-1)!}+\frac{n!}{(n-r)!\,r!}=\frac{n!\,r+n!(n-r+1)}{(n-r+1)!\,r!}=\frac{(n+1)!}{(n+1-r)!\,r!}=\binom{n+1}{r}$.

Combinatorial.

1. $\binom{n+1}{r}$ counts the r-subsets of a set A with $n+1$ elements. Fix $x\in A$.
2. The r-subsets containing x: choose the other $r-1$ elements from the remaining n: $\binom{n}{r-1}$.
3. The r-subsets not containing x: choose all r from the remaining n: $\binom{n}{r}$.
4. The two cases are disjoint and exhaustive, so $\binom{n+1}{r}=\binom{n}{r-1}+\binom{n}{r}$ *(addition rule)*. ■

$\binom{n}{k}=\binom{n}{n-k}$. Choosing the k pupils who go is the same as choosing the $n-k$ who stay.

$k\binom{n}{k}=n\binom{n-1}{k-1}$. Count committees of k from n with a chairperson: choose the committee then the chair, or choose the chair then the other $k-1$ members.

$\sum_{k=0}^{n}\binom{n}{k}=2^n$. Both count the subsets of an n-element set: by size on the left, by Theorem 6.3.1 on the right.

WORKED EXAMPLE **L11 Ex. 12, 13 · T10 Q1**

- $(a+b)^5=a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$.
- $(x-4y)^4=x^4-16x^3y+96x^2y^2-256xy^3+256y^4$ (substitute $b=-4y$ and keep the sign inside the power).
- $\sum_{k=0}^{n}\binom{n}{k}9^k=\sum_{k=0}^{n}\binom{n}{k}1^{n-k}9^k=(1+9)^n=10^n$.
- **T10 Q1.** Term independent of x in $(2x^2+1/x)^9$: the general term is $\binom{9}{r}(2x^2)^{9-r}x^{-r}=\binom{9}{r}2^{9-r}x^{18-3r}$; $18-3r=0$ gives $r=6$: $\binom{9}{6}\cdot 2^3=84\cdot 8=672$.

THEOREM **Counting functions (T9 D3 · T10 D2, D3)**

For finite sets with $|A|=m$ and $|B|=n$:

Functions $A\to B$	**Number**	$m=3,n=2$	$m=n=3$
all functions	n^m	8	27
injections ($m\le n$)	$P(n,m)=n!/(n-m)!$	0	6
surjections ($m\ge n$)	$\sum_{i=0}^{n}(-1)^i\binom{n}{i}(n-i)^m$	6	6
bijections ($m=n$)	$n!$	0	6

Surjections from a 5-element set to a 3-element set: $3^5-3\cdot 2^5+3\cdot 1^5=150$ (inclusion/exclusion on the values missed). **AY25 Q10:** strictly increasing $f:\{0,1\}\to\{0,1,2,3\}$: $\binom{4}{2}=6$; injective $g:\{0,1\}\to\{0,1,2\}$: $P(3,2)=6$.

44 · Probability axioms and expected value

DEFINITION **Probability axioms (L11 s39) · Expected value (L11 s43)**

Let S be a sample space. A **probability function** P from the set of all events in S to $\mathbb{R}$ satisfies, for all events A, B:

1. $0 \le P(A) \le 1$; 2. $P(\varnothing) = 0$ and $P(S) = 1$; 3. if $A \cap B = \varnothing$, then $P(A \cup B) = P(A) + P(B)$.

If the possible outcomes of an experiment are real numbers $a_1, ..., a_n$ occurring with probabilities $p_1, ..., p_n$, the **expected value** is

$$\sum_{k=1}^{n} a_k p_k = a_1 p_1 + a_2 p_2 + \cdots + a_n p_n.$$

THEOREM **Complement · General union (L11 s40–41) · Linearity of expectation (L11 s48)**

- $P(\overline{A}) = 1 - P(A)$. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ for **any** events A, B.
- For random variables X, Y, **dependent or not**: $E[X + Y] = E[X] + E[Y]$; in general $E\left[\sum_{i=1}^{n} c_i X_i\right] = \sum_{i=1}^{n} c_i E[X_i]$.

WORKED EXAMPLE **L11 s41, Ex. 14–16 · T10 Q3, Q4, D1 · AY25 final Q23**

- **Heart or 7:** $\frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$.
- **Fair die:** $(1 + 2 + \cdots + 6)/6 = 3.5$. **Sum of two dice:** $3.5 + 3.5 = 7$ by linearity.
- **Lottery:** 500 000 tickets at \$5; prizes \$1 000 000 (1), \$1000 (10), \$500 (1000), \$10 (10 000). Expected net gain $= (999\,995 + 10 \cdot 995 + 1000 \cdot 495 + 10\,000 \cdot 5 - 488\,989 \cdot 5)/500\,000 = -1.78$ dollars.
- **T10 Q3 (AY20/21 S2).** Loaded die: each big number (4, 5, 6) is twice as likely as each small one, so $P(1) = P(2) = P(3) = 1/9$ and $P(4) = P(5) = P(6) = 2/9$. Maximum of two rolls: $P(\max = 1, ..., 6) = \frac{1}{81}, \frac{3}{81}, \frac{5}{81}, \frac{16}{81}, \frac{24}{81}, \frac{32}{81}$; expected value $398/81$.
- **T10 Q4.** Two balls from $\{1, 2, 2, 8, 8\}$: one ball has expectation $(1 + 2 + 2 + 8 + 8)/5 = 4.2$, so the sum has expectation 8.4 by linearity (the draws are dependent; linearity still applies).
- **T10 D1.** 2 bulbs from 8 with 3 defective: $E[X] = 2 \cdot 3/8 = 0.75$.
- **AY25 Q23.** Sundaes: 20% of customers choose 3 of 10 flavours, 80% choose 2: $0.2\binom{10}{3} + 0.8\binom{10}{2} = 24 + 36 = 60$.

45 · Conditional probability, Bayes' theorem, independence

DEFINITION **Conditional probability (L11 s53) · Independent events (L11 s73) · Pairwise and mutually independent (L11 s79–80)**

For events A, B in S with $P(A) \neq 0$, the **conditional probability of B given A** is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}, \quad \text{so} \quad P(A \cap B) = P(B \mid A) \cdot P(A).$$

A and B are **independent** iff $P(A \cap B) = P(A) \cdot P(B)$.

A, B, C are **pairwise independent** iff (1) $P(A \cap B) = P(A)P(B)$, (2) $P(A \cap C) = P(A)P(C)$, (3) $P(B \cap C) = P(B)P(C)$; they are **mutually independent** iff also (4) $P(A \cap B \cap C) = P(A)P(B)P(C)$. Events $A_1, ..., A_n$ are mutually independent iff the probability of the intersection of **any subset** of them is the product of their probabilities.

THEOREM **Theorem 9.9.1 · Bayes' theorem (L11 s66)**

Suppose S is the union of mutually disjoint events $B_1, ..., B_n$, and A and all B_i have non-zero probability. Then for $1 \le k \le n$:

$$P(B_k \mid A) = \frac{P(A \mid B_k) \cdot P(B_k)}{P(A \mid B_1) \cdot P(B_1) + P(A \mid B_2) \cdot P(B_2) + \cdots + P(A \mid B_n) \cdot P(B_n)}.$$

The denominator is $P(A)$ (total probability).

TRAP **Conditional probability and independence**

- $P(B \mid A)$ and $P(A \mid B)$ are different numbers; Bayes converts one into the other.
- "At least one is a boy" is not "the first is a boy": of BB, BG, GB, GG the condition leaves three outcomes, so $P(\text{both boys} \mid \text{at least one boy}) = 1/3$.
- Independent is not disjoint: disjoint events with positive probabilities are **dependent**.
- Pairwise independence does not give (4), and (4) does not give pairwise independence. Two fair coins with A = first is heads, B = second is heads, C = both the same: every pair is independent, but $P(A \cap B \cap C) = 1/4 \neq 1/8$.

WORKED EXAMPLE **L11 Ex. 17 · Urn with 5 blue and 7 grey balls, two drawn without replacement**

$P(B_1) = \frac{5}{12}$, $P(B_2 \mid B_1) = \frac{4}{11}$, $P\left(B_2 \mid \overline{B_1}\right) = \frac{5}{11}$. Multiply along the branches of the tree:

$$P(B_1 \cap B_2) = \frac{20}{132}, \quad P\left(B_1 \cap \overline{B_2}\right) = \frac{35}{132}, \quad P\left(\overline{B_1} \cap B_2\right) = \frac{35}{132}, \quad P\left(\overline{B_1} \cap \overline{B_2}\right) = \frac{42}{132}.$$

- $P(B_2) = \frac{20}{132} + \frac{35}{132} = \frac{55}{132} = \frac{5}{12}$ *(axiom 3 on the disjoint cases)*.
- $P(B_1 \cup B_2) = \frac{5}{12} + \frac{5}{12} - \frac{20}{132} = \frac{90}{132} = \frac{15}{22}$; so $P(\text{no blue}) = \frac{7}{22}$.
- Expected number of blue balls: $0 \cdot \frac{42}{132} + 1 \cdot \frac{70}{132} + 2 \cdot \frac{20}{132} = \frac{110}{132} = \frac{5}{6} \approx 0.83$.

WORKED EXAMPLE **Bayes · L11 s63–65, Ex. 18 · T10 Q5 · AY25 final Q24(h)**

- **Two urns** (3 blue, 4 grey; 5 blue, 3 grey), one chosen at random: $P(B) = \frac{3}{7} \cdot \frac{1}{2} + \frac{5}{8} \cdot \frac{1}{2} = \frac{59}{112}$ and $P(U_1 \mid B) = (3/14)/(59/112) = \frac{24}{59} \approx 40.7\%$.
- **Medical test.** Disease in 5 per 1000; false positive rate 3%; false negative rate 1%. $P(\text{disease} \mid +) = \frac{0.99 \cdot 0.005}{0.99 \cdot 0.005 + 0.03 \cdot 0.995} \approx 14.2\%$. $P(\text{no disease} \mid -) = \frac{0.97 \cdot 0.995}{0.01 \cdot 0.005 + 0.97 \cdot 0.995} \approx 99.995\%$.
- **T10 Q5 (AY21/22 S2).** $P(D) = 0.001$, $P(+ \mid D) = 0.85$, $P(+) = 0.1$. $P(D \mid +) = (0.85 \cdot 0.001)/0.1 = 0.0085$. False positive: $P\left(+ \mid \overline{D}\right) = \frac{P(\overline{D} \mid +) \cdot P(+)}{P(\overline{D})} = (0.9915 \cdot 0.1)/0.999 \approx 0.0992$.
- **AY25 Q24(h).** $P(S) = 0.2$, $P(O \mid S) = 0.5$, $P(O \mid H) = 0.08$: $P(S \mid O) = 0.1/(0.1 + 0.064) \approx 0.61$. $P(U \mid S) = 0.35$, $P(U \mid H) = 0.04$: $P(S \mid U) = 0.07/(0.07 + 0.032) \approx 0.69$. "Urgent" is the stronger signal.

WORKED EXAMPLE **Independence · L11 Ex. 19, 20 · T9 D1 · T10 Q6**

- **Loaded coin, $P(H) = 0.6$, two tosses:** $P(\text{two heads}) = 0.36$; $P(\text{one head}) = 0.6 \cdot 0.4 + 0.4 \cdot 0.6 = 0.48$; $P(\text{no heads}) = 0.16$; $P(\text{at least one}) = 1 - 0.16 = 0.84$.
- **Ten tosses:** $P(\text{eight heads}) = \binom{10}{8}(0.6)^8(0.4)^2 \approx 0.121$; $P(\text{at least eight}) = \binom{10}{8}(0.6)^8(0.4)^2 + \binom{10}{9}(0.6)^9(0.4) + (0.6)^{10} \approx 0.167$. Probabilities of the form $\binom{n}{k}p^k(1-p)^{n-k}$ are **binomial probabilities**.
- **T9 D1.** 3 blue and 7 white balls, two draws with replacement: $P(\text{blue then white}) = 0.21$; $P(\text{both white}) = 0.49$; $P(\text{second blue}) = 0.3$.
- **T10 Q6 (AY15/16).** A random relation on a 4-element set is reflexive with probability $2^{12}/2^{16} = 1/16$ and symmetric with probability $2^{10}/2^{16} = 1/64$. For $|A| = n$: $1/2^n$ and $2^{(n^2+n)/2}/2^{n^2} = 1/2^{(n^2-n)/2}$.
- **Birthday problem (T9 D4):** $P(\text{some shared birthday among } n) = 1 - (365 \cdot 364 \cdots (365 - n + 1))/365^n$, which first reaches 50% at $n = 23$.

Part X · Graphs and Trees

46 · Graphs, special graphs, degree

DEFINITION **Undirected graph (L12 s7) · Directed graph (L12 s8)**

An **undirected graph** $G = (V, E)$ consists of two finite sets: a non-empty set V of **vertices** and a set E of **edges**, where each edge is associated with a set of one or two vertices called its **endpoints**. Write $e = \{v, w\}$.

An edge **connects** its endpoints and is **incident on** each of them; vertices joined by an edge are **adjacent**; an edge with a single endpoint is a **loop** (its vertex is adjacent to itself); edges with the same endpoints are **parallel**; edges sharing an endpoint are **adjacent edges**.

A **directed graph** (digraph) has directed edges, each associated with an **ordered pair** of vertices: $e = (v, w)$ goes from v to w.

DEFINITION **Simple graph (L12 s19) · Complete graph K_n (L12 s20) · Bipartite, $K_{m,n}$ (L12 s21) · Subgraph (L12 s22)**

- A **simple graph** is an undirected graph with no loops and no parallel edges.
- The **complete graph** K_n ($n > 0$) is the simple graph on n vertices with exactly one edge between each pair of distinct vertices. It has $\binom{n}{2} = n(n-1)/2$ edges.
- A **bipartite graph** is a simple graph whose vertices can be divided into two disjoint sets U and V such that every edge connects a vertex in U to one in V. In the **complete bipartite graph** $K_{m,n}$, with $|U| = m$ and $|V| = n$, every vertex of U is joined to every vertex of V; it has mn edges.
- H is a **subgraph** of G iff every vertex of H is a vertex of G, every edge of H is an edge of G, and every edge of H has the same endpoints as in G.

DEFINITION **Degree, total degree (L12 s23) · Indegree, outdegree (L12 s26) · Complement, self-complementary, triangle (T10)**

The **degree** $\deg(v)$ is the number of edges incident on v, **with a loop counted twice**. The **total degree** of G is the sum of the degrees of all its vertices.

In a digraph, the **indegree** $\deg^-(v)$ is the number of edges ending at v and the **outdegree** $\deg^+(v)$ the number starting at v; $\sum_{v\in V} \deg^-(v) = \sum_{v\in V} \deg^+(v) = |E|$.

The **complement** $\overline{G}$ of a simple graph G has the same vertices; distinct v, w are adjacent in $\overline{G}$ iff they are not adjacent in G. G is **self-complementary** iff $G \cong \overline{G}$. A **triangle** is a simple circuit of length 3.

A **vertex colouring** assigns colours to vertices so that adjacent vertices get different colours (L12 s13).

THEOREM **Theorem 10.1.1 Handshake theorem · Corollary 10.1.2 · Proposition 10.1.3 (L12 s25)**

- For any graph G with vertices $v_1, \ldots, v_n$: $\deg(v_1) + \deg(v_2) + \cdots + \deg(v_n) = 2 \cdot |E|$.
- The total degree of a graph is even.
- In any graph there is an even number of vertices of odd degree.

PROOF **Handshake theorem · Proposition 10.1.3**

1. Let e be an edge of G with endpoints v_i and v_j. If $i \neq j$, then e contributes 1 to $\deg(v_i)$ and 1 to $\deg(v_j)$; if $i = j$ (a loop), it contributes 2 to $\deg(v_i)$ *(definition of degree)*.
2. So every edge contributes exactly 2 to the total degree, which is therefore $2\,|E|$. ■

Proposition 10.1.3. The total degree T is even. Let E be the sum of the even degrees and O the sum of the odd degrees, so $O = T - E$ is even *(difference of even integers)*. A sum of odd integers is even only if the number of terms is even. ■

WORKED EXAMPLE **Degree arguments · T11 D1, D2 · T10 Q8**

- **No graph has degrees 1, 1, 2, 3:** the total would be 7, which is odd.
- **No simple graph has degrees 1, 1, 3, 3 on 4 vertices:** each vertex of degree 3 is adjacent to all three others, so every other vertex has degree at least 2.
- **T11 D1.** A connected simple graph on n vertices has between $n-1$ edges (a tree) and $n(n-1)/2$ edges (K_n).
- **T11 D2.** Simple graphs on n labelled vertices: each of the $\binom{n}{2}$ possible edges is present or absent, so $2^{n(n-1)/2}$; for $n = 3$ there are 8 (in 4 isomorphism classes).
- **T10 Q8.** Non-isomorphic undirected graphs with two vertices and two edges: **4** (two parallel edges; one edge and one loop; two loops at one vertex; one loop at each vertex). The question does not say "simple".
- **K_n and $K_{m,n}$:** every vertex of K_n has degree $n-1$; in $K_{m,n}$ the degrees are n (in U) and m (in V). Check: $n(n-1) = 2 \cdot \binom{n}{2}$ and $mn + nm = 2mn$.

47 · Walks, connectedness, Euler and Hamiltonian circuits

DEFINITION **Walk, trail, path, closed walk, circuit, simple circuit (L12 s32–33)**

A **walk** from v to w is a finite alternating sequence $v_0 e_1 v_1 e_2 \cdots v_{n-1} e_n v_n$ of adjacent vertices and edges with $v_0 = v$, $v_n = w$, and v_{i-1}, v_i the endpoints of e_i. Its **length** is n, the number of edges. The **trivial walk** from v to v is the single vertex v.

	Repeated edge?	Repeated vertex?	Closed?	Notes
Walk	allowed	allowed	allowed	
Trail	no	allowed	allowed	a walk with no repeated edge
Path	no	no	no	a trail with no repeated vertex
Closed walk	allowed	allowed	yes	
Circuit (cycle)	no	allowed	yes	closed walk of length at least 3, no repeated edge
Simple circuit	no	first and last only	yes	

An undirected graph is **cyclic** if it contains a loop or a cycle; otherwise it is **acyclic**. (Other books say "path" for trail and "cycle" for simple circuit; the course follows Epp.)

DEFINITION **Connected (L12 s35) · Connected component (L12 s38)**

Vertices v and w are **connected** iff there is a walk from v to w. G is **connected** iff $\forall v, w \in V$ there is a walk from v to w.

H is a **connected component** of G iff (1) H is a subgraph of G, (2) H is connected, and (3) no connected subgraph of G has H as a subgraph and contains vertices or edges not in H. A connected component is a connected subgraph of largest possible size.

THEOREM **Lemma 10.2.1 (L12 s37)**

Let G be a graph.

- (a) If G is connected, then any two distinct vertices of G can be connected by a **path**.
- (b) If v and w are part of a circuit and one edge is removed from the circuit, there is still a trail from v to w.
- (c) If G is connected and contains a circuit, then an edge of the circuit can be removed without disconnecting G.

DEFINITION **Euler circuit, Eulerian graph (L12 s41) · Euler trail (L12 s43) · Hamiltonian circuit, Hamiltonian graph (L12 s49)**

An **Euler circuit** for G is a circuit that contains every vertex and traverses **every edge exactly once**; a graph with one is **Eulerian**.

An **Euler trail** from v to w ($v \neq w$) starts at v, ends at w, passes through every vertex at least once and traverses every edge exactly once.

A **Hamiltonian circuit** for G is a simple circuit that includes **every vertex** (each exactly once, except that the first and last are the same); a graph with one is **Hamiltonian**.

THEOREM **Theorems 10.2.2–10.2.4 · Corollary 10.2.5 · Proposition 10.2.6 (L12 s41–43, s52)**

- **10.2.2.** If a graph has an Euler circuit, then every vertex has positive even degree. (Contrapositive: an odd-degree vertex rules out an Euler circuit.)
- **10.2.3.** If G is connected and every vertex has positive even degree, then G has an Euler circuit.
- **10.2.4.** G has an Euler circuit **iff** G is connected and every vertex has positive even degree.
- **10.2.5.** There is an Euler trail from v to w iff G is connected, v and w have odd degree, and every other vertex has positive even degree.
- **10.2.6.** If G has a Hamiltonian circuit, then G has a subgraph H that (1) contains every vertex of G, (2) is connected, (3) has as many edges as vertices, and (4) has every vertex of degree 2.

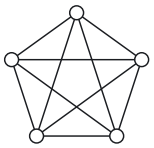
There is no simple criterion for Hamiltonian circuits. To show none exists, use the contrapositive of 10.2.6.

PROOF **Theorem 10.2.2 · An Euler circuit forces every degree to be positive and even**

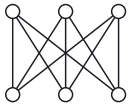
1. Let C be an Euler circuit of G and v any vertex. C contains v, so $\deg(v) > 0$.
2. Follow C once around. Each time it passes through v it arrives on one edge and leaves on another, using two edge-ends at v; if v is the starting vertex, the first departure pairs with the final arrival.
3. C uses every edge exactly once, so the edge-ends at v are split into pairs. Hence $\deg(v)$ is even. ■

WORKED EXAMPLE **Euler and Hamilton · Königsberg · T10 Q7 · AY25 final Q25(b)–(d)**

- **Königsberg** (4 land masses, 7 bridges): degrees $5, 3, 3, 3$. No Euler circuit; no Euler trail either, since four vertices are odd.
- "Every vertex has positive even degree" alone is not enough: two disjoint triangles are not connected.
- $\boldsymbol{K_n}$ $(n \geq 3)$ is Hamiltonian; it is Eulerian iff n is odd.
- $\boldsymbol{K_{m,n}}$ $(m, n \geq 2)$ is Eulerian iff m and n are both even; it is Hamiltonian iff $m = n$ (a circuit alternates between the two sides).
- **Cycle graph** $\boldsymbol{C_n}$ is bipartite iff n is even: C_{12} and C_{8888} are; C_{57}, C_{123}, C_{1231}, C_{12345} are not.
- **Friendship graph** $\boldsymbol{F_n}$ (n triangles sharing a centre, T10 Q7): Eulerian, since the degrees are 2 and $2n$. Not Hamiltonian for $n \geq 2$: passing from one triangle to another repeats the centre.



K_5



$K_{3,3}$

PAST PAPER **AY25 final Q9 · Relations from walks**

On the vertex set of a graph let xRy iff there is a walk from x to y.

Undirected: R is an equivalence relation (reflexive by the trivial walk; symmetric by reversing a walk; transitive by joining walks). Its classes are the vertex sets of the connected components.

Directed: R need not be a partial order. Even with no pair of opposite edges, the directed cycle $a \to b \to c \to a$ gives aRb and bRa with $a \neq b$, so antisymmetry fails. Only (i) is true.

48 · Adjacency matrices and counting walks

DEFINITION **Adjacency matrix (L12 s61–62) · Matrix product (L12 s64) · Identity, powers (L12 s69–70)**

For a graph G with ordered vertices $v_1, \ldots, v_n$, the **adjacency matrix** is the $n \times n$ matrix $A = (a_{ij})$ over the non-negative integers with

- **directed:** a_{ij} = the number of arrows from v_i to v_j;
- **undirected:** a_{ij} = the number of edges connecting v_i and v_j.

The matrix of an undirected graph is **symmetric**: $a_{ij} = a_{ji}$.

For an $m \times k$ matrix A and a $k \times n$ matrix B, the **product** $AB = (c_{ij})$ has $c_{ij} = \sum_{r=1}^{k} a_{ir} b_{rj}$ (row i of A times column j of B). Matrix multiplication is associative but **not commutative**.

I_n is the $n \times n$ **identity matrix** (1 on the main diagonal, 0 elsewhere). **Powers:** $A^0 = I$ and $A^n = AA^{n-1}$ for $n \geq 1$.

THEOREM **Theorem 10.3.2 · Counting walks (L12 s74)**

If A is the adjacency matrix of a graph with vertices $v_1, \ldots, v_m$, then for every positive integer n and all i, j:

the ij-th entry of A^n = the number of walks of length n from v_i to v_j.

WORKED EXAMPLE **L12 s72–73 · Walks of length 2**

G has vertices v_1, v_2, v_3 and edges $e_1 = \{v_1, v_2\}$, a loop e_2 at v_2, and parallel edges $e_3, e_4 = \{v_2, v_3\}$.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 6 & 2 \\ 2 & 2 & 4 \end{pmatrix}.$$

Entry $(2, 2)$ of A^2 is $1 \cdot 1 + 1 \cdot 1 + 2 \cdot 2 = 6$: one walk via v_1 ($v_2 e_1 v_1 e_1 v_2$), one via the loop ($v_2 e_2 v_2 e_2 v_2$), and four via v_3 (choose e_3 or e_4 each way). A walk may repeat edges, and different parallel edges give different walks.

Reading a matrix: a non-zero diagonal entry is a loop; an entry greater than 1 means parallel edges; in a simple graph the row sum of v_i is $\deg(v_i)$, and the diagonal of A^2 lists the degrees.

49 · Isomorphism and planar graphs

DEFINITION **Isomorphic graphs (L12 s79)**
$G = (V_G, E_G)$ is **isomorphic** to $G' = (V_{G'}, E_{G'})$, written $G \cong G'$, iff there are bijections $g : V_G \to V_{G'}$ and $h : E_G \to E_{G'}$ that preserve the edge-endpoint functions: for all $v \in V_G$ and $e \in E_G$,

$$v \text{ is an endpoint of } e \Leftrightarrow g(v) \text{ is an endpoint of } h(e).$$

For simple graphs: $G \cong G'$ iff there is a bijection $\pi : V_G \to V_{G'}$ such that $\{u, v\} \in E_G \Leftrightarrow \{\pi(u), \pi(v)\} \in E_{G'}$.

THEOREM **Theorem 10.4.1 (L12 s81) · Invariants**
Graph isomorphism is an equivalence relation on any set of graphs. (Reflexive: identity maps. Symmetric: inverse bijections. Transitive: compose the bijections.)
Isomorphic graphs agree on every structural property: number of vertices, number of edges, the list of degrees, number of connected components, number of circuits of each length, being bipartite, being Eulerian or Hamiltonian.

METHOD **Isomorphic or not?**
To prove isomorphic: write the bijection on vertices explicitly and check that every edge maps to an edge (and the edge counts agree).
To prove not isomorphic: name one invariant that differs. Compare in this order: vertices, edges, sorted degree list, then circuits of length 3 or 4, then what the vertices of a given degree are adjacent to. Equal degree lists do **not** prove isomorphism.

DEFINITION **Planar graph (L12 s82) · Subdivision (L12 s84)**
A **planar graph** is a graph that **can be** drawn on a plane without edges crossing. One drawing with crossings does not show a graph is non-planar. A planar drawing divides the plane into **faces** (regions), including the outer one.
A **subdivision** of G is obtained by inserting new vertices into edges of G (replace an edge by a path through a new vertex).

THEOREM **Kuratowski's theorem (L12 s83) · Euler's formula (L12 s86)**
- K_5 and $K_{3,3}$ are not planar. A finite graph is planar **iff** it has no subgraph that is a subdivision of K_5 or of $K_{3,3}$.
- For a **connected planar** simple graph with v vertices, e edges and f faces: $f = e - v + 2$.

Examples: K_4 is planar with $v = 4$, $e = 6$, $f = 4$. $K_{3,4}$ is not planar, since it contains $K_{3,3}$ (AY25 Q25(a)). A tree has $f = (v - 1) - v + 2 = 1$ face.

PAST PAPER **T11 Q1 · Self-complementary graphs on $n = 3, 4, 5, 6$ vertices**
If $G \cong \overline{G}$ then G and $\overline{G}$ have the same number of edges, so $\binom{n}{2}$ is even. For $n = 3$ and $n = 6$, $\binom{n}{2} = 3$ and 15: **none**.
$n = 4$ (3 edges): the path on 4 vertices, and only that. $n = 5$ (5 edges): the cycle C_5, and the "bull" (a triangle with a pendant edge at two of its vertices): **two**.

50 · Trees

DEFINITION **Circuit-free, tree, trivial tree, forest (L13 s4) · Leaf, internal vertex (L13 s9)**
A graph is **circuit-free** iff it has no circuits. A simple graph is a **tree** iff it is **circuit-free and connected**. A **trivial tree** is a single vertex. A simple graph is a **forest** iff it is circuit-free and not connected.
In a tree with one or two vertices each vertex is a **terminal vertex** (**leaf**). In a tree with at least three vertices, a vertex of degree 1 is a **leaf** and a vertex of degree greater than 1 is an **internal vertex**.

THEOREM **Lemma 10.5.1 · Theorem 10.5.2 · Lemma 10.5.3 · Theorem 10.5.4 · Lemma 10.5.5 (L13 s7–15, T11)**

- **10.5.1.** Any non-trivial tree has at least one vertex of degree 1 (in fact at least two).
- **10.5.2.** Any tree with n vertices ($n > 0$) has $n - 1$ edges.
- **10.5.3.** If G is connected and C is a circuit in G, removing one edge of C leaves G connected.
- **10.5.4.** If G is a **connected** graph with n vertices and $n - 1$ edges, then G is a tree.
- **10.5.5 (T11).** In a simple graph, if there are two distinct paths from v to a different vertex w, then G contains a cycle.

Connectedness is needed in 10.5.4: a triangle with a pendant edge, together with one isolated vertex, has 5 vertices and 4 edges but is not a tree (L13 s17).

PROOF **Lemma 10.5.1 (L13 s7) · Theorem 10.5.2 by induction (L13 s10–11) · At least two leaves (L13 s12)**

10.5.1. Start at any vertex v and walk along edges, never reusing an edge. A vertex cannot be revisited *(T is circuit-free)*, and T has finitely many vertices, so the walk stops at a vertex with no unused edge: a vertex of degree 1.

10.5.2. Let $P(n) \equiv$ "any tree with n vertices has $n - 1$ edges".

1. $P(1)$: a tree with one vertex has no edges *(a simple graph has no loops)*.
2. Assume $P(k)$ for an arbitrary $k \geq 1$. Let T be any tree with $k + 1$ vertices.
3. T is non-trivial, so it has a vertex v of degree 1 *(Lemma 10.5.1)* let e be the edge at v.
4. Let T' be T with v and e removed. T' has k vertices, is circuit-free (a subgraph of T) and is connected (no path between other vertices passes through v, as $\deg(v) = 1$). So T' is a tree.
5. T' has $k - 1$ edges *(inductive hypothesis)*, so T has $k = (k + 1) - 1$ edges. Thus $P(k + 1)$ is true. ■

Two leaves. Let T be a tree with $n \geq 2$ vertices. Every degree is at least 1 *(connected)*. If at most one vertex had degree 1, the total degree would be at least $1 + 2(n - 1) = 2n - 1$; but it equals $2(n - 1) = 2n - 2$ *(Theorem 10.5.2 and the handshake theorem)*. Contradiction. ■

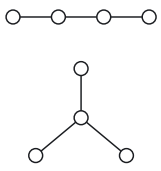
PROOF **Theorem 10.5.4 · A connected graph with n vertices and $n - 1$ edges is a tree (L13 s15–16)**

1. Let G be connected with n vertices and $n - 1$ edges. It suffices to show that G is circuit-free.
2. Suppose G has a circuit. Remove an edge of it; the graph stays connected *(Lemma 10.5.3)*. Repeat while a circuit remains.
3. The result G'' is connected and circuit-free, so it is a tree with n vertices and therefore $n - 1$ edges *(Theorem 10.5.2)*.
4. But at least one edge was removed, so G'' has at most $n - 2$ edges: contradiction.
5. Hence G is circuit-free, and G is a tree. ■

WORKED EXAMPLE **L13 s13 · T11 Q3 · Trees on few vertices**

A tree with 4 vertices has 3 edges and total degree 6, with at least two leaves: the degree lists are $1, 1, 2, 2$ (a path) and $1, 1, 1, 3$ (a star). So there are **two** non-isomorphic trees with 4 vertices.

Labelled copies (T11 Q3): $n = 1$: 1; $n = 2$: 1; $n = 3$: one tree (the path), 3 copies (choose the middle vertex); $n = 4$: the path has $4!/2 = 12$ copies and the star has 4 (choose the centre), 16 in all. In general there are n^{n-2} labelled trees.



PROOF **T11 Q4–Q6 · Edge bounds and the unique-path characterization**

Let $G = (V, E)$ be a simple graph.

1. **Connected ⇒ $|E| \geq |V| - 1$.** While G has a circuit, remove one of its edges; G stays connected *(Lemma 10.5.3)*. The process ends with a connected circuit-free graph on V, a tree with $|V| - 1$ edges *(Theorem 10.5.2)*, all of which are edges of G. **Converse false:** a triangle plus an isolated vertex has $3 \geq 4 - 1$ edges and is not connected.
2. **Acyclic ⇒ $|E| \leq |V| - 1$.** Each connected component is a tree. With $c \geq 1$ components of sizes $n_1, \ldots, n_c$, $|E| = \sum(n_i - 1) = |V| - c \leq |V| - 1$. **Converse false:** the same graph has $3 \leq 3$ edges and contains a cycle.
3. **G is a tree iff there is exactly one path between every pair of distinct vertices.**
   ($\Rightarrow$) G is connected, so any two distinct vertices are joined by a path *(Lemma 10.2.1(a))*. Two distinct paths between them would give a cycle *(Lemma 10.5.5)*, contradicting circuit-freeness.
   ($\Leftarrow$) A path between every pair makes G connected. If G had a circuit, two vertices on it would be joined by two distinct paths (the two ways round), contradicting uniqueness. So G is circuit-free and connected. ■

PAST PAPER **T11 Q7 (AY17/18 exam) · Splitting a pile of n stones**
Splitting a pile of k into k_1 and k_2 earns k_1k_2 dollars. Model the stones as the vertices of K_n, where an edge means "still in the same pile". A split into k_1 and k_2 deletes exactly the k_1k_2 edges between the two parts and earns one dollar per deleted edge. At the end every stone is alone, so all $\binom{n}{2}$ edges have been deleted exactly once. The total is $\boldsymbol{n(n-1)/2}$ **whatever the strategy**.

51 · Rooted trees and binary trees

DEFINITION **Rooted tree, level, height · Child, parent, sibling, ancestor, descendant (L13 s19)**
A **rooted tree** is a tree with one distinguished vertex, the **root**. The **level** of a vertex is the number of edges on the unique path between it and the root (the root has level 0). The **height** of a rooted tree is the maximum level of any vertex.
The **children** of v are the vertices adjacent to v and one level farther from the root; then v is their **parent**; distinct vertices with the same parent are **siblings**. If v lies on the unique path between w and the root ($v \neq w$), then v is an **ancestor** of w and w a **descendant** of v.

DEFINITION **Binary tree, full binary tree · Left and right subtree (L13 s22)**
A **binary tree** is a rooted tree in which every parent has at most two children, each designated a **left child** or a **right child** (not both), with at most one of each. A **full binary tree** is a binary tree in which each parent has **exactly two** children.
The **left subtree** of a parent v is the binary tree whose root is the left child of v and whose vertices are that child and all its descendants; the **right subtree** is defined analogously.

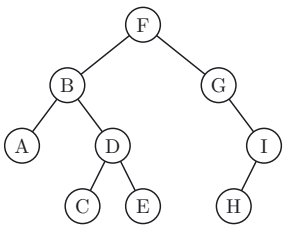
THEOREM **Theorem 10.6.1 Full binary tree theorem (L13 s27) · Theorem 10.6.2 (L13 s29)**
- **10.6.1.** A full binary tree with k internal vertices has $2k+1$ vertices in total and $k+1$ leaves.
- **10.6.2.** A binary tree of height h with t leaves satisfies $t \leq 2^h$; equivalently $\log_2 t \leq h$.

Hence: no full binary tree has 10 internal vertices and 13 leaves (it would have 11 leaves); no binary tree of height 5 has 38 leaves ($38 > 32$). A binary tree of height h has at most $2^{h+1}-1$ vertices.

PROOF **Theorem 10.6.1 (L13 s27) · Theorem 10.6.2 by strong induction (L13 s30–34)**
10.6.1. Every vertex except the root has a parent. Each of the k internal vertices has exactly two children, so $2k$ vertices have a parent. Total: $2k+1$. Leaves: $(2k+1)-k = k+1$. ■
10.6.2. Let $P(h) \equiv$ "every binary tree of height h has at most 2^h leaves".
1. $P(0)$: the tree is a single vertex, a leaf: $t = 1 = 2^0$.
2. Let $k \geq 0$ and assume $P(i)$ for $0 \leq i \leq k$. Let T have height $k+1$, root v and t leaves. Since $k+1 \geq 1$, v has a child.
3. Case 1 (v has one child): its subtree T_L has height k, so $t_L \leq 2^k$ *(inductive hypothesis)*. In the slides the root, having degree 1, is also counted, so $t = t_L + 1 \leq 2^k + 1 \leq 2^k + 2^k = 2^{k+1}$ *($2^k \geq 1$)*.
4. Case 2 (v has two children): the subtrees T_L, T_R have heights $h_L, h_R \leq k$, so $t = t_L + t_R \leq 2^{h_L} + 2^{h_R} \leq 2^k + 2^k = 2^{k+1}$ *(inductive hypothesis)*.
5. In both cases $P(k+1)$ is true. ■

DEFINITION **Tree traversal: BFS and DFS (L13 s36–39)**
Breadth-first search: visit the root, then the vertices level by level, left to right.
Depth-first search, three orders, each applied recursively:
- **Pre-order:** root, left subtree, right subtree.
- **In-order:** left subtree, root, right subtree.
- **Post-order:** left subtree, right subtree, root.

For the tree on the right (L13 s40):
BFS: $F, B, G, A, D, I, C, E, H$
pre-order: $F, B, A, D, C, E, G, I, H$
in-order: $A, B, C, D, E, F, G, H, I$
post-order: $A, C, E, D, B, H, I, G, F$



METHOD **Rebuilding a binary tree from two traversals**
Pre-order + in-order (or **post-order + in-order**) determine the tree. The root is the **first** item of the pre-order (the **last** of the post-order). Find it in the in-order: everything to its left is the left subtree, everything to its right is the right subtree. Split the other traversal into blocks of the same sizes and recurse.
Pre-order + post-order do not determine the tree: a vertex with exactly one child could have it on the left or the right, so there are 2^s trees, where s is the number of one-child vertices.
Expression trees (L13 s24–26): operators at internal vertices, operands at leaves; $((a-b)\cdot c)+(d/e)$ has root $+$.

PAST PAPER **T11 Q8, Q9, D3 · AY25 final Q19, Q21, Q22**
T11 Q8(a). In-order E A C K F H D B G, pre-order F A E K C D H G B. Root F; left part E A C K, right part H D B G. Left subtree: root A, left child E, right child K with left child C. Right subtree: root D, left child H, right child G with left child B. **Post-order: E, C, K, A, H, B, G, D, F.**
T11 Q8(b). In-order D B F E A G C H K, post-order D F E B G K H C A. Root A; left subtree: root B, left child D, right child E with left child F; right subtree: root C, left child G, right child H with right child K. **Pre-order: A, B, D, E, F, C, G, H, K.**
T11 Q9. Binary trees with in-order X Y Z: **5**; with in-order A B C D: **14** (the Catalan numbers: choose the root, then the shapes of the two sides).
AY25 Q21. Pre-order G B E D F C A, in-order B G F D C E A: root G with left child B; right subtree has root E, left child D (children F and C), right child A. Post-order: B, F, C, D, A, E, G.
AY25 Q22. Pre-order A B C D E, post-order C E D B A: A has the single child B; B has children C and D; D has the single child E. Two one-child vertices: $2^2=4$ trees.
AY25 Q19. Unlabelled rooted binary trees, left and right not distinguished: 2 with 3 vertices, 3 with 4 vertices, **6** with 5 vertices.

52 · Spanning trees and minimum spanning trees

DEFINITION **Spanning tree (L13 s45) · Weighted graph, minimum spanning tree (L13 s48)**
A **spanning tree** for a graph G is a subgraph of G that contains every vertex of G and is a tree.
A **weighted graph** is a graph in which each edge e has a positive real **weight** $w(e)$; the **total weight** $w(G)$ is the sum of the weights of all edges. A **minimum spanning tree** (MST) for a connected weighted graph is a spanning tree with the least possible total weight.

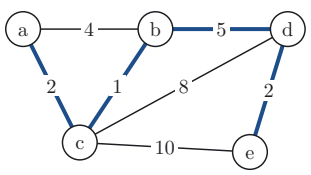
THEOREM **Proposition 10.7.1 (L13 s45)**
(1) Every connected graph has a spanning tree. (2) Any two spanning trees for a graph have the same number of edges ($n-1$ for n vertices).
Counting spanning trees: if G has exactly one circuit, of length k, removing any one of its edges gives a spanning tree: k spanning trees (L13 s46: 3). K_n has n^{n-2}.

METHOD **Algorithm 10.7.1 · Kruskal (L13 s50)**
Input: a connected weighted graph G with n vertices.
1. Initialize T with all vertices of G and no edges.
2. Let E be the set of all edges of G; let $m=0$.
3. While $m<n-1$:
   3.1. find an edge e in E of least weight;
   3.2. delete e from E;
   3.3. if adding e to T does not produce a circuit, add e to T and set $m=m+1$.

Output: T, a minimum spanning tree for G.

METHOD **Algorithm 10.7.2 · Prim (L13 s55)**
Input: a connected weighted graph G with n vertices.
1. Pick a vertex v; let T be the graph with this vertex only.
2. Let V be the set of all vertices of G except v.
3. For $i=1$ to $n-1$:
   3.1. find an edge e of least weight among those connecting T to a vertex in V; let w be its endpoint in V;
   3.2. add e and w to T, and delete w from V.

Output: T, a minimum spanning tree for G.



WORKED EXAMPLE **Kruskal and Prim on one graph**
Edges: $ab=4$, $ac=2$, $bc=1$, $bd=5$, $cd=8$, $ce=10$, $de=2$.
Kruskal: bc (1) added; ac (2) added; de (2) added; ab (4) rejected, since it closes the circuit abc; bd (5) added. Four edges: stop.
Prim from a: add ac (2), then cb (1), then bd (5), then de (2).

Both give the same tree, of weight $1+2+2+5=10$, with the edges added in different orders.

WORKED EXAMPLE **L13 s52, s57 · The airline graph (8 cities)**
Kruskal: Chi–Mil 74, Lou–Cin 83, Lou–Nas 151, Cin–Det 230, StL–Lou 242, StL–Chi 262 added; Chi–Lou 269, Lou–Det 306, Lou–Mil 348 **not added** (each would close a circuit); Min–Chi 355 added. Total weight **1397**.
Prim from Minneapolis: Min–Chi 355, Chi–Mil 74, Chi–StL 262, StL–Lou 242, Lou–Cin 83, Lou–Nas 151, Cin–Det 230. The same tree, total 1397.

TRAP **Minimum spanning trees**
- Kruskal may hold a **disconnected** forest midway; Prim's partial tree is always connected. Marking a Prim step that does not touch the current tree is an error.
- With repeated weights there may be several MSTs, but all have the same total weight; with all weights distinct the MST is unique. To count MSTs, look at each tie and see which choices are interchangeable.
- An MST is not a shortest-path tree. Dijkstra's algorithm (L13 s59 onward) is not examinable this semester.
- Stop after $n-1$ edges. If a question asks for the order of edges added, follow the named algorithm exactly.

Part XI · Exam Playbook and Locators

METHOD **Final exam format and approach (AY25/26 paper)**
Format: 2 hours, open book, calculators allowed, 100 marks. Part A: 23 MCQ at 2 marks each (46). Part B: three structured questions (54), recently Counting and Probability, Graphs and Trees, and Functions; many parts need the answer only, "as a single number".
- **Coverage is the whole module.** Roughly the first third of Part A is logic, sets and relations (§1–25); the rest is functions, induction and recursion, cardinality, counting, graphs and trees (§26–52).
- **New definitions in the question** (matchings, dominating sets, order of a function, left inverses) are routine: copy the definition onto scratch paper and test it on the smallest example.
- **Counting answers:** decide ordered or unordered, repetition or not (table in §56), then check the formula on a case small enough to list.
- **"Single number" answers** get no method marks: recompute by a second route (difference rule against direct cases).
- **Proof parts:** use the class format for induction (§32), the injective/surjective templates (§28), and name the theorem used at each step.

53 · Part A: multiple-choice questions

METHOD **Approach**
- **Judge each of (i)–(iv) on its own** and mark T/F in the margin before reading the options. Several options usually differ in only one item. "None of (A)–(D)" has been the correct answer several times (AY23 Q2, AY24 Q12, AY25 Q2).
- **"For all sets / all relations" claims:** look for a counterexample among $\varnothing$, a singleton, $\{a, b\}$, the empty relation and the identity relation. One counterexample is enough.
- **Relations on small sets:** draw the digraph. **Posets:** draw the Hasse diagram first, then read off the answers.
- **Nested $\varnothing$ / power sets:** write each layer out explicitly, especially for $\mathcal{P}$ of a set containing $\varnothing$.
- **Counting questions:** use the table in §55, or enumerate the $n=2$ case to test an option.

Before choosing an answer: check quantifier connective and domain (§5–7), membership versus subset (§14), empty sets in products (§15), composition order (§20), antisymmetry (§21), class versus quotient (§23), and maximal versus largest (§24), setwise preimage versus inverse (§26), co-domain versus range (§28), basis value of an induction (§32), ordered versus unordered and repetition (§42), $P(A \mid B)$ versus $P(B \mid A)$ (§45), trail versus path and Euler versus Hamilton (§47).

54 · Part B: written answers

Computations: give complete set-roster answers; use "None" when appropriate. For a quotient, choose one representative per distinct class. **Proofs:** state the claim, unpack definitions, number and justify each step, then conclude (§9). For "prove or disprove", state the verdict and supply a proof or one explicit counterexample.

55 · Counting facts

Relations on an n-element set	General	$n=2$	$n=3$
all relations	2^{n^2}	16	512
reflexive	2^{n^2-n}	4	64
symmetric	$2^{n(n+1)/2}$	8	64
reflexive and symmetric	$2^{n(n-1)/2}$	2	8
antisymmetric	$2^n \cdot 3^{n(n-1)/2}$	12	216
asymmetric	$3^{n(n-1)/2}$	3	27
equivalence relations (Bell)	B_n	2	5
partial orders	—	3	19
total orders	$n!$	2	6

Derivation. Each of the n^2 pairs is either in the relation or not; reflexivity fixes the n diagonal pairs; symmetry decides each unordered pair once; antisymmetry allows 3 of the 4 states of each off-diagonal pair; asymmetry also forbids loops. For $n=4$ there are 15 equivalence relations, 219 partial orders and 24 total orders. **AY25 Q4:** on a three-element set there are **5** equivalence relations. **AY25 Q12:** on $\{0,1\}$ there are 2 equivalence relations and 3 partial orders, so $|F|+1=|G|$. Answer B.

Other counts	Value
$\lvert A \times B\rvert$	$\lvert A\rvert \cdot \lvert B\rvert$
$\lvert\mathcal{P}(A)\rvert$	$2^{\lvert A\rvert}$
relations from A to B	$2^{\lvert A\rvert\ \lvert B\rvert}$
partitions of an n-set	B_n : $1, 2, 5, 15, 52$

56 · Formula sheet for Lectures 7–13

Counting	Value
integers from m to n	$n-m+1$
ordered, repetition allowed	n^k
ordered, no repetition	$P(n,k)=n!/(n-k)!$
unordered, no repetition	$\binom{n}{k}=n!/(k!(n-k)!)$
unordered, repetition allowed	$\binom{k+n-1}{k}$
orderings with repeated types	$n!/(n_1!\cdots n_k!)$
circular arrangements of n	$(n-1)!$
$x_1+\cdots+x_n=r$, $x_i\geq 0$	$\binom{r+n-1}{r}$
same, $x_i\geq 1$	$\binom{r-1}{n-1}$
functions $A\to B$, $\lvert A\rvert=m$, $\lvert B\rvert=n$	n^m
injections; bijections ($m=n$)	$P(n,m)$; $n!$
subsets of an n-set	2^n
pigeons: some hole has at least	$\lceil n/m\rceil$

Sums	Closed form
$1+2+\cdots+n$	$n(n+1)/2$
$1^2+2^2+\cdots+n^2$	$n(n+1)(2n+1)/6$
$\sum_{i=0}^{n} r^i$ $(r\neq 1)$	$(r^{n+1}-1)/(r-1)$
$\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$	$(a+b)^n$
$\sum_{k=0}^{n}\binom{n}{k}$	2^n

Probability	Formula
equally likely outcomes	$P(E)=\lvert E\rvert\ /\ \lvert S\rvert$
complement	$P\left(\overline{A}\right)=1-P(A)$
union	$P(A)+P(B)-P(A\cap B)$
conditional	$P(B\mid A)=P(A\cap B)/P(A)$
independent	$P(A\cap B)=P(A)P(B)$
expected value	$\sum a_k p_k$
linearity	$E[X+Y]=E[X]+E[Y]$
binomial	$\binom{n}{k}p^k(1-p)^{n-k}$

Graphs and trees	Fact
total degree	$2\ \lvert E\rvert$
edges of K_n; of $K_{m,n}$	$n(n-1)/2$; mn
simple graphs on n labelled vertices	$2^{n(n-1)/2}$
Euler circuit	connected, all degrees even
Euler trail v to w	connected, only v, w odd
walks of length n from v_i to v_j	entry ij of A^n
connected planar	$f=e-v+2$
tree on n vertices	$n-1$ edges, at least 2 leaves
full binary tree, k internal	$2k+1$ vertices, $k+1$ leaves
binary tree of height h	at most 2^h leaves
labelled trees on n vertices	n^{n-2}

Function properties under composition (§29). f, g injective $\Rightarrow$ $g\circ f$ injective; f, g surjective $\Rightarrow$ $g\circ f$ surjective; $g\circ f$ injective $\Rightarrow$ f injective; $g\circ f$ surjective $\Rightarrow$ g surjective; $(g\circ f)^{-1}=f^{-1}\circ g^{-1}$. **Finite sets (§35):** injection $A\to B$ gives $|A|\leq|B|$; surjection gives $|A|\geq|B|$; bijection gives $|A|=|B|$. For $|A|=|B|$ finite, injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective.

57 · Question locators

Proof methods: §9–12; set inclusion/equality: §17; relation properties: §21. Each worked example is beside its topic.

Past-paper question	Solved in
AY23 Q2, Q4, Q5, Q16	§6, §4, §7, §7
AY23 Q3, Q6, Q7, Q8, Q17	§2, §18, §16, §14, §18
AY23 Q9, Q10, Q12, Q13, Q14, Q15, Q18, Q19	§21, §21, §23, §24, §24, §25, §25, §23
AY24 Q2, Q3, Q4, Q5	§5, §2, §2, §4
AY24 Q6, Q7, Q8, Q16	§15, §18, §18, §18

AY24 Q9, Q10, Q11, Q12, Q13–15, Q17	§23, §21, §20, §20, §25, §20–§25
AY25 Q1, Q2, Q4, Q6, Q7, Q8, Q9, Q10, Q12	§18, §6, §55, §8, §18, §18, §19, §19, §55
AY25 Q11, Q13, Q14, Q15, Q16, Q17	§21, §4, §21, §23, §4; Q17(a)–(e): §20, §13, §23, §23, §25

Tutorial question	Topic	Solved in
T1 Q11	Critique a biconditional proof; contraposition	§9
T2 Q4	Closure under division; rational addition	§11, §12
T2 Q8, Q11	Cases inside a conditional; factor bound	§11
T3 Q3(c)	Cartesian product versus power set	§15
T3 Q4–6	Double inclusion; element method; set identities	§17
T3 Q7	$A \subseteq B \leftrightarrow A \cup B = B$	§18
T4 Q2	Symmetry and the inverse relation	§21
T4 Q6, Q7, Q9	Repeated composition; associativity; parity relation	§20
T4 Q8, Q10	Classes of pairs; equivalence by sign	§23
T5 Q1	Antisymmetry counterexample on strings	§24
T5 Q6	The three conditions defining a closure	§22
T5 Q7	Asymmetric implies antisymmetric	§21

58 · Question locators for Tutorials 6–11 and the final

Tutorial question	Topic	Solved in
T6 D1, D3, D4; Q1, Q2	Is it a function; equality; a wrong proof; images and preimages	§26, §27, §29, §26
T6 Q3, Q6	Inverse of a composition; $g \circ f$ injective	§29
T6 Q4, Q5	Classify four functions; left and right inverses	§28
T6 Q7, Q8; T8 Q10(c)	Order of a bijection and of an input	§29
T6 Q9, Q10; T8 Q10(a), (b)	$f^{-1}(f(X))$; well-defined operations on a quotient	§26, §30
T7 Q1, D2, D3	Mistakes in induction proofs	§32
T7 Q2, Q3, Q4	Sum of squares; Bernoulli; divisibility by 2^{n+2}	§32
T7 Q5, Q6, Q7	\$3 and \$5 notes; sums of powers of 2; three-term recurrence	§33
T7 Q8, Q9; T8 Q9	Fibonacci identities; Tower of Hanoi	§33
T7 Q10, Q11	Membership of recursively defined sets	§34
T8 D1, D2, D3; Q1	Countable or not; bijection $\mathbb{N} \to \mathbb{Z}$	§37, §38
T8 Q2, Q3, Q4, Q5	Unions of countable sets; limits of induction	§37
T8 Q6, Q7, Q8	$X \cup Y \to X$; $\mathcal{P}(A)$ uncountable; $\lvert A\rvert \leq \lvert R\rvert$	§38, §38, §35
T9 D1, D4; Q1, Q2, Q5	Replacement; birthdays; cards; cheque numbers; round tables	§45, §45, §39
T9 Q3, Q4, Q6, Q7	Inclusion/exclusion; consecutive boxes; committees; investments	§40, §42
T9 Q8, Q9; T10 Q10, Q12	Pigeonhole arguments	§41
T10 D1, Q3, Q4; Q5, Q6	Expected value; Bayes; random relations	§44, §45
T10 D2, D3, Q1, Q2	Counting functions; binomial term; choosing block ends	§43, §42
T10 Q7, Q8; T11 D1, D2	Friendship graph; small graphs; edge counts	§47, §46
T11 Q1, Q3, Q4–Q7	Self-complementary graphs; trees; edge bounds; stones	§49, §50
T11 D3, Q8, Q9	Traversals and rebuilding binary trees	§51

AY25/26 final question	Solved in
Q9; Q10; Q11, Q12, Q26(b)	§47; §43; §29
Q13, Q17, Q18; Q14–Q16	§38; §34
Q19, Q21, Q22; Q23	§51; §44
Q24(a)–(e), (g); Q24(f); Q24(h)	§42; §41; §45
Q25(a); Q25(b)–(d)	§49; §47

The folder has no official answers for the AY25/26 final; those solutions are worked here.

59 · Self-test: true or false?

Decide each statement before reading the reason.

Statement		Reason
$\sim(p \leftrightarrow q) \equiv \sim p \leftrightarrow q$	**T**	both are true exactly when p, q differ
$(p \vee q) \to r \equiv (p \to r) \vee (q \to r)$	**F**	it is $\wedge$; $p = T, q = F, r = F$ separates them
$\exists y \in \mathbb{Z}\, \forall x \in \mathbb{Z}\, (x < y)$	**F**	take $x = y$
$\exists x \in \varnothing\, P(x)$	**F**	no witness exists; $\forall x \in \varnothing\, P(x)$ is **T**
$\varnothing \in \{\varnothing\}$ and $\varnothing \subseteq \{\varnothing\}$	**T**	listed element; Theorem 6.2.4
$\{\varnothing\} \subseteq \varnothing$	**F**	$\varnothing \in \{\varnothing\}$ but $\varnothing \notin \varnothing$
$\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$	**T**	$X \subseteq A \cap B \leftrightarrow X \subseteq A \wedge X \subseteq B$
$A \times B = B \times A \Rightarrow A = B$	**F**	$A = \varnothing$, $B = \{1\}$: both products are $\varnothing$
a partition of a finite A has at most $\lvert A\rvert$ components	**T**	components are non-empty and disjoint
symmetric and transitive $\Rightarrow$ reflexive	**F**	the empty relation on $\{a\}$
$R \circ R^{-1}$ is always symmetric	**T**	$(R \circ R^{-1})^{-1} = (R^{-1})^{-1} \circ R^{-1}$
the inverse of a partial order is a partial order	**T**	each property survives reversing pairs
the union of two equivalence relations is one	**F**	$1 \sim 2$ and $2 \sim 3$ from different relations
the intersection of two equivalence relations is one	**T**	each property is preserved by $\cap$
a unique minimal element is smallest, in any poset	**F**	true only for finite posets (§24)
$(\mathbb{Z}^+, \mid)$ has a smallest element	**T**	$1 \mid n$ for all n
there are 4 relations on a one-element set	**F**	$2^{1^2} = 2$: $\varnothing$ and $\{(a, a)\}$
$a \mid b \wedge a \mid c \Rightarrow a \mid (b - c)$	**T**	$b - c = a(r - s)$

60 · Self-test for Lectures 7–13

Decide each statement before reading the reason.

Statement		Reason
$f^{-1}(B)$ is defined only when f is a bijection	**F**	the setwise preimage exists for every function (§26)
$f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 3x + 1$ is a bijection	**F**	injective, but 0 has no preimage
$g \circ f$ injective $\Rightarrow$ g injective	**F**	only f must be injective (§29)
the composition of two bijections is a bijection	**T**	Theorems 7.3.3 and 7.3.4
$f : \mathbb{Z} \to \mathbb{Z}$ and $g : \mathbb{Z} \to \mathbb{Q}$ with the same formula are equal	**F**	different co-domains (Theorem 7.1.1)
$X = f^{-1}(f(X))$ for every $X \subseteq A$	**F**	only $\subseteq$; equality needs injectivity
$[x]_3 \mapsto [x]_2$ is a function $\mathbb{Z}_3 \to \mathbb{Z}_2$	**F**	$[0]_3 = [3]_3$ but $[0]_2 \neq [3]_2$
$2n + 1 < 2^n$ for all $n \geq 1$	**F**	fails at $n = 1, 2$; true from $n = 3$
strong induction proves more statements than weak induction	**F**	1PI, 2PI and WOP are equivalent
every non-empty subset of $\mathbb{Z}$ has a least element	**F**	WOP is about $\mathbb{Z}_{\geq 0}$; take $\mathbb{Z}^-$
an empty sum is 0 and an empty product is 1	**T**	conventions (§31)
$\lvert\mathbb{Z}\rvert = \lvert 2\mathbb{Z}\rvert$	**T**	$n \mapsto 2n$ is a bijection
every infinite subset of $\mathbb{R}$ is uncountable	**F**	$\mathbb{Z}$, $\mathbb{Q}$
every subset of $\mathbb{Q}$ is countable	**T**	Theorem 7.4.3
a countable union of finite sets can be infinite	**T**	$\bigcup\{i\} = \mathbb{Z}^+$; still countable
$\mathcal{P}(\mathbb{N})$ is countable	**F**	diagonal argument (§38)
$P(n, r) = \binom{n}{r} \cdot r!$	**T**	choose a subset, then order it
the number of multisets of size 3 from 4 elements is $\binom{4}{3}$	**F**	$\binom{6}{3} = 20$
$\binom{n}{r} = \binom{n}{n-r}$	**T**	choose which to leave out
disjoint events are independent	**F**	if both have positive probability they are dependent
$E[X + Y] = E[X] + E[Y]$ needs independence	**F**	linearity always holds
$P(A \mid B) = P(B \mid A)$	**F**	equal only if $P(A) = P(B)$ or $P(A \cap B) = 0$

a graph can have exactly three vertices of odd degree	**F**	Proposition 10.1.3
all degrees even ⇒ the graph has an Euler circuit	**F**	it must also be connected
every Eulerian graph is Hamiltonian	**F**	friendship graph F_2
a graph with n vertices and $n-1$ edges is a tree	**F**	only if connected (Theorem 10.5.4)
$K_{3,3}$ is planar	**F**	Kuratowski; K_5 is not planar either
equal degree lists ⇒ isomorphic	**F**	C_6 and two disjoint triangles
pre-order and post-order determine a binary tree	**F**	a one-child vertex is ambiguous (§51)
a full binary tree has an odd number of vertices	**T**	$2k+1$
all spanning trees of a graph have the same number of edges	**T**	Proposition 10.7.1

61 · Symbol table

Symbol	Read as	Symbol	Read as	Symbol	Read as
$\sim p$	not p	$p \wedge q$	p and q	$p \vee q$	p or q
$p \to q$	if p then q	$p \leftrightarrow q$	p iff q	$p \oplus q$	exclusive or
$P \equiv Q$	logically equivalent	$\therefore$	therefore	$\boldsymbol{t}, \boldsymbol{c}$	tautology, contradiction
$\forall$	for all	$\exists$	there exists	$\exists!$	there exists exactly one
$x \in A$	x is an element of A	$A \subseteq B$	A is a subset of B	$A \subsetneq B$	proper subset
$\varnothing$	empty set	$\lvert A \rvert$	cardinality	$\mathcal{P}(A)$	power set
$A \times B$	Cartesian product	$A \setminus B$	set difference	$\overline{A}$	complement
$A \oplus B$	symmetric difference	(a, b)	ordered pair	$\{x \in U : P(x)\}$	set-builder
$d \mid n$	d divides n	$d \nmid n$	d does not divide n	$a \equiv b \pmod n$	congruent mod n
xRy	$(x, y) \in R$	R^{-1}	inverse relation	$S \circ R$	R then S
R^r, R^s, R^t	closures	$[x]_\sim$	class of x	$A/\sim$	quotient: set of classes
$x \preccurlyeq y$	x precedes-or-equals y	$x \preccurlyeq^* y$	a linearization	$(A, \preccurlyeq)$	poset
$\mathbb{Z}^+, \mathbb{Z}_{\geq 0}$	positive / non-negative integers	$\mathbb{N}$	$\{0, 1, 2, ...\}$	n div d, n mod d	quotient, remainder

62 · Symbols for Lectures 7–13

Symbol	Read as	Symbol	Read as	Symbol	Read as
$f : X \to Y$	function from X to Y	$x \mapsto y$	x maps to y	$f(A)$	setwise image
$f^{-1}(B)$	setwise preimage	f^{-1}	inverse function	$g \circ f$	f then g
id_X	identity on X	A^*	strings over A	ε	empty string
$\mathbb{Z}_n$	$\mathbb{Z}$ mod n (L7); $\{1, ..., n\}$ (L9)	$\lfloor x \rfloor, \lceil x \rceil$	floor, ceiling	$n!$	n factorial
$\sum_{k=m}^{n} a_k$	sum of $a_m, ..., a_n$	$\prod_{k=m}^{n} a_k$	product	F_n	Fibonacci number
$\aleph_0$	$\lvert\mathbb{Z}^+\rvert$, aleph-zero	$\lvert A \rvert = \lvert B \rvert$	same cardinality	$P(n, r)$	r-permutations
$\binom{n}{r}$	n choose r	$P(E)$	probability of E	$P(B \mid A)$	B given A
$E[X]$	expected value	$G = (V, E)$	graph	$\deg(v)$	degree
$\deg^-(v), \deg^+(v)$	indegree, outdegree	K_n	complete graph	$K_{m,n}$	complete bipartite graph
$\overline{G}$	complement graph	$G \cong G'$	isomorphic	A^n	n-th matrix power
$w(e), w(G)$	edge weight, total weight	C_n	cycle graph	I_n	identity matrix

63 · Topic locator

§	Definitions / methods	§	Definitions / methods
1–3	Statements, connectives, equivalence, conditional variants	13–14	Sets, notation, intervals, subset, equality, empty set
4	Arguments, validity, soundness, inference rules	15–16	Pairs, products, set operations, disjointness
5–7	Predicates, truth sets, quantifiers, negation, vacuity	17–18	Set proofs, partitions, power sets
8	Quantified inference rules and worked negations	19–20	Relations, domain/range, inverse, composition
9	Proof vocabulary, format, assumptions, biconditionals	21–22	Relation properties and closures
10	Even/odd, prime, rational, divisibility, absolute value	23	Equivalence, classes, quotient, congruence, representatives
11–12	Proof methods and number-theory examples	24–25	Posets, extrema, total/well orders, Hasse, Kahn
26–27	Functions, images, preimages, sequences, strings, equality	28–30	Injective, surjective, inverse, composition, $\mathbb{Z}_n$
31–32	Sums, products, mathematical induction	33–34	Strong induction, well-ordering, recursion, structural induction
35–36	Pigeonhole, finite sets, same cardinality	37–38	Countable and uncountable sets, diagonalization
39–41	Multiplication, addition, inclusion/exclusion, pigeonhole	42–43	Combinations, multisets, Pascal, binomial theorem
44–45	Probability axioms, expectation, Bayes, independence	46–49	Graphs, degree, Euler, Hamilton, matrices, isomorphism, planarity
50–52	Trees, binary trees, traversals, spanning trees, MST	53–58	Exam methods, counting, formula sheet, question locators
59–62	Self-tests and symbols	64–65	Appendix A; slide-numbered theorem locator

Part XII · Appendix A and Theorem Locator

64 · Appendix A · Properties of the Real Numbers

Source: Appendix A, cited in Lecture 1, slide 26; Lecture 3, slide 81; and Lecture 4, slide 4. The complete appendix is reproduced here as a statement reference, including its final least upper bound axiom. The symbols a, b, c, d below denote arbitrary **real numbers**, subject to each stated condition.

Appendix A · Field axioms F1–F6

Addition and multiplication are binary operations on $\mathbb{R}$: if $a, b \in \mathbb{R}$, then $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.

Appendix A label	Statement
Axiom F1 · Commutative laws	$a + b = b + a$; $ab = ba$.
Axiom F2 · Associative laws	$(a + b) + c = a + (b + c)$; $(ab)c = a(bc)$.
Axiom F3 · Distributive laws	$a(b + c) = ab + ac$; $(b + c)a = ba + ca$.
Axiom F4 · Identity elements	There are distinct $0, 1 \in \mathbb{R}$ such that $0 + a = a + 0 = a$ and $1 \cdot a = a \cdot 1 = a$.
Axiom F5 · Additive inverses	For each a, there is $-a \in \mathbb{R}$ with $a + (-a) = (-a) + a = 0$.
Axiom F6 · Reciprocals	For each $a \neq 0$, there is $a^{-1} = 1/a \in \mathbb{R}$ with $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

Appendix A · Theorems T1–T16 (algebra)

Appendix A label	Statement
Theorem T1 · Addition cancellation	If $a + b = a + c$, then $b = c$. The additive identity 0 is unique.

Theorem T2 · Subtraction	For each a, b, exactly one x satisfies $a + x = b$. This x is denoted $b - a$; $0 - a = -a$.
Theorem T3	$b - a = b + (-a)$.
Theorem T4	$-(-a) = a$.
Theorem T5	$a(b - c) = ab - ac$.
Theorem T6	$0 \cdot a = a \cdot 0 = 0$.
Theorem T7 · Multiplication cancellation	If $ab = ac$ and $a \neq 0$, then $b = c$. The multiplicative identity 1 is unique.
Theorem T8 · Division	For $a \neq 0$, exactly one x satisfies $ax = b$. This x is denoted b/a; $1/a$ is the reciprocal of a.
Theorem T9	If $a \neq 0$, then $b/a = b \cdot a^{-1}$.
Theorem T10	If $a \neq 0$, then $\left(a^{-1}\right)^{-1} = a$.
Theorem T11 · Zero product	If $ab = 0$, then $a = 0$ or $b = 0$.
Theorem T12 · Negative signs	$(-a)b = a(-b) = -(ab)$; $(-a)(-b) = ab$. If $b \neq 0$, $-(a/b) = (-a)/b = a/(-b)$.
Theorem T13 · Equivalent fractions	If $b \neq 0$ and $c \neq 0$, then $a/b = (ac)/(bc)$.
Theorem T14 · Addition of fractions	If $b \neq 0$ and $d \neq 0$, then $a/b + c/d = (ad + bc)/(bd)$.
Theorem T15 · Multiplication of fractions	If $b \neq 0$ and $d \neq 0$, then $(a/b)(c/d) = (ac)/(bd)$.
Theorem T16 · Division of fractions	If $b \neq 0$, $c \neq 0$ and $d \neq 0$, then $(a/b)/(c/d) = (ad)/(bc)$.

Appendix A · Order axioms Ord1–Ord3

Appendix A label	Statement
Axiom Ord1	If a and b are positive, then $a + b$ and ab are positive.
Axiom Ord2	If $a \neq 0$, exactly one of a and $-a$ is positive.
Axiom Ord3	0 is not positive.

Appendix A · Order definitions: $a < b$ means $b + (-a)$ is positive; $b > a$ means $a < b$. Also, $a \leq b$ means $a < b$ or $a = b$, and $b \geq a$ means $a \leq b$. A number a is **negative** if $a < 0$, and **nonnegative** if $a \geq 0$.

Appendix A · Theorems T17–T27 (order)

Appendix A label	Statement
Theorem T17 · Trichotomy	Exactly one holds: $a < b$, $b < a$, or $a = b$.
Theorem T18 · Transitivity	If $a < b$ and $b < c$, then $a < c$.
Theorem T19	If $a < b$, then $a + c < b + c$.
Theorem T20	If $a < b$ and $c > 0$, then $ac < bc$.
Theorem T21	If $a \neq 0$, then $a^2 > 0$.
Theorem T22	$1 > 0$.
Theorem T23	If $a < b$ and $c < 0$, then $ac > bc$.
Theorem T24	If $a < b$, then $-a > -b$. In particular, if $a < 0$, then $-a > 0$.
Theorem T25	If $ab > 0$, then a and b are both positive or both negative.
Theorem T26	If $a < c$ and $b < d$, then $a + b < c + d$.
Theorem T27	If $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.

AXIOM **Appendix A · Axiom LUB · Least upper bound**
Every nonempty subset S of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$. Explicitly, let $B = \{u \in \mathbb{R} : \forall s \in S, s \leq u\}$.
If $S \neq \varnothing$ and $B \neq \varnothing$, then there is $\ell \in B$ such that $\forall u \in B, \ell \leq u$. This ℓ is the **least upper bound** (supremum) of S. This axiom holds for $\mathbb{R}$; the analogous statement for subsets and bounds in $\mathbb{Q}$ fails.

Lecture 4, slide 4 · Additional assumptions: equality is reflexive, symmetric and transitive for all objects; no integer lies strictly between 0 and 1; and $\mathbb{Z}$ is closed under addition, subtraction and multiplication. Lecture 1 additionally permits the even/odd dichotomy (Assumption 1, slide 27) and reduction of every rational to lowest terms (Assumption 2, slide 37).

65 · Theorem locator · Slide numbering

Theorems are stated with their proofs or definitions in the listed sections. Labels match the slides. **L** = lecture; **s** = slide.

Slide label	Topic	Source	Section
Theorem 2.1.1	Logical equivalences	L2 s25–26	§2
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